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The effectiveness of the Teddy Bear Hospital in improving health outcomes and wellbeing — a systematic review

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The effectiveness of the Teddy Bear Hospital in improving health outcomes and wellbeing — a systematic review

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Strength and Limitations to this Study

1. This review looks into the effectiveness of the Teddy Bear Hospital (TBH) on children's health outcomes and wellbeing

2. It followed established guidelines and identified 11 interventional studies to review the effectiveness

3. The limitations include a lack of high-quality studies available. The risk of bias for random sequence generation was unclear or high for nine out of 11 studies

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Abstract

Background

The healthcare setting is stressful for many people especially children. Efforts are needed to ease children's anxiety towards it. Teddy Bear Hospital (TBH) and medical play are measures used to facilitate children's exposure and experiences towards healthcare settings. Children will bring their soft toys and act as parents to the 'sick' teddies in the TBH. The objective of this systematic review is to evaluate the effectiveness of TBH in improving children's health outcomes and wellbeing.

Methods

Four electronic databases and reference lists of included studies were searched from inception to May 2020. We included pre-post studies, quasi-experimental, case-control studies, and randomised controlled trials (RCT) that discussed about TBH or medical play as an intervention. Those reported in a foreign language, systematic reviews, studies that involved sick patients, and that used interventions unlike the TBH were excluded. The quality of the included studies was assessed using the Cochrane Collaborations' "Risk of bias" tool.

Results

Eleven studies were included in this systematic review. Five were specifically on the TBH method, while the other six were of the same concept of medial play. Only three studies were randomised control trials. All the studies report more than one outcome, with most being positive outcomes. Three reports lower anxiety levels after the intervention. Two found better healthcare knowledge, with one of them reporting increased feelings of happiness towards visiting a doctor. Two studies found no change of anxiety or feelings, while another

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two found increased levels of fear and lowered mood after the medical play (which involved real medical equipment).

Conclusions

The concept of TBH had mostly positive outcomes. This involves lower anxiety levels and improved healthcare knowledge. Its effectiveness would need to be verified in future studies with a more robust methodology

Systematic review registration: PROSPERO2019 CRD42019106355

Keywords: Teddy Bear Hospital, children, health knowledge, anxiety, education

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BACKGROUND

Children visit healthcare facilities for many reasons. Among them is for regular check-ups, immunisation, or for receiving treatments. [1,2] These visits have been related to negative reactions among them.[2] Literature reports that around 30% of adults are affected by healthcare visits. So much so they are said to have white coat syndrome; rise in blood pressure during medical visits.[3] Like adults, children have a fear of the unknown. They tend to create fantasies and distort information due to lack of accurate ones which could result in fear and anxiety.[4] It has been reported that those children who have been well-informed, cope better and have less fear when faced with the real situation.[5] In view of this, various efforts were made to ease children's anxiety and fear towards healthcare setting and health personnel, one of them being the Teddy Bear Hospital (TBH). [6]

The TBH incorporates pretend hospital play. Children will bring their teddy bears to go through pretend health care services. They also incorporate an element of medical play which allows children to play and explore with medical equipment, mostly in the shape of toys. Medical play is a play that involves a fun activity that involves a medical theme. It is initiated by an adult but led by the child and its main aim is to reduce anxiety in healthcare settings. [1,7]

In the TBH role play, children will act as the teddy bears' parents and bring the sick teddy bear to visit a teddy doctor who is usually a medical student or a volunteer. The setting is in the form of a clinic or hospital where the play area is filled with pretend medical equipment. Most of the TBH done are done to children that are healthy such as kindergartens, or an open day in a specific health care centre as an interventional health education. [5,8–12] The objective

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of this role play intervention is to ease the child's anxiety when they are in the healthcare setting and at the same time to enhance their health knowledge. To our knowledge, there are currently no reviews available to evidently show that the TBH has achieved these outcomes.

The overall objective of this review is to determine the effectiveness of TBH in improving children's health outcomes and wellbeing.

METHODS

Study design

This systematic review was conducted and reported based on the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines. The PRISMA checklist for systematic review can be found in supplementary file 1 (S1). This review was registered with PROSPERO (CRD: 42019106355).

Search strategy

An electronic literature search was carried out in the following databases: MEDLINE (PubMed), CINAHL, Scopus, and Google Scholar from inception to September 2019. We also searched the reference lists and citations of included studies using the forward and backward citation tracking method. Grey literature search was also performed in 'The Grey Literature Report', and OpenGrey. We have rerun the search again for all databases in mid of May 2020 to find for new studies, however, only one new study identified that fitted our inclusion criteria.

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A combination of Medical Subject Heading (MeSH) and free text terms were used to define the search (“child”[MeSH Terms]) OR (“adolescent”[MeSH Terms])) OR (“young adult”[MeSH Terms])) OR (“child*”[Title/Abstract])) OR (“adolescen*”[Title/Abstract])) OR (“preschool child*”[Title/Abstract])) AND (“Teddy bear hospital*”[Title/Abstract])) OR (“Teddy bear clinic*”[Title/Abstract])) OR ("pediatrics/education”[MeSH Terms])) OR (“play therapy/methods”[MeSH Terms). We limited the search to include only journal articles in English. The general search terms for MEDLINE (PubMed) were reviewed by the authors and a medical librarian and adapted for each database.

Inclusion and exclusion criteria

All pre-post studies, quasi-experimental, case-control studies, and randomised controlled trials (RCT) done in both the community and health settings discussing about the Teddy Bear Hospital or medical play interventions involving dolls or soft toys and medical equipment were included. The medical play interventions involving dolls or soft toys and medical equipment were included as it carried the same concept as the TBH. Furthermore, the included studies have to report on the main outcomes which include health outcomes such as health knowledge, and psychological impacts. We excluded studies that were reported in a foreign language, systematic reviews, sick children in hospital, and studies that use different interventions other than the TBH concept of soft toys and medical play equipment.

Study selection

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The articles obtained from the literature search were imported into EndNote© and duplicated articles were removed. The list of remaining articles was then divided into two to be screened by two teams of reviewers. The two teams of reviewers each consisted of two reviewers (DR and AAR) and (NHS and CAT). Each of the reviewers screened through the titles and abstract of the articles independently and then compare with the other team members. Any discrepancies of the inclusion were discussed between the 2 team members and the decision was based on the consensus agreed after discussion. In cases where the consensus unable to reach within the team, it was then brought to discussion with the other team members for discussion. Once identified for relevance, the full papers were then retrieved. These papers were then again independently reviewed by each member in a team independently using a spreadsheet of the same format. The spreadsheets would later be compared. If there were any discrepancies on the articles between the 2 reviewers in a team, the other team would be consulted, and a decision would be made. All the agreed included articles were then later combined into one spreadsheet.

Data extraction

Data were extracted from the relevant studies by one reviewer (AAR) and checked by a second reviewer (CAT) on another spreadsheet. Any disagreements at any stage were resolved through discussion with a third reviewer (NHS or DR). Data on the following subjects were extracted: setting, objective, population, intervention/approach, measuring tools, outcomes of interest, N/N intervention, N/N control, results, and conclusion.

Quality assessment

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The quality of the methodology used in the studies was assessed using The Cochrane Collaborations' "Risk of bias" tool. [13] Each included study was assessed based on a number of characteristics such as a) selection bias (randomisation and allocation concealment) b) performance bias (blinding of participants or personnel), c) detection bias (blinding of outcome assessment), d) attrition bias (incomplete outcome data both long and short term), and e) reporting bias (selective reporting). All studies were critically appraised in duplicates by the two reviewers (AAR and CAT) and presented to the research team members disagreements were discussed, and a consensus was sought until a combined assessment result was deduced.

Data Analysis

Data extracted were all analysed via narrative analysis. Meta-analysis was not performed as the tools for measuring the outcomes vary among the studies, thus unable to pool the results together.

RESULTS

Study selection

A total number of 1154 of studies were yielded from the literature search conducted from in September 2019. 1140 were identified after excluding 14 duplicates. Of those, 1091 were excluded and the remaining 49 studies were available to be retrieved in full text. 10 out of 49 studies were then selected based on the inclusion and exclusion criteria. Among the studies excluded were systematic reviews, sick children in the hospital, and intervention that were not TBH or similar. Search was updated and filtered from the year 2019 to 2020 and found

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one new eligible study to be included. The PRISMA flow diagram and the reasons for exclusion are shown in Figure 1.

Description of included studies

Out of 11 included studies, three were RCTs, [1,14,15] three were non-randomised trials with controlled group [5,8,16] and the remaining were pre-post studies. [9,10,17–19] These studies had a population groups ranging from 30-543 participants within the age group of 2-16 years old. Six of the included studies involved TBH as an intervention [5,8–10,17,18] while the remaining four studies discussed medical play intervention. [1,14–16,19] Among the 11 studies, four were conducted in North America (three in the United States of America, [1,14,15] one in Canada[18]), four in Europe (one in the United Kingdom, [10] one in the Netherlands, [9] one in Germany, [5] one in France [19]) and one each in Singapore, [17] Brazil, [16] and Israel. [8] An overview of the studies is shown in supplementary file 2 (S2).

Quality of studies

The risk of bias for random sequence generation was unclear or high for 10 out of 11 studies, with the exception of one study which was an RCT. [1] All studies had a high risk of bias for allocation concealment and performance bias. Most studies did not do randomisation for their participants while concealing the intervention could not be done with TBH or medical play. With regards to detection bias, only 20% of the studies had a low risk of bias where the assessors were blinded from the group arrangements of the interventions. [1,15] Three studies were found to have a high risk of reporting bias as a proportion of data was excluded from the analysis of results. [9,10,17] All studies had a low risk of bias for selective reporting and no

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other biases were found in all the studies. Figure 2 shows the proportion of studies with low risk, unclear risk, and high risk of bias. The assessment of biases of individual studies is shown in supplementary file number 3 (S3).

Results of individual studies

Children's feelings

Eight studies measured the children's feelings which are level of anxiety and fear as the outcomes. Six studies (One non-randomised trial with the controlled group and four pre-post studies) concluded that the children had more positive feelings with lower anxiety levels after the TBH. [8–10,17,19] Another three studies (two RCTs and one pre-post study) concluded otherwise, where the RCTs [1,15] showed that medical play intervention had lowered the children's mood and increased the level of fear. The pre-post study [18] showed no effect of TBH on the children's level of fear.

Children's behaviour

Two RCTs found that the participants in the medical playgroup were more difficult during triage procedures as compared to the medical information group, typical playgroup, and control group. [1,15] Another two studies (one RCT and one non-randomised trial with the controlled group) showed positive outcomes for medical play intervention[14,16]. They report the children to be more alert and had a greater acceptance of the procedure (immunisation) after the intervention. [14,16]

Children's healthcare knowledge

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Two studies discussed the effect of the TBH on children's healthcare knowledge. Both of the studies (one non-randomised trial with the controlled group and one pre-post study) concluded that children had better healthcare knowledge after the TBH intervention. [5,17]

A detailed summary of the results of individual studies is included in supplementary file number 4 (S4).

DISCUSSION

TBH is a global project with the primary aim to reduce children's fear and anxiety towards healthcare professionals and hospitals through pretend medical play. This strength in this review represents the first effort to summarise the effectiveness of the TBH and its concept of medical play in improving children's health outcomes and wellbeing. This has become a popular concept to educate children and a review is important to look into its effectiveness.

In general, the studies included in this review showed a high level of selection bias and performance bias. However, this finding is unavoidable due to the nature of the intervention which involved children participating in physical activities related to the intervention of TBH or medical play. Detection bias was high as most of the studies did not describe the methods of blinding of outcome assessors.

Our review suggests that medical play intervention has a positive effect on children's health outcomes and wellbeing particularly in terms of reducing anxiety and fear. It can also significantly improve children's healthcare knowledge. However, two studies conducted by Burns-Nader et. al concluded a contradictory effect of medical play intervention on children's health outcomes. [1,15] The disagreements in the results suggest the need for further research to look into the components of medical play intervention that affect the final results of the

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intervention. The medical play in both studies involved real medical equipment which in contrast to the equipment used in TBH which are mostly toys made in the shape of medical equipment.

Further studies should explore whether the physical involvement of the children in role-play or the medical information obtained through the activities serves as the main component in reducing children's fear and anxiety.

We hypothesise that the lack of consistency in the conclusion might also be due to the involvement of third-party personnel in assessing the children's response instead of self-reporting outcomes by the children themselves. The children's distress behaviours were assessed by the nurses during triage, may introduce fear to the children when compared to other studies, where the outcomes were usually reported by the children themselves or assessed by non-medical related personnel.

Significant improvements in healthcare-related knowledge were seen in studies done by Leonhardt et. al and Victorine et. Al [5,10]. This shows that TBH could be a good method to be adopted for delivering programmes among children to raise their awareness on health-related issues. For example, TBH can be used to teach children on healthy living in order to prevent childhood obesity that is on the rise globally. Early intervention in terms of education is important to prevent adverse health risks later in adulthood. [20] The implementation of these programmes, however, will need cooperation from various disciplines like healthcare professionals, parents, teachers, and authority bodies.

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Other clinical implications of this study include the potential use of the TBH as a tool to teach children on personal safety to prevent child sexual abuse in view of the drastic rise in the number of child sexual abuse cases worldwide and nationwide. [21,22]

Limitations

This review has a relatively small number of included studies and most of the studies were pre-post studies. This limitation may impact on the generalisability of the findings in future implementations. Secondly, this study has excluded studies published in a foreign language which may impact on the validity of the results from this review.

Implications of the review on practice and further research

This review found that the concept of the TBH, which involves pretend medical play and using toy medical equipment has lowered the children's anxiety level and improved the children's healthcare knowledge. This could be one of the potential public health strategies in improving health intervention specifically as a health education tool to the pre-school and primary school children in a various topic such as immunisation, healthy lifestyle, and sexual and physical abuse.

The review found that the quality of the studies was less satisfactory as most of the studies were pre-post studies. Future studies would need to be carried out with a more robust methodology to examine the effect of TBH and medical play related intervention.

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CONCLUSION

This review found that the concept of the TBH and medical plays has a positive outcome on the children's emotions and healthcare knowledge in most of the studies. Its effectiveness would need to be verified in future studies with a more robust methodology.

List of abbreviations

TBH: Teddy Bear Hospital; PRISMA: Preferred Reporting Items for Systematic Reviews and Meta-Analyses; RCT: Randomised Controlled Trials

DECLARATIONS

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Availability of data and materials

All data and materials analysed are included in this article and its additional files.

Competing interests

The authors declared to have no competing interests.

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Authors' contributions

AAR, CAT, NHS, and DR designed the study. RHS was responsible for conducting the literature search. All authors participated in the selection of included studies. AAR, CAT, NHS, and DR were responsible for data extraction, study appraisal, data analysis and results interpretation. All authors contributed to drafting the manuscript. The final manuscript was approved by all authors.

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REFERENCES

1. Burns-Nader, E. S. The effects of medical play on reducing fear, anxiety, and procedure distress in school-aged children going to visit the doctor. (University of Alabama Libraries, 2011).
2. Lerwick, J. L. Minimizing pediatric healthcare-induced anxiety and trauma. *World J. Clin. Pediatr.* **5**, 143–150 (2016).
3. Alves, L. M. M., Nogueira, M. S., Godoy, S. de, Hayashida, M. & Cárnio, E. C. Prevalence of white coat hypertension in primary health care. *Arq. Bras. Cardiol.* **89**, 28–35 (2007).
4. Santen, L. & Feldman, T. Teddy bear clinics: a huge community project. *MCN Am J Matern Child Nurs* **19**, 102–106 (1994).
5. Leonhardt, C., Margraf-Stiksrud, J., Badners, L., Szerencsi, A. & Maier, R. F. Does the 'Teddy Bear Hospital' enhance preschool children's knowledge? A pilot study with a pre/post-case control design in Germany. *J. Health Psychol.* **19**, 1250–1260 (2014).

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6. Kaufman, J., Modak, M. & Moylan, S. the Teddy Bear Hospital In Australia. *J Paediatr Child Health* **48**, 541–542 (2012).

7. McCue, K. Medical Play: An Expanded Perspective. *Child. Health Care* **16**, 157–161 (1988).

6. Bloch YH, Toker A. Doctor, is my Teddy Bear Okay? A "Teddy Bear Hospital" as a Method to Reduce Children's Fear of Hospitalization. Isr Med Assoc J. 2008 Sep;10(9):597–9.

9. Ottenheim, M. & Sommeren, R. van. The Teddy Bear Hospital. How to influence the attitude of kindergarten children towards doctors, hospitals and falling ill. /paper/The-Teddy-Bear-Hospital.-How-to-influence-the-of-Ottenheim-Sommeren/778b811ab7b57a3da3a35d58a664ed3a64d7f774 (2018).

10. Victorine, A., Wong, Z., Geilani, J. & Charters, A. Teddy Bear Hospital (TBH)–reducing children's fear of doctors and hospital environment. *PEDIATRICS* **110**, e33 (2002).

11. Ong, L., Chua, K. H., Soh, J. Y. & Aw, M. M. H. Y. Healthcare encounters in young children: Impact of Teddy Bear Hospital, Singapore. *Asia Pac. Sch.* **3**, 24–30 (2018).

12. Dalley, J. S., Creary, P. R., Durzi, T. & McMurtry, C. M. An Interactive Teddy Bear Clinic Tour: Teaching Veterinary Students How to Interact with Young Children. *J Vet Med Educ* **44**, 302–315 (2017).

8. Higgins JPT, Altman DG, Sterne, JAC (editors). Chapter 8: Assessing risk of bias in included studies. In: Higgins JPT, Green S (editors). Cochrane Handbook for Systematic Reviews of Interventions. Version 5.1.0 [updated March 2011]. The Cochrane Collaboration, 2011. Available from www.cochrane-handbook.org.

14. Brown, J. C. Group medical play for reducing stress and improving mood in children going to visit the pediatrician. (University of Alabama Libraries, 2012).

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15. Burns-Nader, S., Hernandez-Reif, M. & Thoma, S. J. Play and Video Effects on Mood and Procedure Behaviors in School-Aged Children Visiting the Pediatrician. *Clin. Pediatr. (Phila.)* **52**, 929–935 (2013).

12. Pontes JE, Tabet E, Folkmann MÁ dos S, Cunha ML da R, Almeida F de A. Therapeutic play: preparing the child for the vaccine. Einstein São Paulo. 2015 Jun;13(2):238–42.

17. Ong, L., Chua, K. H., Soh, J. Y. & Aw, M. M. H. Y. Healthcare encounters in young children: Impact of Teddy Bear Hospital, Singapore. *Asia Pac. Sch.* **3**, 24–30 (2018).

18. Dalley, J. S. & McMurtry, C. M. Teddy and I get a check up: A pilot educational intervention teaching children coping strategies for managing procedure-related pain and fear. *Pain Res. Manag.* (2015).

19. Morel, B. *et al.* Impact on child and parent anxiety level of a teddy bear-scale mock magnetic resonance scanner. *Pediatr. Radiol.* **50**, 116–120 (2020).

20. Reilly, J. J. Childhood Obesity: An Overview: Childhood Obesity. *Child. Soc.* **21**, 390–396 (2007).

21. Pereda, N., Guilera, G., Forns, M. & Gómez-Benito, J. The prevalence of child sexual abuse in community and student samples: A meta-analysis. *Clin. Psychol. Rev.* **29**, 328–338 (2009).

22. Singh, H. A., Yiing, W. W. & Nurani, H. N. K. Prevalence of childhood sexual abuse among Malaysian paramedical students. *Child Abuse Negl.* **20**, 487–492 (1996).

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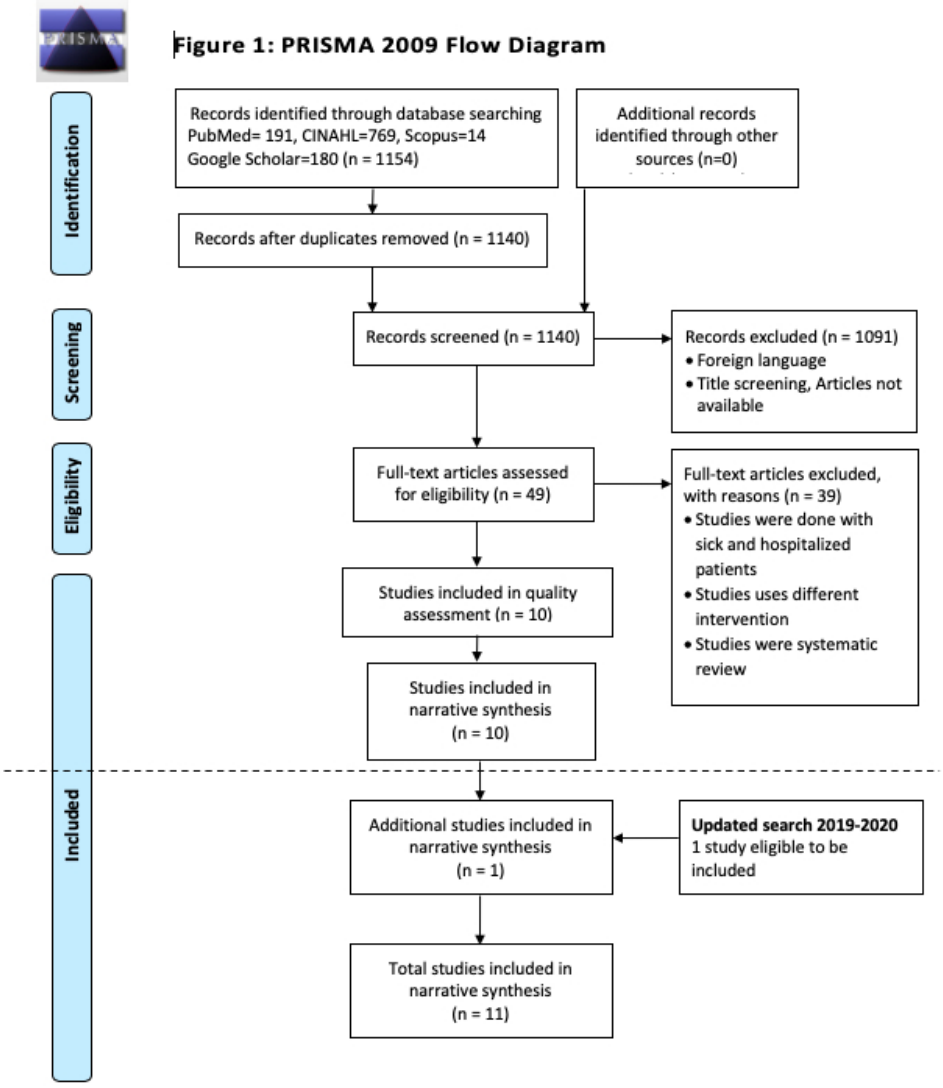
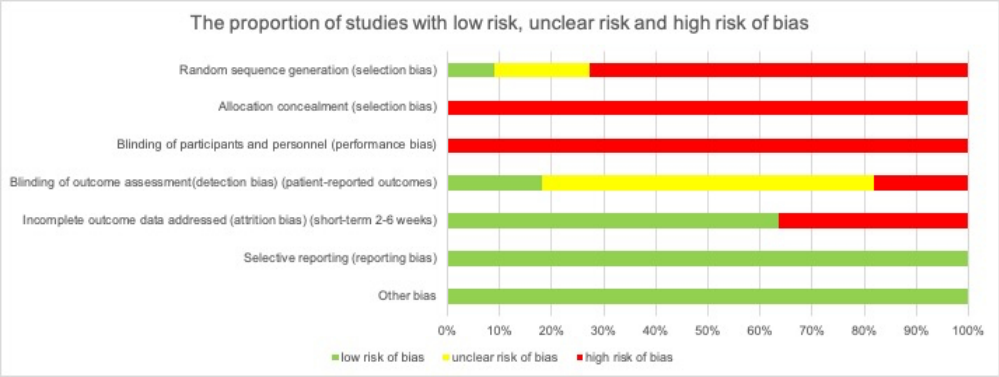


figure 1: PRISM FLOW DIAGRAM

226x252mm (72 x 72 DPI)



The proportion of studies with low risk, unclear risk, and high risk of bias

251x94mm (72 x 72 DPI)



PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	1-2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	3
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	3
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	4
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	4-5
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	4
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	4
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	5
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	5
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	N/A
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	6
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	N/A
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I_2) for each meta-analysis.	N/A

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	N/A
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	N/A
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	6-7
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	7
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	7-8
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	8-9
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	N/A
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	N/A
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	N/A
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	9-10
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	11
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	11
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	12

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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Supplementary file number 2 (S2): Overview of included studies.

Study (Author, Year, Country)	Population	Setting	Intervention	Outcome measured	Results	Outcome assessment
Randomised Controlled Trials						
1. Burns-Nader ES, 2011, USA (10)	Children aged 5.5 – 12 years	Paediatric clinic	3 intervention groups: 1. Medical play group (doll and authentic medical equipment) 2. Medical information video group 3. Typical play group	1. Children's anxiety level 2. Children's level of fear 3. Children's distress behaviours	• No effect of types of activity on children's anxiety level • Medical play group had increased level of fear after the activity and after doctor's visit • Medical play group was more difficult (higher mean score in nurses' overall rating of difficulty) during triage procedure as compared to medical information video group, typical play group and control group.	Self-report Self-report Third-party evaluation

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			Control: viewed video on safari life			
2. Burns-Nader S, 2013, USA (11)	Children aged 5.5 – 12 years	Paediatric clinic	3 intervention groups: 1. Medical play group (doll and authentic medical equipment) 2. Medical information video group 3. Typical play group Control: viewed video on safari life	1. Children's mood 2. Children's behaviours during procedures	• Medical play lowered children's mood after the activity and after the doctor's visit • Medical play group was more difficult (higher mean score in nurses' overall rating of difficulty) during triage procedure as compared to medical information video group, typical play group and control group.	Self-report Third-party evaluation
3. Brown JC, 2012, USA (9)	Children aged 3 - 5 years	Paediatric clinic	Medical collage group (medical play) Control: No intervention	1. Children's behaviours: - State - Affect - Activity - Vocalisation	• Medical collage (medical play using medical equipments) group was more alert, and had more activity and more vocalizations after the intervention. Levels of activity and vocalization went back down after doctor's visit	Third-party evaluation

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				- Fidgeting/Squirming 2. Children's feelings	• No significant change on children's feelings	Self-report
Non-randomised trials with controlled group						
4.Leonhardt C, 2014, Germany (5)	Children aged 4.5 - 5.5 years	Kindergarten	TBH Control: No intervention	Children's medical knowledge	• Children in TBH group had better knowledge than the control group one week after the intervention	Self-report
5. Bloch YH, 2008, Israel (6)	Children aged 3 – 6.5 years	TBH at healthcare centre	TBH Control: No intervention	Children's anxiety about hospitalisation	• Children in TBH group had lower levels of anxiety than the control group one week after the intervention	Self-report
6. Pontes JE, 2015, Brazil (12)	Children aged 3 - 6 years and 11 months	Paediatric clinic	Instructional therapeutic play (medical play) Control: No intervention	1. Children's body movement during procedure 2. Children's verbal expression during procedure 3. Children's emotional expression during procedure	• Instructional therapeutic play (medical play) group had greater acceptance of the procedure as compared to control group: - Staying still - Collaborating - Smiling - Being at ease	Third-party evaluation

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Pre-post Studies						
7. Morel B 2019, France	Children aged 4-16 years	Radiology waiting room	Medical play: Mock MRI in teddy bear size scale	1.Children's anxiety levels pre and post mock period in the waiting room 2.Children's anxiety levels pre and post mock period in the preparation room 3.Children's anxiety levels pre and post mock period after the examination	• the anxiety level estimated by children was significantly lower after the explanations in the post-mock period as compared to the level estimated in the waiting room (no difference for waiting room versus after explanation in the pre-mock condition) • In pre- and post-mock periods, a significant difference between anxiety score in the waiting room and after the exam was also observed • a difference was also observed in the preparation room between anxiety score in the waiting room and the score after explanation only in girls	Self report
8. Ottenheim M, 2018, Netherlands (13)	Children aged 4 - 7 years	TBH at healthcare centre	TBH	1. Children's attitude towards visiting a hospital 2. Children's attitude towards doctors 3. Children's attitude towards falling ill	• Reduced anxiety level towards doctors after TBH • Reduced anxiety level towards falling ill after introductory lesson and after TBH • Attitude towards visiting a hospital, and attitude towards hospitals had no significant finding	Self-report

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				4. Children's attitude towards hospitals		
9. Ong L, 2018, Singapore (14)	Children aged 5 - 8 years	TBH at kindergarten and primary schools	TBH	1. Childhood feelings towards healthcare services 2. Willingness of health-seeking behaviour 3. Children's medical knowledge	• Children's feeling of happiness towards visiting the doctor increased after TBH • Children who felt happy to visit the hospital increased after TBH • Increased healthcare knowledge after TBH	Self-report
10. Dalley JS, 2016, Canada (15)	Children aged 5 - 10 years	TBH at healthcare centre	TBH	1. Children's level of fear 2. Children's level of expected pain 3. Children's coping strategies	• No effect on level of fear • Decreased mean expected pain ratings after TBH • Higher frequency of intervention-taught coping strategies after TBH	Self-report
11. Victorine A, 2002, UK (7)	Children aged 2 – 12 years	TBH at hotel	TBH	1. Children's attitude towards doctor 2. Children's attitude towards hospital	• Increased proportion of children with positive feelings towards doctor and hospital after TBH **No further test was done to determine whether results were statistically significant*	Self-report

A summary of the author, year of publication, country of publication, population, setting, intervention, outcome measured, results and outcome assessment of each included studies.

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Supplemetary file number 3 (S3) Quality Assesment

cochrane risk of bias	Ottenheim 2018		Lynn 2018	
entry	**Judgement**	**Support for judgment**	**Judgement**	**Support for judgment**
Random sequence generation (selection bias)	High Risk	No randomisation - 794 kindergarten children (age 4-7) participated in the Teddy Bear Hospital at Leiden University Medical Centre (Leiden, the Netherlands). Data was collected on 543 children attending elementary schools in the Leiden region.	High Risk	No randomisation - cross sectional descriptive study of children taking part in TBH sessions between 28 March and 20 May 2016. All children aged 5 to 8 years old were recruited from kindergartens and primary schools which TBH sessions were organized. (only children with parental consent were recruited)
Allocation concealment (selection bias)	High Risk	No allocation concealment - Questionnaire is given to children who participated in TBH	High Risk	No allocation concealment - Questionnaire is given to children who participated in TBH
Blinding of participants and personnel (performance bias)	High Risk	No blinding - arranged for all participating schools a box with materials and a concrete lesson for the teachers with the intention that this lesson was given before the visit to the Teddy Bear Hospital.	High Risk	No blinding - Questionnaire was administered right before TBH and immediately after TBH

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Blinding of outcome assessment(detection bias) (patient-reported outcomes)	Unclear Risk	The study did not address blinding of outcome assessors	Unclear Risk	The study did not address blinding of outcome assessors
Blinding of outcome assessment (detection bias) (all-cause mortality)	N/A	N/A	N/A	N/A
Incomplete outcome data addressed (attrition bias) (short-term 2-6 weeks)	Unclear Risk	The four questions of the questionnaire were analysed separately so no effect on the interpretation of the results are expected however missing data percentage ranges from 17.5%-19.89%. Data analysis was done only for completed questionnaire.	High Risk	We excluded children who had incomplete data, such as missing answers to any of the multiple choice question in the survey. Children who completed the multiple choice component but did not answer the open-ended questions were still included. (Excluded 82 out of 334 - 24.55%)
Incomplete outcome data addressed (attrition bias) (long-term >6 weeks)	N/A	N/A	N/A	N/A
Selective reporting (reporting bias)	Low Risk	All outcomes of interests were reported	Low Risk	All outcomes of interests were reported
Other bias	Low Risk	None	High Risk	Marion M. Aw is the clinical mentor for the medical students in charge of the Teddy Bear Hospital, Singapore. Ong Lynn and Chua Khoon Han were both involved in TBH as part of the organising committee during the course of this study.

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Dalley 2016		Pontes 2015		L
Judgement	**Support for judgment**	**Judgement**	**Support for judgment**	**Judgement**
High Risk	No randomisation - A convenience sample was recruited from groups of children already scheduled to attend a Teddy Bear Clinic Tour at the Hill's Pet Nutrition Primary Healthcare Centre of the Ontario Veterinary College (OVC)	High Risk	No randomisation - subjects were not randomly selected, and therefore this is not a true experimental study.	High Risk
High Risk	No allocation concealment - Questionnaire is given to children who participated in TBH	Unclear Risk	Method of concealment was not described - Children were distributed to an experimental group, with 30 children taking part in an instructional therapeutic play session before vaccination, and to a control group, with 30 children who did not receive any kind of previous emotional	High Risk
High Risk	No blinding - Data were collected from participating children immediately before and immediately afer the tour	High Risk	Blinding is not posible - children in experimental group have to participate in instructional therapeutic play sessions.	High Risk

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Unclear Risk	No mention about blinding - Two undergraduate research assistants independently coded all of the participants' responses. Disagreements were resolved by discussion and consensus.	High Risk	Reactions were observed by researcher.	Unclear Risk
N/A	N/A	N/A	N/A	N/A
Low Risk	A total of 71 participants had parental consent and assented to participate; no data were excluded from the final analysis.	Low Risk	No data were excluded from the final analysis	Low Risk
N/A	N/A	N/A	N/A	N/A
Low Risk	All outcomes of interests were reported	Low Risk	All outcomes of interests were reported	Low Risk
Low Risk	None	Low Risk	None	Low Risk

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eonhardt 2014		Burns 2013	
Support for judgment	**Judgement**	**Support for judgment**	**Judgement**
No randomisation - Randomization was not possible because the TBH is an established local project and any kindergarten in the area is free to decide whether it will take part or not.	Unclear Risk	The study did not specify how randomisation was done - Child participants were randomly assigned to 1 of 4 groups.	Unclear Risk
No allocation concealment - experimental group was recruited from kindergarten who registered to participate in TBH and control group from non-participating facilities	Unclear risk	Method of concealment was not described	Unclear risk
No blinding - Children were assessed in small groups in the kindergarten during the week prior to the intervention and 1 week after visiting the TBH. Control children were interviewed twice as well and at a similar time interval.	High Risk	Blinding is not possible - children were either involved in a medical play group, a medical information video group, a typical play group or a control group whose participants viewed a video on safari life.	High Risk

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No mention about blinding - The interviews with the children were conducted by two trained students who were supervised by an experienced psychologist.	Low Risk	The nurses who rated the children's behaviour were unaware of the group assignments	High Risk
N/A	N/A	N/A	N/A
For reasons of non-appropriate age and outlier data, 8 children were excluded from final analysis. (<15%)	Low Risk	No data were excluded from the final analysis	Low Risk
N/A	N/A	N/A	N/A
All outcomes of interests were reported	Low Risk	All outcomes of interests were reported	Low Risk
None	Low Risk	None	Low Risk

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Brown 2012	Burns Nader 2011		\
Support for judgment	**Judgement**	**Support for W2**	**Judgement**
The study did not specify how randomisation was done - Children were randomly assigned to participate in a medical collage activity or a control group.	Low Risk	Assignment to groups was determined using random sample numbers generated by a computer software program	High Risk
Method of concealment was not described	Unclear risk	Method of concealment was not described	High Risk
Blinding is not possible - study group has to participate in medical collage activity	High Risk	Blinding is not possible - children were either involveld in a medical play group, a medical information video group, a typical play group or a control group whose participants viewed a video on safari life.	High Risk

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Reactions were observed by researcher. (Behaviour Observation Scale)	Low Risk	A researcher, blind to the children's group assignment, rated the children's affect and behaviors. The developmental interview was administered to the parent by a researcher blind to the children's group assignment.	Unclear Risk
N/A	N/A	N/A	N/A
No data were excluded from the final analysis	Low Risk	No data were excluded from the final analysis	High Risk
N/A	N/A	N/A	N/A
All outcomes of interests were reported	Low Risk	All outcomes of interests were reported	Low Risk
None	Low Risk	None	Low Risk

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Victorine 2002		Yuval 2008	
Support for judgment	**Judgement**	**Support for judgment**	**Judgement**
No randomisation - We estimated a total of about 124 children attended this Teddy Bear Hospital. Every child was given a questionnaire to fill-in prior to attending Teddy Bear Hospital and after the experience.	High Risk	No randomisation - Forty-one preschool children, age 3–6.5 years from three kindergartens in Beer Sheva, Israel, took part in this study. Fifty preschool children, age matched and from a similar residential area, served as the control group.	high risk
No allocation concealment - Questionnaire is given to children who participated in TBH	High Risk	No allocation concealment - intervention group was recruited from kindergarten who participated in TBH and control group from a similar residential area	high risk
No blinding - Every child was given a questionnaire to fill-in prior to attending Teddy Bear Hospital and after the experience.	High Risk	No blinding - a simple one-item visual analog scale of anxiety about hospitalization was assessed individually one day prior to the intervention and again a week after the intervention when the children were in kindergarten.	high risk

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The study did not address blinding of outcome assessors	Unclear Risk	The study did not address blinding of outcome assessors	unclear risk
N/A	N/A	N/A	N/A
Total 124 questionnaires, questionnaires numbered 86-124 were not found. Out of the 85 questionnaires, 28 of the questionnaires were not filled in fully. Therefore, only 57 fully completed questionnaires were considered for this study. (Excluded 67 out of 124 - >15%)	Low Risk	Three children in the intervention group (7%) and four in the control group (8%) were excluded from the study because of one or more previous hospitalizations.	high risk
N/A	N/A	N/A	N/A
All outcomes of interests were reported	Low Risk	All outcomes of interests were reported	Low risk
None	Low Risk	None	low risk

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Morel 2019
Support for judgment
no randomisation or risk
This prospective controlled trial was performed from March 2018 to March 2019. All children ages 4 to 16 years who presented to our ambulatory tertiary center for an MRI scan were included, provided that they had a normal level of understanding expected for their age and had parents who agreed to participate in the research.
no allocation concealment. Ninety-one children (46 girls and 45 boys) and their parents were included (44 in the pre-mock and 47 in the post-mock period). Questionnaires were given randomly in 20% of the cases.
no blinding. All children ages 4 to 16 years who presented to our ambulatory tertiary center for an MRI scan were included, provided that they had a normal level of understanding expected for their age and had parents who agreed to participate in the research.

	The study did not address blinding of outcome assessors
	N/A
	Questionnaires were given randomly in 20% of the cases. Participation rate was 95%- no reasons to why this happened.
	N/A
	All outcomes of interests were reported
	none

cochrane risk of bias	Ottenheim 2018		Lynn 2018	
entry	**Judgement**	**Support for judgment**	**Judgement**	**Support for judgment**
Random sequence generation (selection bias)	High Risk	Participants were selected based on their attendance to teddy bear hospital. No randomisation	High Risk	All the children from all the schools and kindergardens that has participated in TBH were recruited with the exception of children without parental consent. No randomisation.
Allocation concealment (selection bias)	High Risk	The same questions were used thrice, from pre exposure to TBH, to post TBH lecture and post TBH participation. No allocation concealment.	Low risk	Each children were assigned a coded number without identifier.

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Blinding of participants and personnel (performance bias)			High Risk	A lesson regarding the concept of TBH was provided to children immediately after TBH session to assess knowledge acquired and their change of feelings towards healthcare.
	High Risk	No Blindng was applied. All the participants of TBH was included in selection with a prior lesson before participation. Despite that, a significant improvement was noticed only after the intervention which is T2. For question one, significant difference were seen between T2 and T1, T2 and T0, but no difference were seen between T1 and T0. Outcome is not influenced by the lack of blinding.		
Blinding of outcome assessment(detection bias) (patient-reported outcomes)	Unclear risk	Insufficient details for judgement	Unclear Risk	The study did not adress this outcome.
Blinding of outcome assessment (detection bias) (all-cause mortality)	N/A	N/A	N/A	N/A

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Incomplete outcome data addressed (attrition bias) (short-term 2-6 weeks)	Low risk	There were missing datas in terms of non balanced participants between each questions. But the questions were analysed separately and the insufficient details is not enough to produce clinically relevant impact on the study .	High Risk	We excluded children who had incomplete data, such as missing answers to any of the multiple choice question in the survey. Children who completed the multiple choice component but did not answer the open-ended questions were still included.
Incomplete outcome data addressed (attrition bias) (long-term >6 weeks)	N/A	N/A	N/A	N/A
Selective reporting (reporting bias)	Low Risk	All the outcomes were reported in a prespecifed way	Low Risk	All outcomes were reported in a prespecified way
Other bias	Low Risk	None	Low Risk	None

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Dalley 2016		Pontes 2015		L
Judgement	**Support for judgment**	**Judgement**	**Support for judgment**	**Judgement**
High Risk	A convenience sample was recruited from groups of children already scheduled to attend a Teddy Bear Clinic Tour at the Hill's Pet Nutrition Primary Healthcare Centreof theOntarioVeterinaryCollege (OVC).	High Risk	subjects were not randomly selected, and therefore this is not a true experimental study. No randomization	High Risk
Unclear risk	The allocation concealment method was not sufficiently described.	High Risk	Data were recorded in a form with information about identification of child and the reactions during vaccination. No allocation concealment was practiced	Low Risk

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High Risk	The data was collected from participants before and immediately after the tour.	High Risk	After informing children and their legal guardians about the study and their participation, those selected for the experimental group were invited to play. No blinding	High Risk
Unclear risk	Insufficient details for judgement	High Risk	The children's reaction were observed by the researches themselve while vaccinating.	High Risk
N/A	N/A	N/A	N/A	N/A

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Low Risk	A total of 71 participants had parental consent and assented to participate; no data were excluded from the final analysis.	Low Risk	No incomplete datas were detected.	High Risk
N/A	N/A	N/A	N/A	N/A
Low Risk	All outcomes were reported in a prespecified way	Low Risk	All outcomes were reported after a statistical analysis through Fisher	Low Risk
Low Risk	None	Low Risk	None	Low Risk

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.eonhardt 2014		Burns 2013	
Support for judgment	**Judgement**	**Support for judgment**	**Judgement**
Randomization was not possible because the TBH is an established local project and any kindergarten in the area is free to decide whether it will take part or not. No randomization	High Risk	Seventy-two caregiver–child dyads were recruited to participate, following institutional review board approval. The children (n = 33 males) were attending a scheduled, or walk-in, doctor's appointment at a pediatric clinic in a city in the southeast region of the United States, for either a well checkup (ie, school physical, vaccinations, etc) or for an ill checkup (ie, respiratory infection, skin rash, etc). No	Unclear Risk
Children were assessed in small groups (up to 5 children) in the kindergarten during the week prior to the intervention and 1 week after visiting the TBH. No individual identifiers involved.	Low Risk	Child participants were randomly assigned to 1 of 4 groups: (a) a medical play group, where children were given a doll and authentic medical materials to play with (ie, stethoscope, otoscope, tongue depressor, etc); (b) a medical information video group, in which children were shown a taped medical play session; (c) a typical play group, where children played a board game; or (d) a control group whose participants viewed a	Low Risk

Children were assessed in small groups (up to 5 children) in the kindergarten during the week prior to the intervention and 1 week after visiting the TBH. No individual identifiers involved. Control children were interviewed twice as well and at a similar time interval (about 21 days). No blinding	Low Risk	Upon participating in their assigned activity, the children underwent a doctor's checkup where a trained nurse will conduct a behavioural assessment on the children.	Low Risk
The interviews with the children were conducted by two trained students who were supervised by an experienced psychologist.	High Risk	Because there will be trained nurse assigned to conduct the behavioural assessment on the children.	High Risk
N/A	N/A	N/A	N/A

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For reasons of nonappropriate age and outlier data, 8 children were excluded from final analysis (N = 131). Attritioan bias	Unclear risk	The exclusion of data was not mentioned.	Unclear risk
N/A	N/A	N/A	N/A
All outcomes were reported in a prespecified way.	Low Risk	All outocmes were reported in a prespecified way	Low Risk
None	Low Risk	None	Low Risk

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Brown 2012	Burns Nader 2011		
Support for judgment	**Judgement**	**Support for judgment**	**Judgement**
A total of 30 children ranging in age from three to five were recruited with parental consent and participant (verbal) assent from a pediatric clinic in Tuscaloosa, AL. But there was no explanantion on the mode of selection.	High Risk	No randomisation. The children (n= 33 males) were attending a scheduled, or walk-in, doctor's appointment at a general medical school pediatric clinic.	High Risk
The children were randomly assigned to participate in medical collage activity or control group.	Low Risk	Child participants were randomly assigned to one of four groups: 1) a medical play group, where children were given a doll and authentic medical equipment, 2) a medical information video group, in which children were shown a taped medical play session, 3) a typical play group (i.e., played Connect Four), or 4) a control group whose participants viewed a video on safari life. Assignment to groups was determined using random sample numbers generated by a computer software program	Low Risk

A behavioural Observation Scale was used by researchers to evaluate the child's behaviour pre and post study group and control group acitvity participation.	Low Risk	The child's pulse reading (Indication of anxiety) and Behaviour rating were conducted before and after the group activities were conuducted in the clinical exam room.	High Risk
The researchers evaluate the children's behaviour by using Behavioural Observation Scale.	Low Risk	A researcher, blind to the children's group assignment, rated the children's affect and behaviors before the assigned activity, during triage, and after the doctor's visit.	Low Risk
N/A	N/A	N/A	N/A

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No incomplete datas were detected	Low Risk	All data were included in final analysis	High Risk
N/A	N/A	N/A	N/A
All outcomes were reported in a prespecified way.	Low Risk	All outcomes were reported in prespecified way.	Low Risk
None	Low Risk	None	Low Risk

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Victorine 2002		Yuval 2008	
Support for judgment	**Judgement**	**Support for judgment**	**Judgement**
No randomisation. As a child comes in, a teddy doctor in the reception area will register the child and help them fill in part one of the questionnaire. Participants are selected based on attendance.	High Risk	No randomisation. Forty-one preschool children, age 3–6.5 years (mean 5.1 ± 0.7 years), from three kindergartens in Beer Sheva, Israel, took part in this study. Fifty preschool children, age matched and from a similar residential area, served as the control group.	high risk
All the questionnaires are numbered which means each pariticipants are allocated a number and identifiers are not used.	Low Risk	The paticipants are divivded into intervention group and control group.	high risk

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The child participants were required to fill in 2 questionairres pre and post intervention. No blinding was applied.	Low Risk	Assessment included a simple one-item visual analog scale of anxiety about hospitalization. The scale included five facial ex┐pressions indicating a happy face on one side of the scale and a very distressed face on the other. Children had to choose the face that best described the way they would feel if they were about to be hospitalized [7]. This was assessed individually one day prior to the intervention and again a week after the intervention when the children were in kindergarten.	high risk
Questionairres were used to assess the effects of TBH to reduce child's fear of doctors and hospitals.	Unclear Risk	Insifficient details for judgement	unclear risk
N/A	N/A	N/A	N/A

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All the questionnaires are numbered, unfortunately questionnaires numbered 86-124 were not found. Thus we have only 85 filled questionnaires. Out of the 85 questionnaires, 28 of the questionnaires were not filled in fully, some only completed part 1 and not part 2 while others were scribbled on. Therefore, only 57 fully completed questionnaires were considered for this study.	Low Risk	Three children in the intervention group and four in the control group were excluded from the study because of one or more previous hospitalization.	high risk
N/A	N/A	N/A	N/A
All outcomes were reported in prespecified way.	Low Risk	All outcomes were reported in prespecified way.	Low risk
None	Low Risk	None	low risk

ew only

Morel 2019
Support for judgment
no randomisation or risk
no allocation concealment.

no blinding. All children ages 4 to 16 years who presented to our ambulatory tertiary center for an MRI scan were included	
Insifficient details for judgement	
N/A	

Questionnaires were given randomly	
N/A	
All outcomes of interests were reported	
none	

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cochrane risk of bias		Ottenheim 2018		
entry		**Judgement**	**Support for judgment**	**Judgement**
Random sequence generation (selection bias)	R1	High Risk	No randomisation - 794 kindergarten children (age 4-7) participated in the Teddy Bear Hospital at Leiden University Medical Centre (Leiden, the Netherlands). Data was collected on 543 children attending elementary schools in the Leiden region.	High Risk
	R2	High Risk	Participants were selected based on their attendance to teddy bear hospital. No randomisation	High Risk
Allocation concealment (selection bias)	R1	High Risk	No allocation concealment - Questionnaire is given to children who participated in TBH	High Risk

For peer review only

	R2	High Risk	The same questions were used thrice, from pre exposure to TBH, to post TBH lecture and post TBH participation. No allocation concealment.	Low risk
Blinding of participants and personnel (performance bias)	R1	High Risk	No blinding - arranged for all participating schools a box with materials and a concrete lesson for the teachers with the intention that this lesson was given before the visit to the Teddy Bear Hospital.	High Risk
	R2	High Risk	No Blindng was applied. All the participants of TBH was included in selection with a prior lesson before participation. Despite that, a significant improvement was noticed only after the intervention which is T2. For question one, significant difference were seen between T2 and T1, T2 and T0, but no difference were seen between T1 and T0. Outcome is not influenced by the lack of blinding.	High Risk

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Blinding of outcome assessment(detection bias) (patient-reported outcomes)	R1	Unclear Risk	The study did not address blinding of outcome assessors	Unclear Risk
	R2	Unclear risk	Insufficient details for judgement	Unclear Risk
Blinding of outcome assessment (detection bias) (all-cause mortality)	R1	N/A	N/A	N/A
	R2	N/A	N/A	N/A
Incomplete outcome data addressed (attrition bias) (short-term 2-6 weeks)	R1	High Risk	The four questions of the questionnaire were analysed separately so no effect on the interpretation of the results are expected however missing data percentage ranges from 17.5%-19.89%. Data analysis was done only for completed questionnaire.	High Risk
	R2	Low risk	There were missing datas in terms of non balanced participants between each questions. But the questions were analysed separately and the insufficient details is not enough to produce clinically relevant impact on the study .	High Risk
Incomplete outcome data addressed (attrition bias) (long-term >6 weeks)	R1	N/A	N/A	N/A
	R2	N/A	N/A	N/A
Selective reporting (reporting bias)	R1	Low Risk	All outcomes of interests were reported	Low Risk
	R2	Low Risk	All the outcomes were reported in a prespecifed way	Low Risk

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Other bias	R1	Low Risk	None	High Risk
	R2	Low Risk	None	Low Risk

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Lynn 2018		Dalley 2016	
Support for judgment	**Judgement**	**Support for judgment**	**Judgement**
No randomisation - cross sectional descriptive study of children taking part in TBH sessions between 28 March and 20 May 2016. All children aged 5 to 8 years old were recruited from kindergartens and primary schools which	High Risk	No randomisation - A convenience sample was recruited from groups of children already scheduled to attend a Teddy Bear Clinic Tour at the Hill's Pet Nutrition Primary Healthcare Centre of the Ontario Veterinary College (OVC)	High Risk
All the children from all the schools and kindergardens that has participated in TBH were recruited with the exception of children without parental consent. No randomisation.	High Risk	A convenience sample was recruited from groups of children already scheduled to attend a Teddy Bear Clinic Tour at the Hill's Pet Nutrition Primary Healthcare Centreof theOntarioVeterinaryCollege (OVC).	High Risk
No allocation concealment - Questionnaire is given to children who participated in TBH	High Risk	No allocation concealment - Questionnaire is given to children who participated in TBH	Unclear Risk

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Each children were assigned a coded number without identifier.	Unclear risk	The allocation concealment method was not sufficiently described.	High Risk
No blinding - Questionnaire was administered right before TBH and immediately after TBH	High Risk	No blinding - Data were collected from participating children immediately before and immediately afer the tour	High Risk
A lesson regarding the concept of TBH was provided to children immediately after TBH session to assess knowledge acquired and their change of feelings towards healthcare.	High Risk	The data was collected from participants before and immediately after the tour.	High Risk

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The study did not address blinding of outcome assessors	Unclear Risk	No mention about blinding - Two undergraduate research assistants independently coded all of the participants' responses. Disagreements were resolved by discussion and consensus.	High Risk
The study did not adress this outcome.	Unclear risk	Insufficient details for judgement	High Risk
N/A	N/A	N/A	N/A
N/A	N/A	N/A	N/A
We excluded children who had incomplete data, such as missing answers to any of the multiple choice question in the survey. Children who completed the multiple choice component but did not answer the open-ended questions were still included. (Excluded 82 out of 334 - 24.55%)	Low Risk	A total of 71 participants had parental consent and assented to participate; no data were excluded from the final analysis.	Low Risk
	Low Risk	A total of 71 participants had parental consent and assented to participate; no data were excluded from the final analysis.	Low Risk
We excluded children who had incomplete data, such as missing answers to any of the multiple choice question in the survey. Children who completed the multiple choice component but did not answer the open-ended questions were still included.			
N/A	N/A	N/A	N/A
N/A	N/A	N/A	N/A
All outcomes of interests were reported	Low Risk	All outcomes of interests were reported	Low Risk
All outcomes were reported in a prespecified way	Low Risk	All outcomes were reported in a prespecified way	Low Risk

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Marion M. Aw is the clinical mentor for the medical students in charge of the Teddy Bear Hospital, Singapore. Ong Lynn and Chua Khoon Han were both involved in TBH as part of the organising committee during the course of this study.	Low Risk	None	Low Risk
None	Low Risk	None	Low Risk

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Pontes 2015	Leonhardt 2014	
Support for judgment	**Judgement**	**Support for judgment**
No randomisation - subjects were not randomly selected, and therefore this is not a true experimental study.	High Risk	No randomisation - Randomization was not possible because the TBH is an established local project and any kindergarten in the area is free to decide whether it will take part or not.
subjects were not randomly selected, and therefore this is not a true experimental study. No randomization	High Risk	Randomization was not possible because the TBH is an established local project and any kindergarten in the area is free to decide whether it will take part or not. No randomization
Method of concealment was not described - Children were distributed to an experimental group, with 30 children taking part in an instructional therapeutic play session before vaccination, and to a control group, with 30 children who did not receive any kind of previous emotional preparation.	High Risk	No allocation concealment - experimental group was recruited from kindergarten who registered to participate in TBH and control group from non-participating facilities

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Data were recorded in a form with information about identification of child and the reactions during vaccination. No allocation concealment was practiced	Low Risk	Children were assessed in small groups (up to 5 children) in the kindergarten during the week prior to the intervention and 1 week after visiting the TBH. No individual identifiers involved.
Blinding is not posible - children in experimental group have to participate in instructional therapeutic play sessions.	High Risk	No blinding - Children were assessed in small groups in the kindergarten during the week prior to the intervention and 1 week after visiting the TBH. Control children were interviewed twice as well and at a similar time interval.
After informing children and their legal guardians about the study and their participation, those selected for the experimental group were invited to play. No blinding	High Risk	Children were assessed in small groups (up to 5 children) in the kindergarten during the week prior to the intervention and 1 week after visiting the TBH. No individual identifiers involved. Control children were interviewed twice as well and at a similar time interval (about 21 days). No blinding

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Reactions were observed by researcher.	Unclear Risk	No mention about blinding - The interviews with the children were conducted by two trained students who were supervised by an experienced psychologist.
The children's reaction were observed by the researches themselve while vaccinating.	High Risk	The interviews with the children were conducted by two trained students who were supervised by an experienced psychologist.
N/A	N/A	N/A
N/A	N/A	N/A
No data were excluded from the final analysis	Low Risk	For reasons of non-appropriate age and outlier data, 8 children were excluded from final analysis. (<15%)
No incomplete datas were detected.	High Risk	For reasons of nonappropriate age and outlier data, 8 children were excluded from final analysis (N = 131). Attritioan bias
N/A	N/A	N/A
N/A	N/A	N/A
All outcomes of interests were reported	Low Risk	All outcomes of interests were reported
All outcomes were reported after a statistical analysis through Fisher Exact	Low Risk	All outcomes were reported in a prespecified way.

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None	Low Risk	None	
None	Low Risk	None	

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Burns 2013		
Judgement	**Support for judgment**	**Judgement**
Unclear Risk	The study did not specify how randomisation was done - Child participants were randomly assigned to 1 of 4 groups.	Unclear Risk
High Risk	Seventy-two caregiver–child dyads were recruited to participate, following institutional review board approval. The children (n = 33 males) were attending a scheduled, or walk-in, doctor's appointment at a pediatric clinic in a city in the southeast region of the United States, for either a well checkup (ie, school physical, vaccinations, etc) or for an ill checkup (ie, respiratory infection, skin rash, etc). No randomisation.	Unclear Risk
High Risk	Method of concealment was not described	High Risk

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Low Risk	Child participants were randomly assigned to 1 of 4 groups: (a) a medical play group, where children were given a doll and authentic medical materials to play with (ie, stethoscope, otoscope, tongue depressor, etc); (b) a medical information video group, in which children were shown a taped medical play session; (c) a typical play group, where children played a board game; or (d) a control group whose participants viewed a video on safari life.	Low Risk
High Risk	Blinding is not possible - children were either involved in a medical play group, a medical information video group, a typical play group or a control group whose participants viewed a video on safari life.	High Risk
Low Risk	Upon participating in their assigned activity, the children underwent a doctor's checkup where a trained nurse will conduct a behavioural assessment on the children.	Low Risk

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Low Risk	The nurses who rated the children's behaviour were unaware of the group assignments	High Risk
High Risk	Because there will be trained nurse assigned to conduct the behavioural assessment on the children.	High Risk
N/A	N/A	N/A
N/A	N/A	N/A
Low Risk	No data were excluded from the final analysis	Low Risk
Unclear risk	The exclusion of data was not mentioned.	Unclear risk
N/A	N/A	N/A
N/A	N/A	N/A
Low Risk	All outcomes of interests were reported	Low Risk
Low Risk	All outocmes were reported in a prespecified way	Low Risk

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Low Risk	None	Low Risk
Low Risk	None	Low Risk

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Brown 2012	Burns Nader 2011	
Support for judgment	**Judgement**	**Support for judgment**
The study did not specify how randomisation was done - Children were randomly assigned to participate in a medical collage activity or a control group.	Low Risk	Assignment to groups was determined using random sample numbers generated by a computer software program
A total of 30 children ranging in age from three to five were recruited with parental consent and participant (verbal) assent from a pediatric clinic in Tuscaloosa, AL. But there was no explanantion on the mode of selection.	High Risk	No randomisation. The children (n= 33 males) were attending a scheduled, or walk-in, doctor's appointment at a general medical school pediatric clinic.
Method of concealment was not described	High Risk	Method of concealment was not described

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The children were randomly assigned to participate in medical collage activity or control group.	Low Risk	Child participants were randomly assigned to one of four groups: 1) a medical play group, where children were given a doll and authentic medical equipment, 2) a medical information video group, in which children were shown a taped medical play session, 3) a typical play group (i.e., played Connect Four), or 4) a control group whose participants viewed a video on safari life. Assignment to groups was determined using random sample numbers generated by a computer software program
Blinding is not possible - study group has to participate in medical collage activity	High Risk	Blinding is not possible - children were either involveld in a medical play group, a medical information video group, a typical play group or a control group whose participants viewed a video on safari life.
A behavioural Observation Scale was used by researchers to evaluate the child's behaviour pre and post study group and control group acitvity participation.	Low Risk	The child's pulse reading (Indication of anxiety) and Behaviour rating were conducted before and after the group activities were conuducted in the clinical exam room.

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Reactions were observed by researcher. (Behaviour Observation Scale)	Low Risk	A researcher, blind to the children's group assignment, rated the children's affect and behaviors. The developmental interview was administered to the parent by a researcher blind to the children's group assignment.
The researchers evaluate the children's behaviour by using Behavioural Observation Scale.	Low Risk	A researcher, blind to the children's group assignment, rated the children's affect and behaviors before the assigned activity, during triage, and after the doctor's visit.
N/A	N/A	N/A
N/A	N/A	N/A
No data were excluded from the final analysis	Low Risk	No data were excluded from the final analysis
No incomplete datas were detected	Low Risk	All data were included in final analysis
N/A	N/A	N/A
N/A	N/A	N/A
All outcomes of interests were reported	Low Risk	All outcomes of interests were reported
All outcomes were reported in a prespecified way.	Low Risk	All outcomes were reported in prespecified way.

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None	Low Risk	None	
None	Low Risk	None	

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Victorine 2002		
Judgement	**Support for judgment**	**Judgement**
High Risk	No randomisation - We estimated a total of about 124 children attended this Teddy Bear Hospital. Every child was given a questionnaire to fill-in prior to attending Teddy Bear Hospital and after the experience.	High Risk
High Risk	No randomisation. As a child comes in, a teddy doctor in the reception area will register the child and help them fill in part one of the questionnaire. Participants are selected based on attendance.	High Risk
High Risk	No allocation concealment - Questionnaire is given to children who participated in TBH	High Risk

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Low Risk	All the questionnaires are numbered which means each pariticipants are allocated a number and identifiers are not used.	Low Risk
High Risk	No blinding - Every child was given a questionnaire to fill-in prior to attending Teddy Bear Hospital and after the experience.	High Risk
High Risk	The child participants were required to fill in 2 questionairres pre and post intervention. No blinding was applied.	Low Risk

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Unclear Risk	The study did not address blinding of outcome assessors	Unclear Risk
Low Risk	Questionairres were used to assess the effects of TBH to reduce child's fear of doctors and hospitals.	Unclear Risk
N/A	N/A	N/A
N/A	N/A	N/A
High Risk	Total 124 questionnaires, questionnaires numbered 86-124 were not found. Out of the 85 questionnaires, 28 of the questionnaires were not filled in fully. Therefore, only 57 fully completed questionnaires were considered for this study. (Excluded 67 out of 124 - >15%)	Low Risk
High Risk	All the questionnaires are numbered, unfortunately questionnaires numbered 86-124 were not found. Thus we have only 85 filled questionnaires. Out of the 85 questionnaires, 28 of the questionnaires were not filled in fully, some only completed part 1 and not part 2 while others were scribbled on. Therefore, only 57 fully completed questionnaires were considered for this study.	Low Risk
N/A	N/A	N/A
N/A	N/A	N/A
Low Risk	All outcomes of interests were reported	Low Risk
Low Risk	All outcomes were reported in prespecified way.	Low Risk

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Low Risk	None	Low Risk
Low Risk	None	Low Risk

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Yuval 2008		Morel 2019
Support for judgment	**Judgement**	**Support for judgment**
No randomisation - Forty-one preschool children, age 3–6.5 years from three kindergartens in Beer Sheva, Israel, took part in this study. Fifty preschool children, age matched and from a similar residential area, served as the control group.	high risk	no randomisation or risk This prospective controlled trial was performed from March 2018 to March 2019. All children ages 4 to 16 years who presented to our ambulatory tertiary center
No randomisation. Forty-one preschool children, age 3–6.5 years (mean 5.1 ± 0.7 years), from three kindergartens in Beer Sheva, Israel, took part in this study. Fifty preschool children, age matched and from a similar residential area, served as the control group.	high risk	no randomisation or risk
No allocation concealment - intervention group was recruited from kindergarten who participated in TBH and control group from a similar residential area	high risk	no allocation concealment. Ninety-one children (46 girls and 45 boys) and their parents were included (44 in the pre-mock and 47 in the post-mock period). Questionnaires were given randomly in 20% of the cases.

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The paticipants are divivded into intervention group and control group.	high risk	no allocation concealment.
No blinding - a simple one-item visual analog scale of anxiety about hospitalization was assessed individually one day prior to the intervention and again a week after the intervention when the children were in kindergarten.	high risk	no blinding. All children ages 4 to 16 years who presented to our ambulatory tertiary center for an MRI scan were included, provided that they had a normal level of understanding expected for their age and had parents who agreed to participate in the research.
Assessment included a simple one-item visual analog scale of anxiety about hospitalization. The scale included five facial ex¬pressions indicating a happy face on one side of the scale and a very distressed face on the other. Children had to choose the face that best described the way they would feel if they were about to be hospitalized [7]. This was assessed individually one day prior to the intervention and again a week after the intervention when the children were in kindergarten.	high risk	no blinding. All children ages 4 to 16 years who presented to our ambulatory tertiary center for an MRI scan were included

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The study did not address blinding of outcome assessors	unclear risk	The study did not address blinding of outcome assessors
Insifficient details for judgement	unclear risk	Insifficient details for judgement
N/A	N/A	N/A
N/A	N/A	N/A
Three children in the intervention group (7%) and four in the control group (8%) were excluded from the study because of one or more previous hospitalizations.	high risk	Questionnaires were given randomly in 20% of the cases. Participation rate was 95%- no reasons to why this happened.
Three children in the intervention group and four in the control group were excluded from the study because of one or more previous hospitalization.	high risk	Questionnaires were given randomly
N/A	N/A	N/A
N/A	N/A	N/A
All outcomes of interests were reported	Low risk	All outcomes of interests were reported
All outcomes were reported in prespecified way.	Low risk	All outcomes of interests were reported

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None	low risk	none
None	low risk	none

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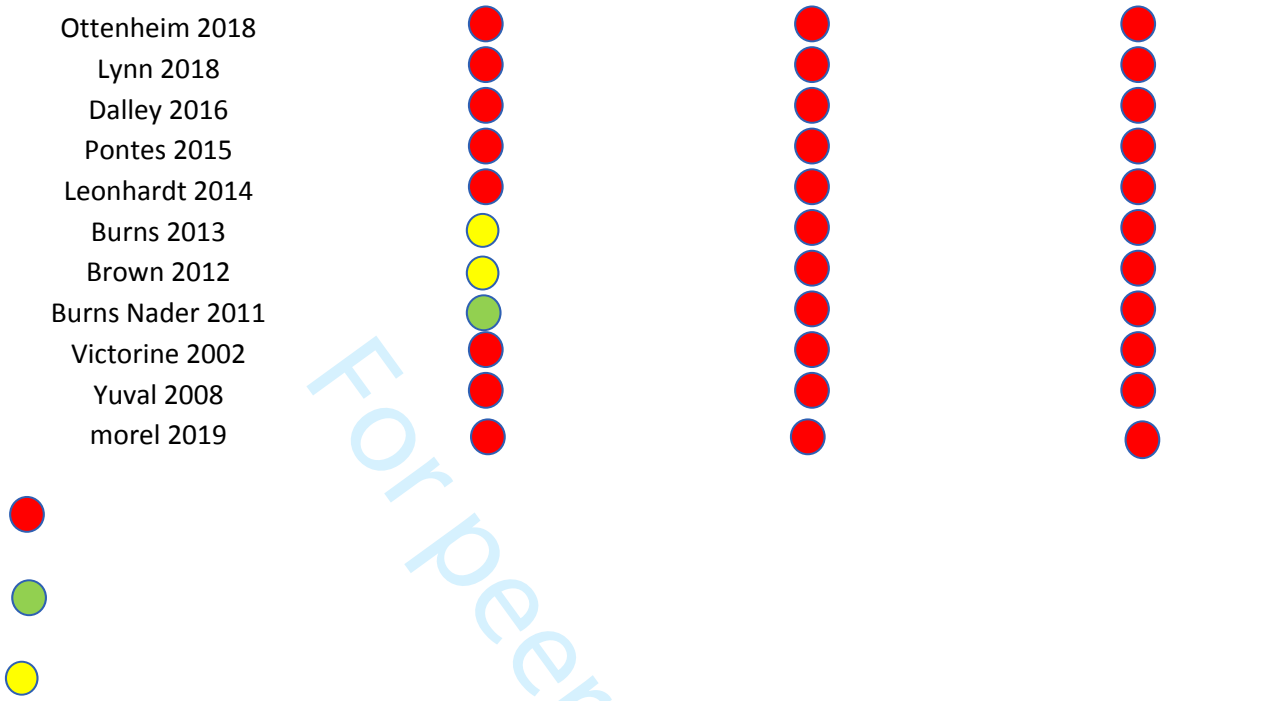
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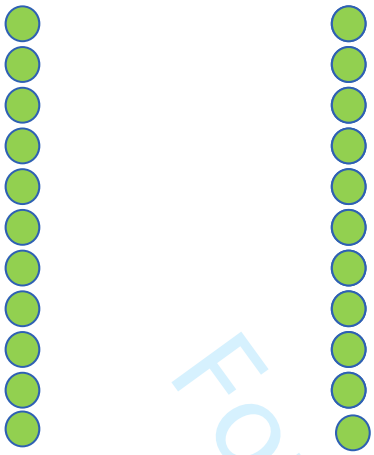
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Blinding of outcome assessment(detection bias) (patient-reported outcomes)	Blinding of outcome assessment (detection bias) (all-cause mortality)	Incomplete outcome data addressed (attrition bias) (short-term 2-6 weeks)	Incomplete outcome data addressed (attrition bias) (long-term >6 weeks)
	N/A		N/A
	N/A		N/A
	N/A		N/A
	N/A		N/A
	N/A		N/A
	N/A		N/A
	N/A		N/A
	N/A		N/A
	N/A		N/A
	N/A		N/A
	N/A		N/A

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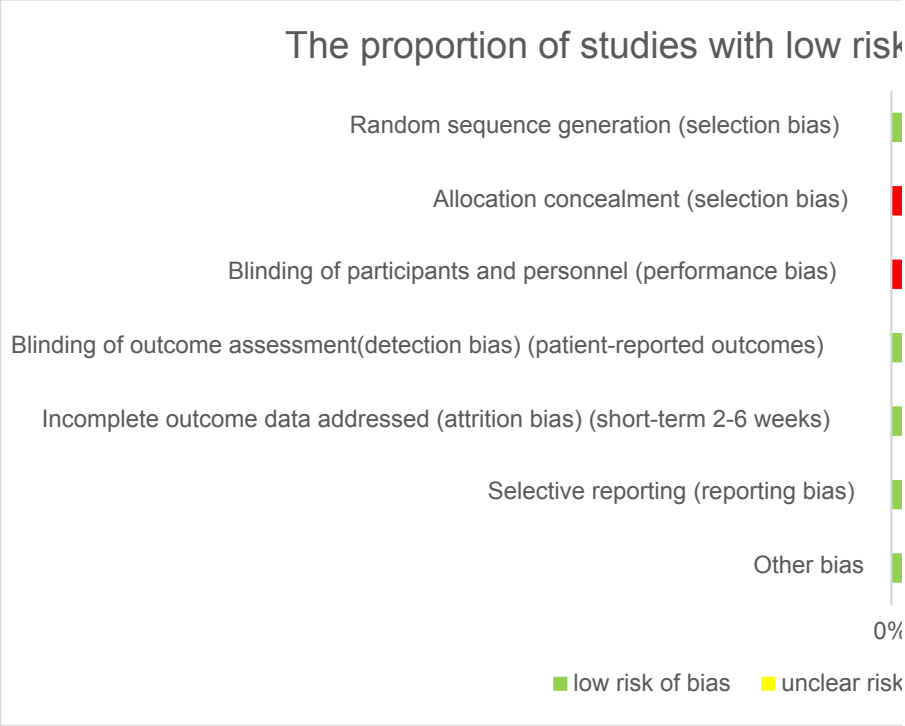
Selective reporting (reporting bias)

Other bias



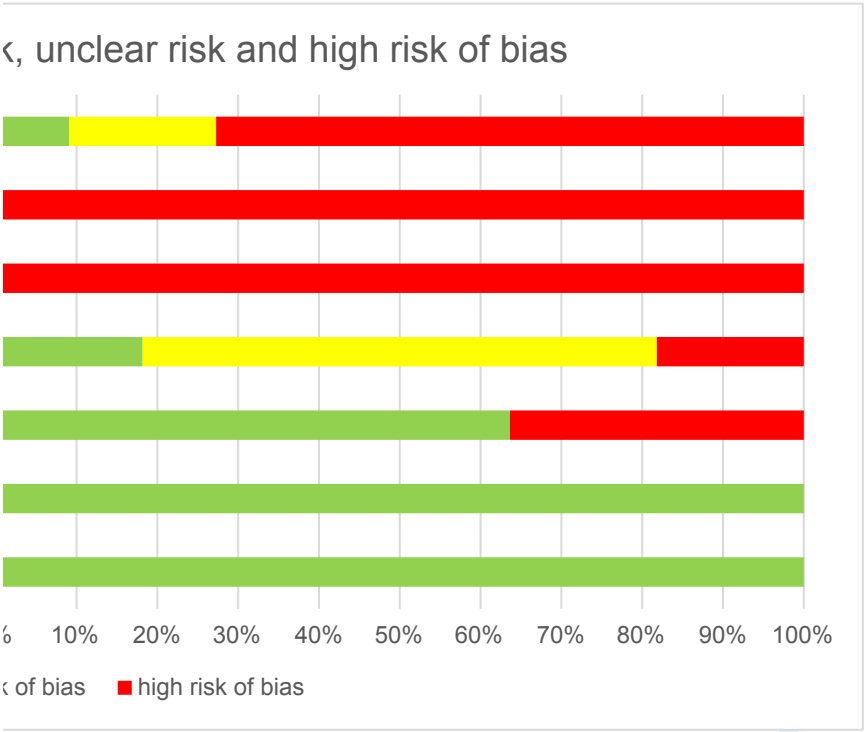
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	Other bias	Selective reporting (reporting bias)	Incomplete outcome data addressed (attrition bias) (short-term 2-6 weeks)
low risk of bias	11	11	7
unclear risk of bias	0	0	0
high risk of bias	0	0	4



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Blinding of outcome assessment(detection bias) (patient-reported outcomes)	Blinding of participants and personnel (performance bias)	Allocation concealment (selection bias)	Random sequence generation (selection bias)
2	0	0	1
7	0	0	2
2	11	11	8



none

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Supplementary file number 4 (S4) : Detailed summary of the results of individual stu

Study (Author, Year, Country)	Outcomes
Burns-Nader ES, 2011, USA	
	Anxiety level
	Level of fear
	Children's distress behaviours
Brown JC, 2012, USA	
	Children's behaviour and feelings

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Ottenheim M, 2018, Netherlands	1. Children's attitude towards visiting a hospital 2. Children's attitude towards doctors 3. Children's attitude towards falling ill 4. Children's attitude towards hospitals
Ong L, 2018, Singapore	Childhood feelings towards healthcare services & willingness of health-seeking behaviour:
	Children's healthcare knowledge
Dalley JS, 2016, Canada	Level of fear
	Level of expected pain
	Coping strategies
Pontes JE, 2015, Brazil	1. Children's body movement during procedure 2. Children's verbal expression during procedure

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	3. Children’s emotional expression during procedure
Leonhardt C, 2014, Germany	Children's medical knowledge
Burns-Nader S, 2013, USA	Children's mood
	Children's behaviour during procedure

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Victorine A, 2002, UK	1. Children’s attitude towards doctor 2. Children’s attitude towards hospital 3. Children’s attitude towards TBH
Bloch YH, 2008, Israel	Children’s anxiety about hospitalisation

children's anxiety

Morel 2019, France

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Tools	Results
The Child Drawing: Hospital (Higher score indicates higher anxiety) Heart rate	Not a significant effect of type of activity on the childre in their drawings
Fear Scale (5-points likert scale) (Higher score indicates higher level of fear)	1. Increase in fear for children in the medical play group activity (1.67), suggesting that children in the medical p fearful after medical play
	2. Increase in fear for children in the medical play activi the doctor's visit (1.94), suggesting that children in the fearful after the doctor's visit compared to the baseline
	3. Decrease in fear for children who watched the medic baseline (2.28) to post activity (1.31) and from baseline (1.50), suggesting that the medial information video wa children's fear after the activity and after the doctor's v
Behaviour Observation Scale (BOS) (Higher score indicates less alert) Children's Hospital of Eastern Ontario Pain Scale (CHEOPS) (Higher score indicates more difficulty)	Children in the typical play group improved in their stat (1.94) to post-activity (1.72)
	The video control group shifted from an active alert sta inactive alert state (1.72) after watching the safari video occurred from baseline (1.28) to after the doctor's visit were most alert before the activity
	The medical play group has higher mean than the rest o nurses's report on overall difficulty suggests that childre were more difficult than any other group while the nurs procedures
Behaviour Observation Scale (BOS) (Higher score indicates more alert) Faces Scale (4-points likert scale) (Higher score indicates more positive mood)	The intervention group showed significant differences between time points pre and post one and post one and post two. These differences were seen between pre and post one state (Z = -2.00, p = .046), pre and post one activity (Z = -2.83, p = .005), post one and post two activity (Z = -2.45, p = 0.14), pre and post one vocalization (Z = -2.83, p = .005), and post one and post two vocalization (Z = -3.05, p = .002). This indicates that after the intervention, the participants became more alert, and had more activity and more vocalizations. After the intervention, levels of activity and vocalization went back down.

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Seven-point likert scale (Higher score indicates more positive attitude)	Significant difference in anxiety level for doctors be moments. Gender, if children visited a hospital befo between the covariates did not differ significantly. ' comparison showed significant differences betwee (p=0.002); T2 (5.34) and T1 (5.00) (p=0.018) but not (5.00) (p=1.00).
	Significant difference in anxiety level for falling ill b moments. Gender, if children visited a hospital befo between the covariates did not differ significantly. ' comparison showed significant differences betwee (p=0.000); T0 (2.72) and T2 (3.25) (p=0.000) but not (3.25) (p=1.00).
	Gender had a significant effect on anxiety levels for girls reporting on average higher levels of anxiety (I compared to boys.
5-point likert scale (Higher score indicates more positive mood)	Children's feeling of happiness towards visiting the to 94.4% (p-value <0.001) with the median changin
	Children who felt happy to visit the hospital increas value 0.035) when measured on the same 5-point s
3 health-care related questions	increased from 81 (32.1%) before attending TBH to
The Children's Fear Scale (5-points likert scale) (Higher score indicates higher level of fear)	Children's fear ratings did not signifcantly change fr
The Faces Pain Scale - Revised (0-10 metric) (Higher score indicates higher level of expected pain)	Mean expected pain ratings decreased signifcantly 1.27) to post intervention at Time 2 (M = 1.64, SD =
14-item coding system (Higher score indicates more coping strategies adapted)	Children reported a higher frequency of interventic Time 2 (M = 0.90, SD = 0.72) versus Time 1 (M = 0.7 = .034, and r = −.06, representing a signifcant, albei
Child's reaction checklist	Most reactions related to a greater acceptance of t staying still, smiling, collaborating, and being at eas experimental group. Fisher's exact test showed tha significant for standing still (p=0.006) and collabora (p=0.015), with significance level at 5%.

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	(p < 0.025), with significance level at 5%. Some of the reactions that show less acceptance of observed in the experimental group, such as hitting pulling the hand of the professional, clinging to the
Children's Medical Knowledge Scale (ChildMedKnow) (Higher score indicates better knowledge)	Children who visited the TBH had a significantly bet health and disease (ChildMedKnow at time point 2) control group (F(1, 118) = 141.07, p < 0.001). The ef η2 = 0.55). There was an influence of the baseline s 213.55, p < 0.001, partial η2 = 0.65).
	Children who visited the TBH improved their knowl of the body significantly (F(1, 121) = 79.43, p < 0.00 from the intervention group knew the organs of the the TBH (M = 4.5, SD = 1.1) than children in the con 1.2). Baseline M = 1.93
	The ANCOVA also revealed here a significant baseli 121) = 111.05, p < 0.001, partial η2 = .48).
Mood Scale (5-poins likert scale) (Higher score indicates less positive mood)	A significant interaction was found between time a children's self-reported mood, F(5.92,122.31) = 3.4 tests were conducted for post hoc comparisons wit lower scores in mood (ie, shifting from the directio the direction of happy) for children in the medical p postactivity and from baseline to after the doctor's play activity lowered the children's mood after the doctor's visit (1.94). Baseline mean = 1.44
	Higher scores in mood (ie, shifting from the directic direction of very happy) for children who watched t video from baseline to post activity and from baseli suggesting that the medical information video impr the activity (1.31) and after the doctor's visit (1.5).
Nurse's rating on Children's behaviours	Significant effect for children's leg behaviors, F(3) = children's overall difficulty during the triage proced Post hoc results for leg behaviors revealed significa medical play group and the nonmedical informatio medical play group having a higher mean, suggestir to display distress behaviors with their legs than th video group during triage.

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(Higher score indicates more distress)	The medical play group was also significantly differ information video, typical play, and nonmedical inf the nurses' overall rating of how difficult the childr procedures. The medical play group had a higher m activity groups. This would suggest that children in more difficult than any other group while the nurse procedures.
7-point likert scale (Higher score indicates more positive mood)	1. Of the 57 children, 13 of them gave a score of 6 on th they enjoyed Teddy Bear Hospital while 44 of them gav 2. When asked if Teddy enjoyed Teddy Bear Hospital, 4 5, 19 children gave a score of 6 and the balance 34 gave 3. Of the 57 children, when asked if Teddy Bear Hospita their fears, 55 of them answered yes and 2 not. Of the t expressed no fear of doctors and hospital at all from the 4. All the children answered yes when asked if they wou again, if teddy would attend Teddy Bear Hospital again, friends to Teddy Bear Hospital and if teddy will bring te Hospital.
One-item visual analog scale of anxiety about hospitalization (5 facial expressions) (Higher score indicates higher level of anxiety)	A repeated measures ANOVA revealed that time signifc [F(1,89) = 14.34, P < 0.001]. While baseline levels of anxiety were not different betw children in the teddy bear hospital group reported signi than the control group at follow-up
1.The ambiance of the preparation room was evaluated with a four-point Likert scale (relaxing, neutral, frightening, scary). 2.The anxiety level was self-estimated with a visual analogue scale (from 0,completely relaxed, to 100, extremely stressed) at three different moments: in the waiting room, in the preparation room after an explanation by the MRI technologists and at the issue of the MR acquisition images	1. Post hoc tests (Scheffé) showed that the anxiety leve significantly lower after the explanations in the post-mc compared to the level estimated in the waiting room (no difference for w explanation in the pre-mock condition). 2. In pre- and post-mock periods, a significant differenc waiting room and after the exam was also observed (pr mock period: P<0.0000001).

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	Positive/Negative Outcome	Pre Scores
n's level of anxiety, as depicted	NO EFFECT	
o, from baseline (1.44) to post- lay activity became more	NEGATIVE (MEDICAL PLAY)	1.44
ty from baseline (1.44) to after medical play activity were more	NEGATIVE (MEDICAL PLAY)	1.44
cal information video from (2.28) to after the doctor's visit as effective in reducing visit.	POSITIVE (MEDICAL VIDEO)	2.28
	POSITIVE (MEDICAL VIDEO)	2.28
e, or alertness, from baseline	POSITIVE (TYPICAL PLAY)	1.94
te at baseline (1.28) to a more o. The decrease in alertness also (1.67), suggesting that they	NEGATIVE (NON-MEDICAL VIDEO)	1.28
	NEGATIVE (NON-MEDICAL VIDEO)	1.28
of the activity groups on en in the medical play group se was performing triage	NEGATIVE (MEDICAL PLAY)	
State	POSITIVE	Inactive alert: 5 (Pre) Active alert: 10 (Pre)
Activity	POSITIVE	Moderate: 11 (Pre) High: 2 (Pre)
Vocalisation	POSITIVE	Low: 4 (Pre) High: 4 (Pre)

etween the three measure ore or the interaction The Bonferroni pairwise n T2 (5.34) and T0 (5.00) t between T0 (5.00) and T1	POSITIVE	5.00
etween the three measure ore or the interaction The Bonferroni pairwise n T0 (2.72) and T1 (3.23) t between T1 (3.23) and T2	POSITIVE	2.72
r going to the hospital ill with low on the Likert scale)		Boy: 4.42 Girl: 4.40
doctor increased from 82.5% g from 2 to 1.	POSITIVE	82.50%
sed from 70.2% to 73.4% (p- scale.	POSITIVE	70.20%
185 (73.4%) after attending	POSITIVE	32.10%
rom pre to post intervention		
from Time 1 (M = 2.00, SD = = 1.35)	POSITIVE	2.00
on-taught coping strategies at 70, SD = 0.76), t(70) = −2.17, p it small effect	POSITIVE	0.7
he procedure, such as se, were more frequent in the at the differences were ating spontaneously	POSITIVE	

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f the procedure were not g, kicking, pushing, biting, ir parents, asking for		
tter knowledge of their body,) than the children from the ffect size was large (partial score as well (F(1, 118) =	POSITIVE	
ledge of the different organs 01, partial η2 = 0.41). Children eir body better after visiting ntrol group (M = 3.1, SD =	POSITIVE	1.93
ine score as covariate (F(1,		
nd type of activity on 3, P < .01. Paired samples t thin groups and revealed n of very happy to more in play group, from baseline to visit, suggesting the medical activity (1.67) and after the	NEGATIVE (MEDICAL PLAY)	1.44 1.44
on of happy to more in the the medical information ine to after the doctor's visit, roved children's mood after Baseline mean = 2.28	POSITIVE (MEDICAL VIDEO)	2.28 2.28
= 2.85, P < .05, and the dures, F(3) = 5.63, P < .01. nt difference between the n video control, with the ng that they were more likely e nonmedical information	NEGATIVE (MEDICAL PLAY)	

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ent than the medical ormation video groups on en were during the triage nean than the rest of the the medical play group were e was performing triage	NEGATIVE (MEDICAL PLAY)	
ne Likert scales when asked if e a score of 7. of the children gave a score of e a score of 7. l have helped them to reduce two that said not, both e beginning. uld attend Teddy Bear Hospital , if the children will invite their ddy's friend to Teddy Bear	POSITIVE	Likert score 7 (most happy and comforable) Doctor: 0 Hospital: 0
cantly interacted with group veen groups [t(89) = 0.4, NS], ifcantly lower levels of anxiety	POSITIVE	TBH: 6.5 Control: 7.0
l estimated by children was ock period (P<0.0001) as vaiting room versus after e between anxiety score in the e-mock period: P<0.001, post-	POSITIVE	

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T1	T2
MEAN SCORES	
1.67 (post-activity)	
	1.94 (after doctor's visit)
1.31 (post-activity)	
	1.5 (after doctor's visit)
1.72 (post-activity)	
1.72 (post-activity)	
	1.67 (after doctor's visit)
FREQUENCIES	
Inactive alet: 1 (Post-one) Active alert: 14 (Post-one)	
Moderate: 5 (Post-One) High: 9 (Post-One)	Moderate: 11 (Post-Two) High: 3 (Post-Two)
Low: 1 (Post-one) High: 9 (Post-one)	Low: 7 (Post-two) High: 4 (Post-two)
MEAN SCORES	

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5.00	5.34
3.23	3.25
Boy: 4.83 Girl: 4.14	Boy: 4.63 Girl: 4.24
FREQUENCY PERCENTAGE	
94.40%	
73.40%	
73.40%	
MEAN SCORES	
1.64	
0.9	

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MEAN SCORES	
Intervenion: 4.5 Control: 3.1	
MEAN SCORES	
1.67 (Post-intervention)	
	1.94 (Post-doctor's visit)
1.31 (Post-intervention)	
	1.5 (Post-doctor's visit)

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FREQUENCY	
Likert score 7 (most happy and comforable) Doctor: 32 Hospital: 32	
MEAN SCORES	
TBH: 5.0 Control: 8.0	

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Remarks
Relationship between variables
Pre - before intervention Post one - right after intervention Post two - after doctor's visit

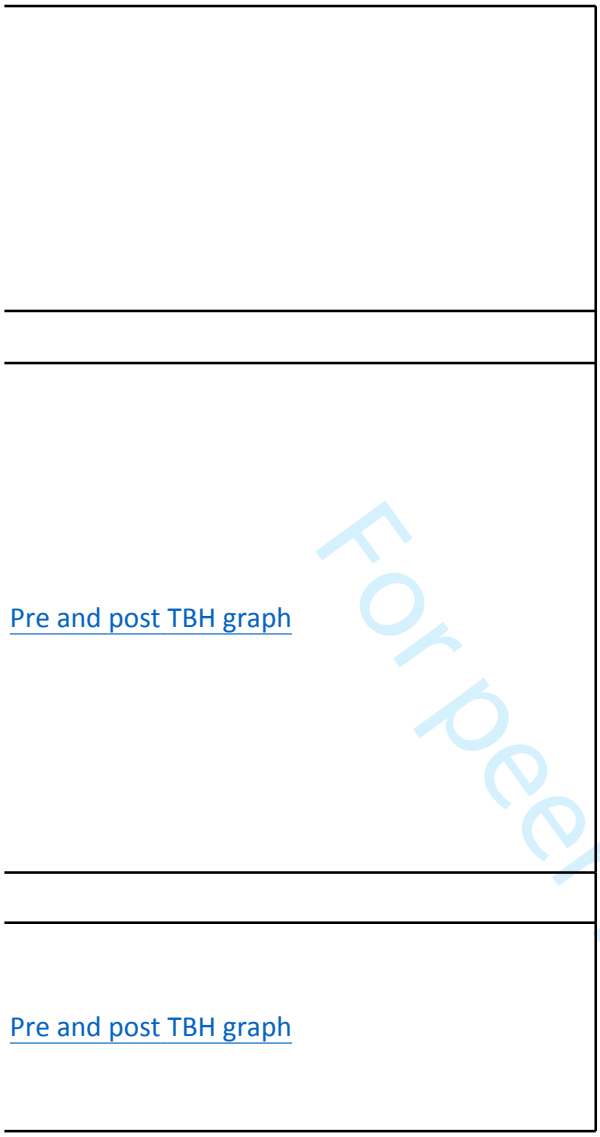
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T0: before the lesson with the materials provided by the IFMSANL. T1: after the lesson with the materials provided by the IFMSA-NL but before the actual visit to the Teddy Bear Hospital. T2: after the visit to the Teddy Bear Hospital.
Time 1 - immediately before the tour Time 2 - immediately after the tour (45 minutes later)

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Time 1 - during the week prior to the intervention Time 2 - 1 week after visiting the TBH

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Pre post mock and no mock (of teddy sized MRI) at 3 different times

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Burns Nader 2011.pdf

Table 5
Mean Scores (sd) and Between Group Differences for Nurse Rating of Child Distress Scale

Variable	Groups				
	Medical Play	Medical Play Video	Typical Play	Video Control	*F*
Facial	0.56 (.71)$_a$	0.39 (.61)$_a$	0.39 (.50)$_a$	0.33 (.59)$_a$	0.45
Verbal	0.61 (.61)$_a$	0.56 (.62)$_a$	0.44 (.51)$_a$	0.33 (.49)$_a$	0.88
Torso	1.28 (.46)$_a$	1.11 (.32)$_a$	1.17 (.38)$_a$	1.06 (.24)$_a$	1.25
Touch	1.28 (.46)$_a$	1.17 (.38)$_a$	1.17 (.38)$_a$	1.00 (0)$_a$	1.87
Legs	1.33 (.49)$_a$	1.11 (.32)$_{ab}$	1.17 (.38)$_{ab}$	1.00 (0)$_b$	2.85
Total	6.06 (1.8)$_a$	5.39 (1.88)$_a$	5.33 (1.33)$_a$	4.72 (1.18)$_a$	2.15
Overall Stress	1.00 (1.14)$_a$	0.56 (1.25)$_a$	0.56 (.78)$_a$	0.28 (.46)$_a$	1.75
Overall Pain	0.33 (1.19)$_a$	0.39 (.61)$_a$	0.50(.86)$_a$	0.39 (1.24)$_a$	0.09
Overall Difficulty	0.56 (.78)$_a$	0.17 (.51)$_b$	0.00 (0)$_b$	0.00 (0)$_b$	5.63

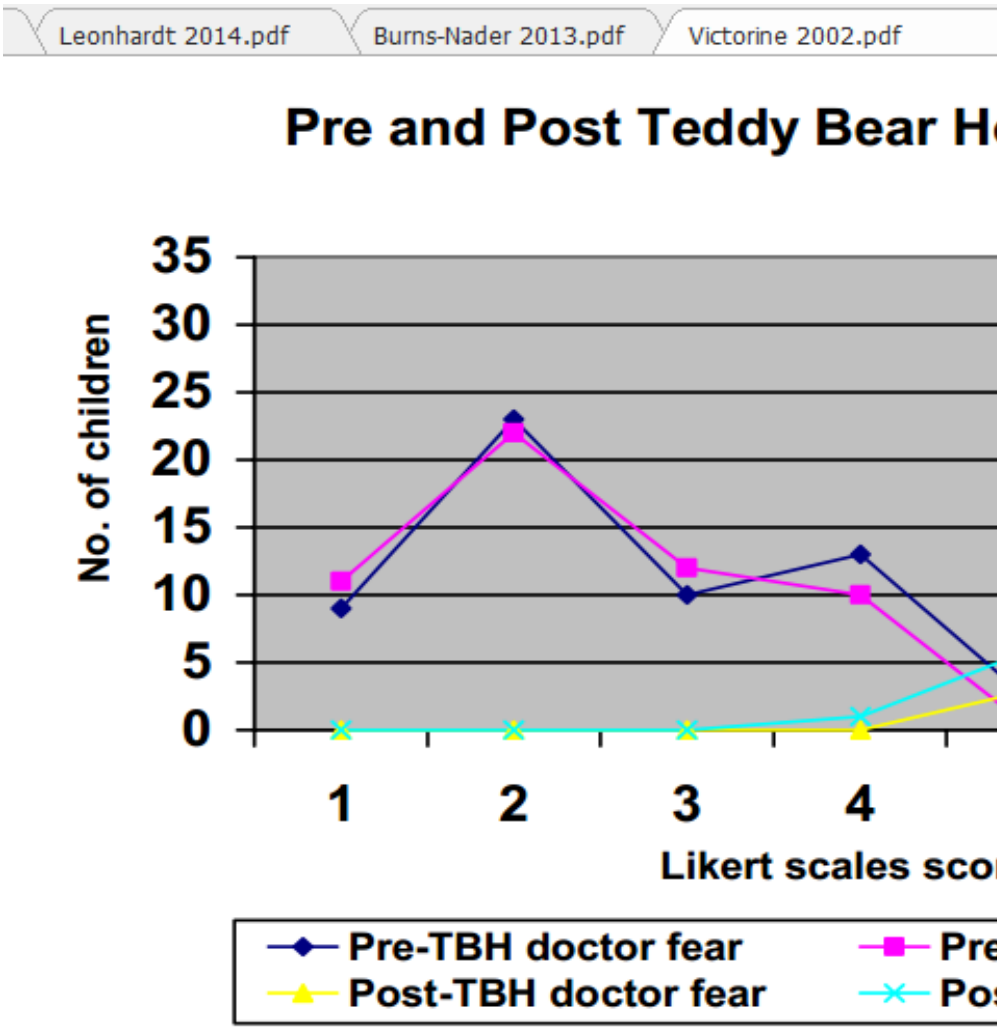
Note. Mean scores that do not have the same subscript are significantly different at the .05 level. Lower sc
* p < .05, ** p < .0

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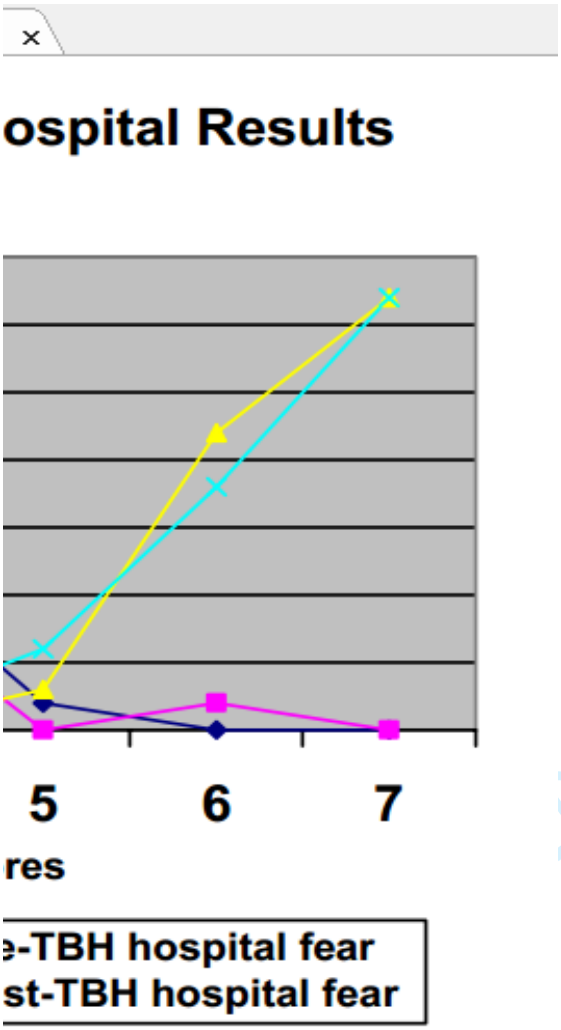
	p
	.72
	.46
	.30
	.14
	.04*
	.10
	.17
	.97
	.002**

cores are more optimal.

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Leonhardt 2014.pdf
Burns-Nader 2013.pdf
Victorine 2002.pdf
Pre and Post Teddy Bear H
No. of children
35
30
25
20
15
10
5
0
1
2
3
4
Likert scales sco
Pre-TBH doctor fear
Pre
Post-TBH doctor fear
Pos

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nhardt 2014.pdf Burns-Nader 2013.pdf Victorine 2002.pdf Yu

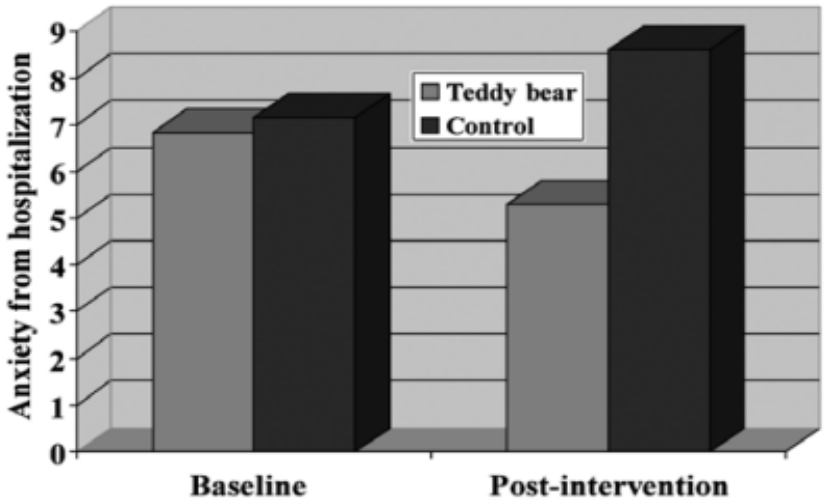


Figure 2. Effects of time of measurement on children's anxiety regarding hospitalization

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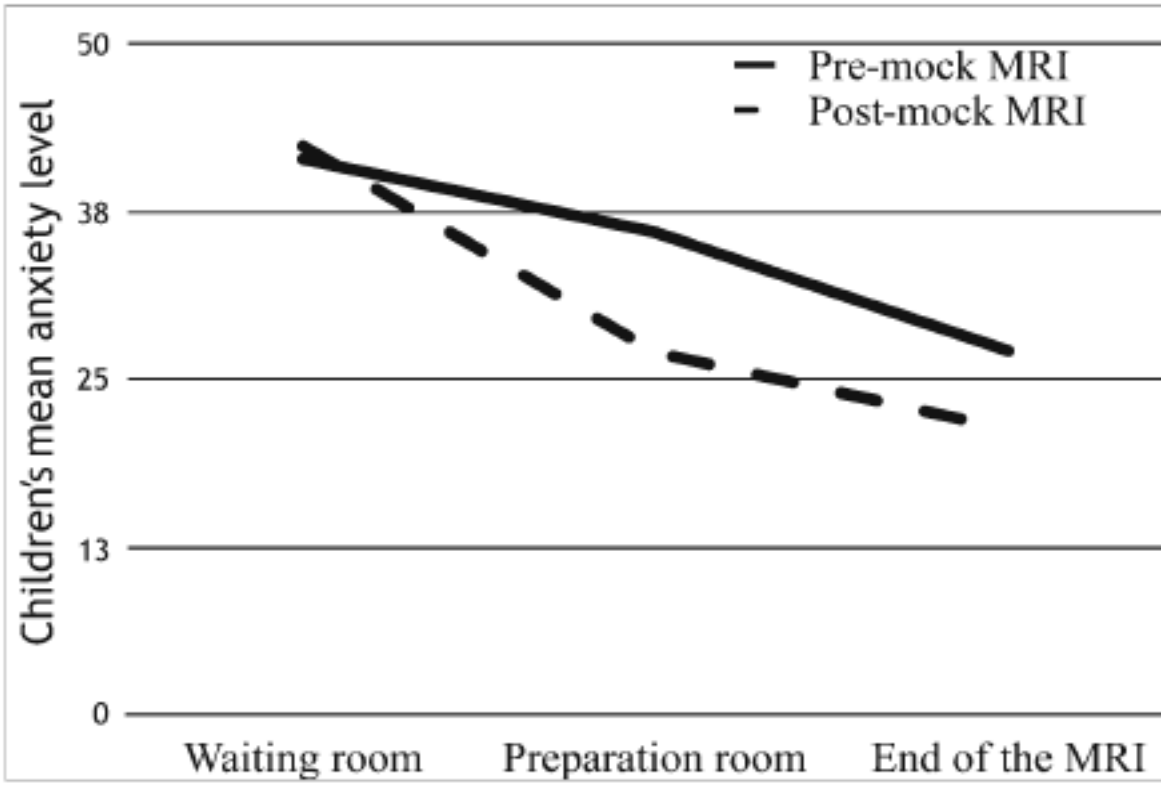


Fig. 2 Mean child anxiety level evaluation in the waiting room preparation room and at the end of the MRI

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The effectiveness of pretend medical play using the Teddy Bear Hospital (TBH) concept in improving children's health outcomes and wellbeing— a systematic review

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Strength and Limitations to this Study

1. This review looks into the effectiveness of pretend medical play using the Teddy Bear Hospital (TBH) concept on children's health outcomes and wellbeing

2. It followed established guidelines and identified 10 interventional studies to review the effectiveness

3. The limitations include a lack of high-quality studies available. The risk of bias for random sequence generation was unclear or high for nine out of 10 studies

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Abstract

Background

The healthcare setting is stressful for many people especially children. Efforts are needed to ease children's anxiety towards it. Medical play using the Teddy Bear Hospital (TBH) concept are measures used to facilitate children's exposure and experiences towards healthcare settings. Children will bring their soft toys and act as parents to the 'sick' teddies in a pretend hospital or clinic play setting. The objective of this systematic review is to evaluate the effectiveness of the TBH in improving children's health outcomes and wellbeing.

Methods

Reference lists of included studies from four electronic databases PubMed, CINAHL, Scopus, and Google Scholar were searched from inception till November 2020. We included pre-post studies, quasi-experimental, case-control studies, and randomised controlled trials (RCT) that discussed about medical play using the TBH concept as an intervention. Studies that involved sick patients, and that used interventions unlike the TBH were excluded. The quality of the included studies were assessed using the Cochrane Collaborations' "Risk of bias" tool.

Results

Ten studies were included in this systematic review. Five were specifically on the TBH method, while the others were of the same concept using medical play. Only three studies were randomised control trials. All the studies report more than one outcome, most being positive outcomes. Two reports lower anxiety levels after the intervention. Two found better healthcare knowledge, with one of them reporting increased feelings of happiness towards visiting a doctor. Two studies found no change of anxiety or feelings, while another two found

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increased levels of fear and lowered mood after the medical play (which involved real medical equipment).

Conclusions

The concept of TBH had mostly positive outcomes. This involves lower anxiety levels and improved healthcare knowledge. Its effectiveness would need to be verified in future studies with a more robust methodology

Systematic review registration: PROSPERO2019 CRD42019106355

Keywords: Teddy Bear Hospital, children, health knowledge, anxiety, education

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BACKGROUND

Children visit healthcare facilities for many reasons. Among them are for regular check-ups, immunisation, or for receiving treatments. [1,2] These visits have been related to negative reactions among them.[2] Literature reports that around 30% of adults are affected by healthcare visits. So much so, they are said to have white coat syndrome; a rise in blood pressure during medical visits.[3] Like adults, children have a fear of the unknown. They tend to create fantasies and distort information due to lack of accurate ones which could result in fear and anxiety.[4] It has been reported that children who have been well-informed of a certain condition, cope better and have less fear when faced with the real situation.[5] In view of this, various efforts were made to ease children's anxiety and fear towards healthcare setting and health personnel. The Teddy Bear Hospital (TBH) is one of such efforts, using the concept of medical play. [6]

The TBH incorporates pretend hospital play. Children will bring their teddy bears or soft toy to go through pretend health care services. They also incorporate an element of medical play which allows children to play and explore with medical equipment, mostly in the shape of toys but sometimes using real medical equipment. Medical play is a play involving a fun activity that involves a medical theme. It is initiated by an adult but led by the child and its main aim is to reduce anxiety in healthcare settings. [1,7]

In the TBH role play, children will act as the teddy bears' parents and bring the sick teddy bear to visit a teddy doctor who is usually a medical student or a volunteer. The setting is in the form of a clinic or hospital where the play area is filled with pretend or suitable real

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medical equipment. Most TBH are aimed at healthy children such as those in kindergartens, or during an open day in a specific health care centre for health education purposes. [5,8–12] The objective of this role play intervention is to ease the child's anxiety when they are in the healthcare setting, and at the same time to enhance their health knowledge. To our knowledge, there are currently no reviews available to evidently show the TBH has achieved these outcomes.

The overall objective of this review is to determine the effectiveness of medical play in the TBH concept in improving children's health outcomes and wellbeing.

METHODS

Study design

This systematic review was conducted and reported based on the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines. [13]This review was registered with PROSPERO (CRD: 42019106355). The search strategy can be found in supplementary file 1 (S1)

Search strategy

An electronic literature search was carried out in the following databases: MEDLINE (PubMed), CINAHL, Scopus, and Google Scholar from inception to November 2020. A combination of Medical Subject Heading (MeSH) and free text terms were used for the keywords 'Teddy-Bear Hospital', 'Teddy-Bear Clinic', 'medical play' and child. These keywords needed to appear in conjunction in the title, abstract, or full text of the article. We also conducted an additional search on the reference of relevant articles.

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Inclusion and exclusion criteria

All pre-post studies, quasi-experimental, case-control studies, and randomised controlled trials (RCT) done in both the community and health settings discussing about the Teddy Bear Hospital or medical play interventions involving dolls or soft toys and medical equipment were included. The medical play interventions involving dolls or soft toys and medical equipment were included as it carried the same concept as the TBH. Furthermore, the included studies have to report on the main outcomes which include health outcomes such as health knowledge, and psychological impacts. We excluded studies that reported sick children in hospital, and studies that use different interventions other than the TBH concept of soft toys and medical play equipment. As mentioned earlier, since most of the TBH done are done to children that are healthy such as kindergartens, or an open day in a specific health care centre as an interventional health education; hence we excluded sick children. The concept of TBH is to focus on healthy children going for well-visits as these visits are common and the experiences may influence children's medical fears. [12] Furthermore, the psychology of sick hospitalised children will most likely differ to those that are not.[14]

Study selection

The articles obtained from the literature search were imported into EndNote© and duplicated articles were removed. The list of remaining articles was then divided into two to be screened by two teams of reviewers. The two teams of reviewers each consisted of two reviewers (DR and AAR) and (NHS and CAT). Each of the reviewers screened through the titles and abstract of the articles independently and then compare with the other team members.

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Any discrepancies of the inclusion were discussed between the 2 team members and the decision was based on the consensus agreed after discussion. In cases where the consensus unable to reach within the team, it was then brought to discussion with the other team members for discussion. Once identified for relevance, the full papers were then retrieved. These papers were then again independently reviewed by each member in a team independently using a spreadsheet of the same format. The spreadsheets would later be compared. If there were any discrepancies on the articles between the 2 reviewers in a team, the other team would be consulted, and a decision would be made. All the agreed included articles were then later combined into one spreadsheet.

Data extraction

Data were extracted from the relevant studies by one reviewer (AAR) and checked by a second reviewer (CAT) on another spreadsheet. Any disagreements at any stage were resolved through discussion with a third reviewer (NHS or DR). Data on the following subjects were extracted: setting, objective, population, intervention/approach, measuring tools, outcomes of interest, N/N intervention, N/N control, results, and conclusion.

Quality assessment

The quality of the methodology used in the studies was assessed using The Cochrane Collaborations' "Risk of bias" tool. [15] Each included study was assessed based on a number of characteristics such as a) selection bias (randomisation and allocation concealment) b) performance bias (blinding of participants or personnel), c) detection bias (blinding of outcome assessment), d) attrition bias (incomplete outcome data both long and short term), and e) reporting bias (selective reporting). All studies were critically appraised in duplicates by the two reviewers (AAR and CAT) and presented to the research team members

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disagreements were discussed, and a consensus was sought until a combined assessment result was deduced.

Data Analysis

Data extracted were all analysed via narrative analysis. Meta-analysis was not performed as the tools for measuring the outcomes vary among the studies, thus unable to pool the results together.

RESULTS

Study selection

A total number of 16,306 of studies were yielded from the literature search conducted from inception to November 2020. We identified 12,522 articles after excluding 3784 duplicates. Of those, 12,473 were excluded. The excluded articles were titles and abstracts not within the scope of the study. The remaining 49 studies were available to be retrieved in full text. 10 out of 49 studies were then selected based on the inclusion and exclusion criteria. Among the studies excluded were sick children in the hospital, and intervention that were not TBH or similar and outcomes not in line with the study objectives. The PRISMA flow diagram and the reasons for exclusion are shown in Figure 1.

Description of included studies

Out of 10 included studies, three were RCTs, [1,16,17] three were non-randomised trials with controlled group [5,8,18] and the remaining were pre-post studies. [9,10,19,20] These studies had a population groups ranging from 30-543 participants within the age group of 2-12 years old. Six of the included studies involved TBH as an intervention [5,8–12] while the remaining 4 studies discussed medical play intervention. [1,16–18] The medical play involved used similar concept whereby 3 involved dolls but with real medical equipment [1,17,18], while another study used

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medical play with a collage of medical equipment .[16] Among the 10 studies, 4 were conducted in North America (three in the United States of America, [1,16,17] one in Canada[20]), 4 in Europe (1 in the United Kingdom, [10] 1 in the Netherlands, [9] and 1 in Germany[5]) and 1 each in Singapore, [19] Brazil, [18] and Israel. [8] The studies were done between 2002 -2018, involving mostly medical or veterinary students in the TBH settings and researchers in the medical play setting. An overview of the studies is shown in supplementary file 2 (S2).

Quality of studies

The risk of bias for random sequence generation was unclear or high for 9 out of 10 studies, with the exception of 1 study which was an RCT. [1] All studies had a high risk of bias for allocation concealment and performance bias. Most studies did not do randomisation for their participants while concealing the intervention could not be done with TBH or medical play. With regards to detection bias, only 20% of the studies had a low risk of bias where the assessors were blinded from the group arrangements of the interventions. [1,17] Three studies were found to have a high risk of reporting bias as a proportion of data was excluded from the analysis of results. [9,10,19] All studies had a low risk of bias for selective reporting and no other biases were found in all the studies. Figure 2 shows the proportion of studies with low risk, unclear risk, and high risk of bias. The assessment of biases of individual studies is shown in supplementary file number 3 (S3).

Results of individual studies

Children's feelings

Overall the studies had mixed results. Seven studies measured the children's feelings which are level of anxiety and fear as the outcomes. Four studies (1 non-randomised trial with the controlled group and 3 pre-post studies) concluded that the children had more positive

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feelings with lower anxiety levels after the TBH. [8–10,19] Another 3 studies (2 RCTs and 1 pre-post study) concluded otherwise, where the RCTs [1,17] showed that medical play intervention had lowered the children's mood and increased the level of fear. The pre-post study [20] showed no effect of TBH on the children's level of fear.

Children's behaviour

Two RCTs found that the participants in the medical playgroup were more difficult during triage procedures as compared to the medical information group, typical playgroup, and control group. [1,17] Another 2 studies (1 RCT and one non-randomised trial with the controlled group) showed positive outcomes for medical play intervention[16,18]. They report the children to be more alert and had a greater acceptance of the procedure (immunisation) after the intervention. [16,18] This also shows mixed results.

Children's healthcare knowledge

Two studies discussed the effect of the TBH on children's healthcare knowledge. Both of the studies (1 non-randomised trial with the controlled group and 1 pre-post study) concluded that children had better healthcare knowledge after the TBH intervention. [5,19] Therefore, the 2 studies that looked into children's knowledge showed positive results.

A detailed summary of the results of individual studies is included in supplementary file number 4 (S4).

Others

Although the outcomes on parents were not discussed in this study, it is worth mentioning that 1 study reported that their level of worry significantly correlated with their perceived children level of worry to see the doctor. [1] The other studies showed either no significant

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associations of parental talking to the children; and perception of the children's' worry to see the doctor between the control and intervention group. [16] The other 8 studies either did not involve parents or did not report any findings on the parents.

DISCUSSION

TBH is a popular project that has been done around the world with the primary aim to reduce children's fear and anxiety towards healthcare professionals and hospitals through pretend medical play. This strength of this review represents the first effort to summarise the effectiveness of the TBH and its concept of medical play in improving children's health outcomes and wellbeing. The TBH has become a popular concept to educate children and a review is important to look into its effectiveness.

In general, the studies included in this review showed a high level of selection bias and performance bias. However, this finding is unavoidable due to the nature of the intervention which involves children participating in physical activities related to the intervention of TBH or medical play. Detection bias was high as most of the studies did not describe the methods of blinding of outcome assessors.

Our review suggests that medical play using the TBH concept has a positive effect on children's health outcomes and wellbeing particularly in terms of reducing anxiety and fear. This is most likely because of the playful activities that are important in child development and learning especially on how to interact socially. Therefore, medical play helps them be more open to healthcare personnel and its surroundings. [18,20] Exploration of medical items in medical play has been reported to allow children to be able to express themselves better,

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leading to better psychological wellbeing, specifically in relation to the healthcare setting. [16,17] This method of simulating a scenario is very common and is used mainly to teach those in the medical and healthcare line. [21,22] For children specifically, the term 'experiential learning 'is more commonly used and has proven to be effective. For example children actively participating in growing fruits and vegetables in their school gardens showed greater improvements in nutrition knowledge, while exhibiting an increase in fruit and vegetable eating preferences, compared to those that were only taught on nutrition but not exposed to the gardening activity. [23,24]

It is also worth mentioning that the medical play is supervised play, which can be another factor to the reduction of anxiety. Another reason for this is the non-threating environment of the TBH where it can be seen as "control over a possible threatening situation". [8] TBH via medical play can also significantly improve children's healthcare knowledge. This is not surprising because as mentioned earlier, children learn well via playful activities and in addition to that, the role-playing component adds on to the effectiveness of learning.[5,18]

However, two studies conducted by Burns-Nader et. al concluded a contradictory effect of medical play intervention on children's health outcomes. This involved reported increase fear and lower mood by the children after the play therapy and after the doctor's visit compared to baseline. [1,17] The disagreements in the results suggest the need for further research to look into the components of medical play intervention that affect the final results of the intervention. The medical play in both studies involved real medical equipment without any

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toys which in contrast to the equipment used in TBH that are mostly toys made in the shape of medical equipment.

Further studies should explore whether the physical involvement of the children in role-play or the medical information obtained through the activities serves as the main component in reducing children's fear and anxiety.

We hypothesise that the lack of consistency in the conclusion might also be due to the involvement of third-party personnel in assessing the children's response instead of self-reporting outcomes by the children themselves. The children's distress behaviours were assessed by the nurses during triage, may introduce fear to the children when compared to other studies, where the outcomes were usually reported by the children themselves or assessed by non-medical related personnel.

Significant improvements in healthcare-related knowledge were seen in studies done by Leonhardt et. al and Victorine et. Al .[5,10] This shows that TBH could be a good method to be adopted for delivering programmes among children to raise their awareness on health-related issues. For example, TBH can be used to teach children on healthy living in order to prevent childhood obesity that is on the rise globally. Early intervention in terms of education is important to prevent adverse health risks later in adulthood. [25] The implementation of these programmes, however, will need cooperation from various disciplines like healthcare professionals, parents, teachers, and authority bodies.

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Other clinical implications of this study include the potential use of the TBH as a tool to teach children on personal safety to prevent child sexual abuse in view of the drastic rise in the number of child sexual abuse cases worldwide and nationwide. [26,27]

Limitations

This review has a relatively small number of included studies and most of the studies were pre-post studies. This limitation may impact on the generalisability of the findings in future implementations. There are also issues of consent for the children as they themselves are unable to consent to their involvement in the interventions. Apart from that, as mentioned earlier, some studies showed negative effect on the children.

Implications of the review on practice and further research

This review found that the concept of the TBH, which involves pretend medical play and using toy and sometimes real medical equipment has lowered the children's anxiety level and improved the children's healthcare knowledge. This could be one of the potential public health strategies in improving health intervention specifically as a health education tool to the pre-school and primary school children in a various topic such as immunisation, healthy lifestyle, and sexual and physical abuse.

The review found that the quality of the studies was less satisfactory as most of the studies were pre-post studies. Future studies would need to be carried out with a more robust methodology to examine the effect of the TBH concept of medical play intervention.

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CONCLUSION

This review found that the TBH concept of medical play has a positive outcome on the children's emotions and healthcare knowledge in most of the studies. Its effectiveness would need to be verified in future studies with a more robust methodology.

List of abbreviations

TBH: Teddy Bear Hospital; PRISMA: Preferred Reporting Items for Systematic Reviews and Meta-Analyses; RCT: Randomised Controlled Trials

DECLARATIONS

Patient and Public Involvement Statement

No patients or public involved.

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Availability of data and materials

All data and materials analysed are included in this article and its additional files.

Competing interests

The authors declared to have no competing interests.

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Authors' contributions

AAR, CAT, NHS, and DR designed the study. RHS was responsible for conducting the literature search. All authors participated in the selection of included studies. AAR, CAT, NHS, and DR were responsible for data extraction, study appraisal, data analysis and results interpretation. All authors contributed to drafting the manuscript. The final manuscript was approved by all authors.

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REFERENCES

1. Burns-Nader, E. S. The effects of medical play on reducing fear, anxiety, and procedure distress in school-aged children going to visit the doctor. (University of Alabama Libraries, 2011).
2. Lerwick, J. L. Minimizing pediatric healthcare-induced anxiety and trauma. *World J. Clin. Pediatr.* **5**, 143–150 (2016).
3. Alves, L. M. M., Nogueira, M. S., Godoy, S. de, Hayashida, M. & Cárnio, E. C. Prevalence of white coat hypertension in primary health care. *Arq. Bras. Cardiol.* **89**, 28–35 (2007).
4. Santen, L. & Feldman, T. Teddy bear clinics: a huge community project. *MCN Am J Matern Child Nurs* **19**, 102–106 (1994).
5. Leonhardt, C., Margraf-Stiksrud, J., Badners, L., Szerencsi, A. & Maier, R. F. Does the 'Teddy Bear Hospital' enhance preschool children's knowledge? A pilot study with a pre/post-case control design in Germany. *J. Health Psychol.* **19**, 1250–1260 (2014).

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

6. Kaufman, J., Modak, M. & Moylan, S. the Teddy Bear Hospital In Australia. *J Paediatr Child Health* **48**, 541–542 (2012).

7. McCue, K. Medical Play: An Expanded Perspective. *Child. Health Care* **16**, 157–161 (1988).

8. Bloch, Y. & Toker, A. Doctor, is my Teddy Bear Okay? A "Teddy Bear Hospital" as a Method to Reduce Children's Fear of Hospitalization. *Isr. Med. Assoc. J.* **10**, 597–9 (2008).

9. Ottenheim, M. & Sommeren, R. van. The Teddy Bear Hospital. How to influence the attitude of kindergarten children towards doctors, hospitals and falling ill. /paper/The-Teddy-Bear-Hospital.-How-to-influence-the-of-Ottenheim-Sommeren/778b811ab7b57a3da3a35d58a664ed3a64d7f774 (2018).

10. Victorine, A., Wong, Z., Geilani, J. & Charters, A. Teddy Bear Hospital (TBH)–reducing children's fear of doctors and hospital environment. *PEDIATRICS* **110**, e33 (2002).

11. Ong, L., Chua, K. H., Soh, J. Y. & Aw, M. M. H. Y. Healthcare encounters in young children: Impact of Teddy Bear Hospital, Singapore. *Asia Pac. Sch.* **3**, 24–30 (2018).

12. Dalley, J. S., Creary, P. R., Durzi, T. & McMurtry, C. M. An Interactive Teddy Bear Clinic Tour: Teaching Veterinary Students How to Interact with Young Children. *J Vet Med Educ* **44**, 302–315 (2017).

13. Moher, D., Liberati, A., Tetzlaff, J., Altman, D. G. & Group, T. P. Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. *PLOS Med.* **6**, e1000097 (2009).

14. Rennick, J. E. & Rashotte, J. Psychological outcomes in children following pediatric intensive care unit hospitalization: a systematic review of the research. *J. Child Health Care* **13**, 128–149 (2009).

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15. Chapter 8: Assessing risk of bias in included studies. In: Higgins JPT, Green S (editors). Cochrane Handbook for Systematic Reviews of Interventions. Version 5.1.0. in.

16. Brown, J. C. Group medical play for reducing stress and improving mood in children going to visit the pediatrician. (University of Alabama Libraries, 2012).

17. Burns-Nader, S., Hernandez-Reif, M. & Thoma, S. J. Play and Video Effects on Mood and Procedure Behaviors in School-Aged Children Visiting the Pediatrician. *Clin. Pediatr. (Phila.)* **52**, 929–935 (2013).

18. Pontes, J. E. D., Tabet, E., Folkmann, M. Á. dos S., Cunha, M. L. da R. & Almeida, F. de A. Therapeutic play: preparing the child for the vaccine. *Einstein São Paulo* **13**, 238–242 (2015).

19. Ong, L., Chua, K. H., Soh, J. Y. & Aw, M. M. H. Y. Healthcare encounters in young children: Impact of Teddy Bear Hospital, Singapore. *Asia Pac. Sch.* **3**, 24–30 (2018).

20. Dalley, J. S. & McMurtry, C. M. Teddy and I get a check up: A pilot educational intervention teaching children coping strategies for managing procedure-related pain and fear. *Pain Res. Manag.* (2015).

21. Jeffries, P. R., Bambini, D., Hensel, D., Moorman, M. & Washburn, J. Constructing Maternal-Child Learning Experiences Using Clinical Simulations. *J. Obstet. Gynecol. Neonatal Nurs.* **38**, 613–623 (2009).

22. Schreiber, J. *et al.* Experiential Learning With Children: An Essential Component of Professional Physical Therapy Education. *Pediatr. Phys. Ther.* **27**, 356–367 (2015).

23. Parmer, S. M., Salisbury-Glennon, J., Shannon, D. & Struempler, B. School Gardens: An Experiential Learning Approach for a Nutrition Education Program to Increase Fruit and Vegetable Knowledge, Preference, and Consumption among Second-grade Students. *J. Nutr. Educ. Behav.* **41**, 212–217 (2009).

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24. Heim, S., Stang, J. & Ireland, M. A Garden Pilot Project Enhances Fruit and Vegetable Consumption among Children. *J. Am. Diet. Assoc.* **109**, 1220–1226 (2009).

25. Reilly, J. J. Childhood Obesity: An Overview: Childhood Obesity. *Child. Soc.* **21**, 390–396 (2007).

26. Pereda, N., Guilera, G., Forns, M. & Gómez-Benito, J. The prevalence of child sexual abuse in community and student samples: A meta-analysis. *Clin. Psychol. Rev.* **29**, 328–338 (2009).

27. Singh, H. A., Yiing, W. W. & Nurani, H. N. K. Prevalence of childhood sexual abuse among Malaysian paramedical students. *Child Abuse Negl.* **20**, 487–492 (1996).

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Figures legends

Figure 1: PRISMA flowchart detailing the search strategy used.

Figure 2: The proportion of studies with low risk, unclear risk and risk of bias

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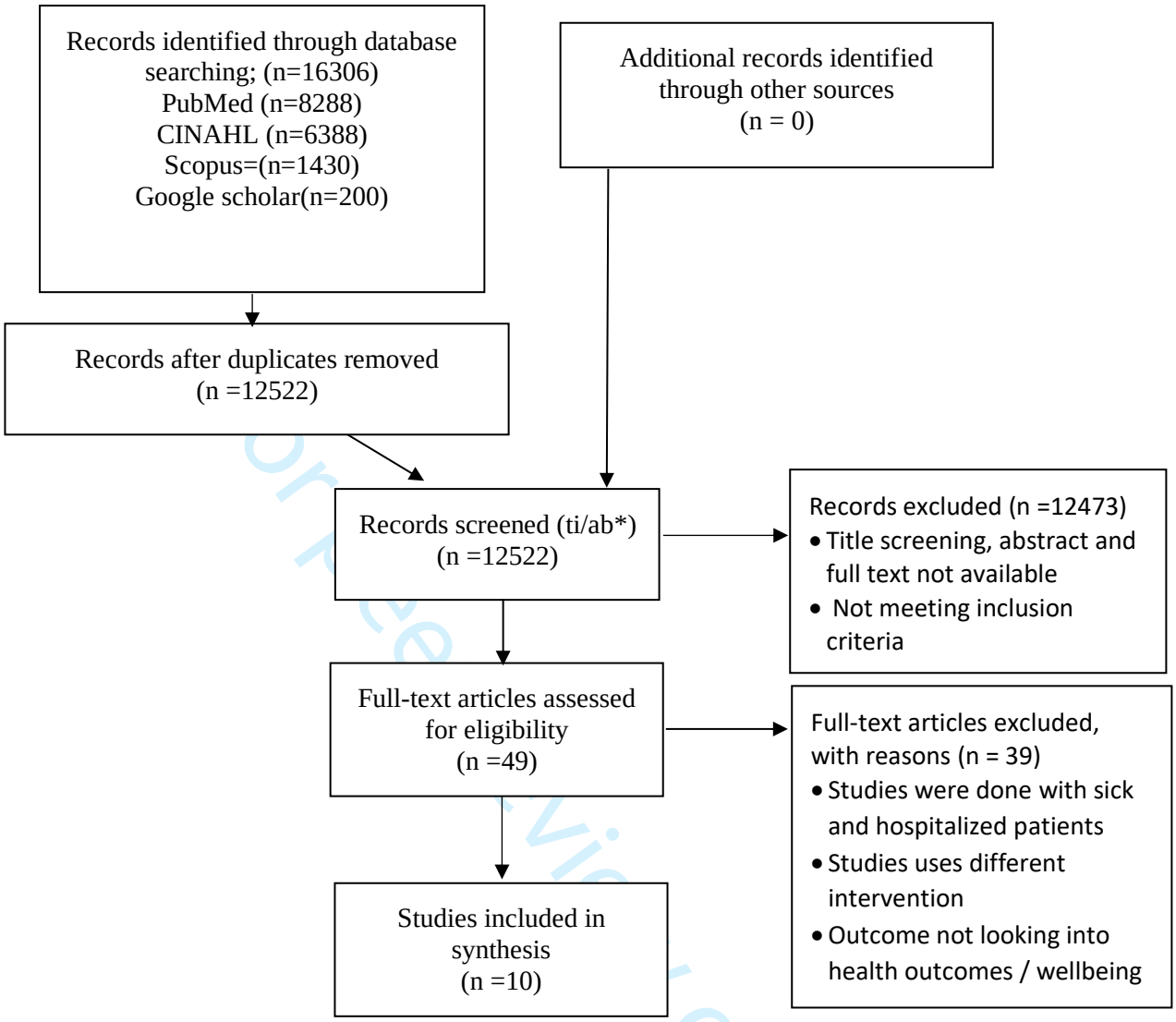


Figure 1: PRISMA flowchart detailing the search strategy used.

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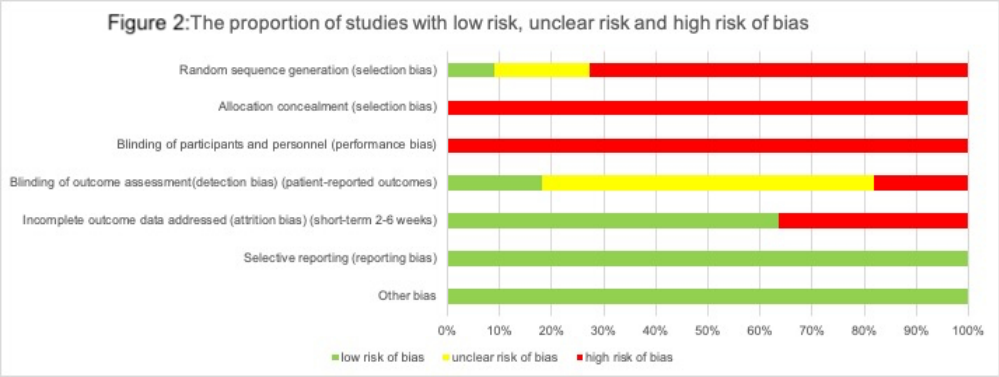


Figure 2: The proportion of studies with low risk, unclear risk and risk of bias

251x94mm (72 x 72 DPI)

Supplementary file number 1 (S1): **Search strategy (S1)**

1.	Pubmed	((((("Teddy bear hospital"[Title/Abstract] OR Teddy bear clinic*[Title/Abstract] OR "medical play"[Title/Abstract] OR ("Play and Playthings"[Mesh])) OR "teddy bear"[Title/Abstract] OR soft toy*[Title/Abstract] OR dolls[Title/Abstract])) AND ((((((("child"[MeSH Terms]) OR "child, preschool"[MeSH Terms]) OR child*[Title/Abstract] OR preschool child*[Title/Abstract] OR preschooler*[Title/Abstract] OR "play therapy/methods"[MeSH Major Topic]) OR pediatric*[Title/Abstract] OR boys[Title/Abstract] OR girls[Title/Abstract] OR kindergar*[Title/Abstract] OR toddler*[Title/Abstract] OR "pediatrics/education"[MeSH Major Topic]))))))	8288

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Supplementary file number 2 (S2): Overview of included studies.

Study (Author, Year, Country)	Population	Setting	Intervention	Outcome measured	Results	Outcome assessment
Randomised Controlled Trials						
Burns-Nader ES, 2011, USA (10)	Children aged 5.5 – 12 years	Paediatric clinic	3 intervention groups: 1. Medical play group (doll and authentic medical equipment) 2. Medical information video group 3. Typical play group	1. Children's anxiety level 2. Children's level of fear 3. Children's distress behaviours	• No effect of types of activity on children's anxiety level • Medical play group had increased level of fear after the activity and after doctor's visit • Medical play group was more difficult (higher mean score in nurses' overall rating of difficulty) during triage procedure as compared to medical information video group, typical play group and control group.	Self-report Self-report Third-party evaluation

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			Control: viewed video on safari life			
Burns-Nader S, 2013, USA (11)	Children aged 5.5 – 12 years	Paediatric clinic	3 intervention groups: 1. Medical play group (doll and authentic medical equipment) 2. Medical information video group 3. Typical play group Control: viewed video on safari life	1. Children's mood 2. Children's behaviours during procedures	• Medical play lowered children's mood after the activity and after the doctor's visit • Medical play group was more difficult (higher mean score in nurses' overall rating of difficulty) during triage procedure as compared to medical information video group, typical play group and control group.	Self-report Third-party evaluation
Brown JC, 2012, USA (9)	Children aged 3 - 5 years	Paediatric clinic	Medical collage group (medical play) Control: No intervention	1. Children's behaviours: - State - Affect - Activity - Vocalisation	• Medical collage (medical play) group was more alert, and had more activity and more vocalizations after the intervention. Levels of activity and vocalization went back down after doctor's visit	Third-party evaluation

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				- Fidgeting/Squirming 2. Children's feelings		Self-report
Non-randomised trials with controlled group						
Leonhardt C, 2014, Germany (5)	Children aged 4.5 - 5.5 years	Kindergarten	TBH Control: No intervention	Children's medical knowledge	• Children in TBH group had better knowledge than the control group one week after the intervention	Self-report
Bloch YH, 2008, Israel (6)	Children aged 3 – 6.5 years	TBH at healthcare centre	TBH Control: No intervention	Children's anxiety about hospitalisation	• Children in TBH group had lower levels of anxiety than the control group one week after the intervention	Self-report
Pontes JE, 2015, Brazil (12)	Children aged 3 - 6 years and 11 months	Paediatric clinic	Instructional therapeutic play (medical play) Control: No intervention	1. Children's body movement during procedure 2. Children's verbal expression during procedure 3. Children's emotional expression during procedure	• Instructional therapeutic play (medical play) group had greater acceptance of the procedure as compared to control group: - Staying still - Collaborating - Smiling - Being at ease	Third-party evaluation

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Pre-post Studies						
Ottenheim M, 2018, Netherlands (13)	Children aged 4 - 7 years	TBH at healthcare centre	TBH	1. Children's attitude towards visiting a hospital 2. Children's attitude towards doctors 3. Children's attitude towards falling ill 4. Children's attitude towards hospitals	• Reduced anxiety level towards doctors after TBH • Reduced anxiety level towards falling ill after introductory lesson and after TBH	Self-report
Ong L, 2018, Singapore (14)	Children aged 5 - 8 years	TBH at kindergarten and primary schools	TBH	1. Childhood feelings towards healthcare services 2. Willingness of health-seeking behaviour 3. Children's medical knowledge	• Children's feeling of happiness towards visiting the doctor increased after TBH • Children who felt happy to visit the hospital increased after TBH • Increased healthcare knowledge after TBH	Self-report
Dalley JS, 2016, Canada (15)	Children aged 5 - 10 years	TBH at healthcare centre	TBH	1. Children's level of fear 2. Children's level of expected pain 3. Children's coping strategies	• No effect on level of fear • Decreased mean expected pain ratings after TBH • Higher frequency of intervention-taught coping strategies after TBH	Self-report

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Victorine A, 2002, UK (7)	Children aged 2 – 12 years	TBH at hotel	TBH	1. Children's attitude towards doctor 2. Children's attitude towards hospital	• Increased proportion of children with positive feelings towards doctor and hospital after TBH **No further test was done to determine whether results were statistically significant*	Self-report

A summary of the author, year of publication, country of publication, population, setting, intervention, outcome measured, results and outcome assessment of each included studies.

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Supplementary file number 3 (S3): Quality Assessment of Studies.

Cochrane risk of bias		Ottenheim 2018		Lynn 2018		Dalley 2016		Pastor 2015		Leonhardt 2014		Burns 2013		Brown 2012		Burns-Nader 2011		Victorino 2002		Yuval 2008	
entry		Judgement	Support for judgement	Judgement	Support for judgement	Judgement	Support for judgement	Judgement	Support for judgement	Judgement	Support for judgement	Judgement	Support for judgement	Judgement	Support for judgement	Judgement	Support for judgement	Judgement	Support for judgement	Judgement	Support for judgement
Random sequence generation (selection bias)	D1	High Risk	No randomization - TRA kindergarten (age 4-7) participated in the Teddy Bear Hospital at Leiden University Medical Centre (Leiden, the Netherlands). Data was collected on 143 children attending elementary schools in the Leiden region.	High Risk	No randomization - cross sectional descriptive study Children who took part in TBH sessions between 28 March and 20 May 2016. All children aged 5 to 8 years old were recruited from kindergartens and primary schools which	High Risk	No randomization - A convenience sample was recruited from groups of children already scheduled to attend a Teddy Bear Clinic Tour at the Hill's Pet Nutrition Primary Healthcare Centre of the Ontario Veterinary College (OVC)	High Risk	No randomization - subjects were not randomly selected, and therefore this is not a true experimental study.	High Risk	No randomization - the allocation was not possible because the TBH is an established local project and any kindergarten in the area is free to decide whether it will take part or not. No randomization	Unclear Risk	The study did not specify how randomization was done - Children were randomly assigned to participate in a medical college activity or a control group.	Unclear Risk	The study did not specify how randomization was done - Children were randomly assigned to participate in a medical college activity or a control group.	Low Risk	Assignment to groups was determined using random sample numbers generated by a computer software program	High Risk	No randomization - The children in 1 to 3 classes were attending a scheduled, or walk-in, doctor's appointment at a general medical school pediatric clinic.	High Risk	No randomization - As a child comes in, a baby doctor in the reception area will register the child and help them fill in part one of the questionnaire. Participants are selected based on attendance.	High Risk	No randomization - Forty-one preschool children, age 4-6 years (mean ± 1 ± 0.7 years), from three kindergartens in Beer Sheva, Israel, took part in this study. Fifty preschool children, age matched and from a similar residential area, served as the control group.
	D2	High Risk	Participants were selected based on their attendance to Teddy Bear hospital. No randomization	High Risk	All the children from all the schools and kindergartens that has participated in TBH were recruited with the exception of children without parental consent. No randomization.	High Risk	A convenience sample was recruited from groups of children already scheduled to attend a Teddy Bear Clinic Tour at the Hill's Pet Nutrition Primary Healthcare Centre of the Ontario Veterinary College (OVC).	High Risk	subjects were not randomly selected, and therefore this is not a true experimental study. No randomization	High Risk	Pre-formulation was not possible because the TBH is an established local project and any kindergarten in the area is free to decide whether it will take part or not. No randomization	High Risk	Seventy-two caregiver-child dyads were recruited to participate, following institutional review board approval. The children (n = 33 males) were attending a scheduled, or walk-in, doctor's appointment at a pediatric clinic in a large urban region of the United States, for either a well-checkup (ie, school physical, vaccinations, etc) or for an ill-checkup (ie, respiratory infection, skin rash, etc). No randomization.	Unclear Risk	A total of 60 children ranging in age from three to	High Risk	No randomization. The children (n = 33 males) were attending a scheduled, or walk-in, doctor's appointment at a general medical school pediatric clinic.	High Risk	No randomization. As a child comes in, a baby doctor in the reception area will register the child and help them fill in part one of the questionnaire. Participants are selected based on attendance.	High Risk	No randomization. Forty-one preschool children, age 4-6 years (mean ± 1 ± 0.7 years), from three kindergartens in Beer Sheva, Israel, took part in this study. Fifty preschool children, age matched and from a similar residential area, served as the control group.
Allocation concealment (selection bias)	D3	High Risk	No allocation concealment - Questionnaire is given to children who participated in TBH	High Risk	No allocation concealment - Questionnaire is given to children who participated in TBH	High Risk	No allocation concealment - Questionnaire is given to children who participated in TBH	Unclear Risk	Method of concealment was not described - Children were distributed to an experimental group, with 30 children taking part in an activity, therapeutic play session before vaccination, and to a control group, with 30 children who did not receive any kind of previous emotional preparation.	High Risk	No allocation concealment - experimental group was recruited from kindergartens who registered to participate in TBH and control group from non-participating facilities.	High Risk	Method of concealment was not described	High Risk	Method of concealment was not described	High Risk	Method of concealment was not described	High Risk	No allocation concealment - Questionnaire is given to children who participated in TBH	High Risk	No allocation concealment - intervention group was recruited from kindergartens who participated in TBH and control group from a similar residential area
	D4	High Risk	The same questions were used in this, from pre-exposure to TBH, to post TBH lecture and post TBH participation. No allocation concealment.	Low risk	Each children were assigned a coded number without identifier.	Unclear risk	The allocation concealment method was not sufficiently described.	High Risk	Data were recorded in a Excel with information about identification of child and the reactions during vaccination. No allocation concealment was practiced	Low Risk	Children were assessed in small groups (up to 5 children) in the kindergarten during the week prior to the intervention and 1 week after visiting the TBH. No individual identifiers involved.	Low Risk	Child participants were randomly assigned to one of 4 groups: (a) a medical play group, where children were given adult and authentic medical materials to play with (ie, stethoscope, otoscope, tongue depressor, etc.); (b) a medical information video group, in which children were shown a taped medical play session; (c) a typical play group (i.e., played Connect Four), and (d) a control group whose participants viewed a video on safari life. Assignment to groups was determined using random sample numbers generated by a computer software program	Low Risk	The children were randomly assigned to participate in medical college activity or control group.	Low Risk	Child participants were randomly assigned to one of four groups: 1) a medical play group, where children were given adult and authentic medical equipment, 2) a medical information video group, in which children were shown a taped medical play session, 3) a typical play group (i.e., played Connect Four), and 4) a control group whose participants viewed a video on safari life. Assignment to groups was determined using random sample numbers generated by a computer software program	Low Risk	All the questionnaires are numbered which mean each participants are allocated a number and identifiers are not used.	Low Risk	The participants are divided into intervention group and control group.
Blinding of participants and personnel (performance bias)	D5	High Risk	No blinding - arranged for all participating schools a box with materials and a concrete lesson plan. Teachers with the intention that the c lesson was given before the visit to the Teddy Bear Hospital.	High Risk	No blinding - Questionnaire was administered right before TBH and immediately after TBH	High Risk	No blinding - Data were collected from participating children immediately before and immediately after the tour	High Risk	Blinding is not possible - children in experimental group have to participate in instructional therapeutic play sessions.	High Risk	No blinding - Children were assessed in small groups in the kindergarten during the week prior to the intervention and 1 week after visiting the TBH. Control children were interviewed twice as well and at a similar time interval.	High Risk	Blinding is not possible - children were either involved in a medical play group, a medical information video group, a typical play group (control group) whose participants viewed a video on safari life.	High Risk	Blinding is not possible - study group has to participate in medical college activity	High Risk	Blinding is not possible - children were either involved in a medical play group, a medical information video group, a typical play group (control group) whose participants viewed a video on safari life.	High Risk	No blinding - Every child was given a questionnaire to fill in, prior to attending Teddy Bear Hospital and after the experience.	High Risk	No blinding - a simple one-item visual analog scale of anxiety about hospitalization was assessed individually one day prior to the intervention and again a week after the intervention when the children were in kindergarten.
	D6	High Risk	No Blinding was applied . All the participants of TBH was included in selection with a prior lesson before participation. Despite that, a significant difference was observed between the intervention which is T2 . For question one, significant difference were seen between T2 and T1. T2 and T3, but no difference were seen between T1 and T3. Outcome is not influenced by the lack of blinding.	High Risk	Unease regarding the concept of TBH was provided to child ren immediately after TBH session to assess knowledge acquired and their changed feelings towards healthcare.	High Risk	The data was collected from participants before and immediately after the tour.	High Risk	After informing children and their legal guardians about the study and their participation, those selected for the experimental group were invited to play. No blinding	Low Risk	Children were assessed in small groups (up to 5 children) in the kindergarten during the week prior to the intervention and 1 week after visiting the TBH. No individual identifiers involved. Control children were interviewed twice as well and at a similar time interval (about 21 days). No blinding	Low Risk	Upon participating in their assigned activity, the children underwent a doctor's checkup where a trained nurse will conduct a behavioural assessment on the children.	Low Risk	A behavioural Observation Scale was used by researchers to evaluate the child's behaviour pre and post study group and control group activity participation.	Low Risk	The child's pulse reading (indication of anxiety) and Behavioural rating were conducted before and after the group activities were consulted in the clinical exam room.	High Risk	The child participants were required to fill in a 5 anxiometre pre and post intervention. No blinding was applied.	Low Risk	Assessment included a single one-item visual analog scale of anxiety about hospitalization. The scale included five facial expressions indicating a happy face on one side of the scale and a very distressed face on the other. Children had to choose the face that best described the way they would feel if they were required to be hospitalized [7]. This was assessed individually one day prior to the intervention and again a week after the intervention when the children were in kindergarten.
Blinding of outcome assessment (detection bias) (patient-reported outcomes)	D7	Unclear Risk	The study did not address blinding of outcome assessors	Unclear Risk	The study did not address blinding of outcome assessors	Unclear Risk	No mention about blinding - Two independent research assistants independently coded all of the participants' responses. Disagreements were resolved by discussion and consensus.	High Risk	Reactions were observed by researcher.	Unclear Risk	No mention about blinding - The interviews with the children were conducted by two trained students who were supervised by an experienced psychologist.	Low Risk	The nurse who rated the children's behaviour were unaware of the group assignments	High Risk	Reactions were observed by researcher. Behaviour Observation Scale)	Low Risk	A researcher, blind to the children's group assignment, rated the children's affect and behaviors. The developmental interview was administered to the parent by a researcher blind to the children's group assignment.	Unclear Risk	The study did not address blinding of outcome assessors	Unclear Risk	The study did not address blinding of outcome assessors
	D8	Unclear risk	Insufficient details for judgement	Unclear Risk	The study did not address this outcome	Unclear risk	Insufficient details for judgement	High Risk	The children's reaction were observed by the researcher themselves with no action at ing.	High Risk	The interviews with the children were conducted by two trained students who were supervised by an experienced psychologist.	High Risk	Because there will be trained nurse assigned to conduct the behavioural assessment on the children.	High Risk	The researchers evaluate the child non's behaviour by using Behavioural Observation Scale.	Low Risk	A researcher, blind to the children's group assignment, rated the children's affect and behaviors. before the assigned activity, during triage, and after the doctor's visit.	Low Risk	Questionnaires were used to assess the affect of TBH to reduce child's fear of doctors and hospitals.	Unclear Risk	Insufficient details for judgement
Blinding of outcome assessment (detection bias) (all-cause mortality)	D9	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	D10	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Incomplete outcome data addressed (attrition bias) (short-term 2-6 weeks)	D11	High Risk	The four questions of the questionnaire were analysed separately so no effect on the interpretation of the results are expected however missing data percentages ranges from 17.5% -19.89%. Data analysis was done only for completed questionnaire.	High Risk	We excluded children who had incomplete data, such as missing answers to any of the multiple-choice question in the survey. Children who completed the multiple-choice component but did not answer the open-ended questions were still included. (Excluded 42 out of 194 - 19.59%)	Low Risk	A total of 71 participants had parental consent and assented to participate; no data were excluded from the final analysis.	Low Risk	No data were excluded from the final analysis	Low Risk	Four reasons of in-appropriate agreed outlier data, 8 children were excluded from final analysis. (=16%)	Low Risk	No data were excluded from the final analysis	Low Risk	No data were excluded from the final analysis	Low Risk	No data were excluded from the final analysis	High Risk	Total 126 questionnaire, questionnaires numbered 86-126 were not found. Out of the 85 questionnaires, 28 of the questionnaire were not filled in fully. Therefore, only 57 fully completed questionnaires were considered for this study. (Excluded 47 out of 126 - =37%)	Low Risk	Three children on the intervention group (=5%) and four in the control group (8%) were excluded from the study because of one or more previous hospitalizations.
	D12	Low risk	There were missing data in terms of non balanced participation between each questions. But the questions were analysed separately and the insufficient details is not enough to produce clinically relevant impact on the study .	High Risk	We excluded children who had incomplete data, such as missing answers to any of the multiple choice question in the survey. Children who completed the multiple choice component but did not answer the open-ended questions were still included.	Low Risk	A total of 71 participants had parental consent and assented to participate; no data were excluded from the final analysis.	Low Risk	No incomplete data was detected	High Risk	Four reasons of in-appropriate agreed outlier data, 8 children were excluded from final analysis (N = 141). Attrition bias.	Unclear risk	The exclusion of data was not mentioned	Unclear risk	No incomplete data was detected	Low Risk	All data were included in final analysis	High Risk	All the questionnaires are numbered, unfortunately questionnaires numbered 86-126 were not found. Thus we have only 85 filled questionnaires. Out of the 85 questionnaires, 28 of the questionnaires were not filled in fully, some only completed part 1 and not part 2 while others were scribbled on. Therefore, only 57 fully completed questionnaires were considered for this study.	Low Risk	Three children on the intervention group and four in the control group were excluded from the study because of one or more previous hospitalizations.
Incomplete outcome data addressed (attrition bias) (long-term >6 weeks)	D13	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	D14	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Selective reporting (reporting bias)	D15	Low Risk	All outcomes of interests were reported	Low Risk	All outcomes of interests were reported	Low Risk	All outcomes of interests were reported	Low Risk	All outcomes of interests were reported	Low Risk	All outcomes of interests were reported	Low Risk	All outcomes of interests were reported	Low Risk	All outcomes of interests were reported	Low Risk	All outcomes of interests were reported	Low Risk	All outcomes of interests were reported	Low Risk	All outcomes of interests were reported
	D16	Low Risk	All the outcomes were reported in a prespecified way	Low Risk	All outcomes were reported in a prespecified way	Low Risk	All outcomes were reported in a prespecified way	Low Risk	All outcomes were reported after a statistical analysis through Fisher Exact Test.	Low Risk	All outcomes were reported in a prespecified way.	Low Risk	All outcomes were reported in a prespecified way	Low Risk	All outcomes were reported in a prespecified way	Low Risk	All outcomes were reported in prespecified way	Low Risk	All outcomes were reported in prespecified way	Low Risk	All outcomes were reported in prespecified way
Other bias	D17	Low Risk	None	High Risk	Marion M. Aw is the clinical mentor for the medical students in charge of the Teddy Bear Hospital, Singapore. Ong Lynn and Chua Khoon Hee were both involved in TBH as part of the organising committee during the course of this study.	Low Risk	None	Low Risk	None	Low Risk	None	Low Risk	None	Low Risk	None	Low Risk	None	Low Risk	None	Low Risk	None
	D18	Low Risk	None	Low Risk	None	Low Risk	None	Low Risk	None	Low Risk	None	Low Risk	None	Low Risk	None	Low Risk	None	Low Risk	None	Low Risk	None
Decided after discussion																					

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Supplementary file number 4 (list) - Detailed summary of the results of individual studies

Study (Author, Year, Country)	Intervention	Staff	Subjects	Outcomes	Tools	Results	Positive/Negative Outcome	Pre Scores	T1	T2	Remarks
								MEAN SCORES			
Burns-Nader 65, 2011, USA	·An medical play program, where children were given a doll and authentic medical equipment, 2) a medical information video group, in which children were shown a taped medical play session, 3) a typical play group (i.e., played Connect Four), or 4) a control group whose participants viewed a video on safari life.	·5 nurses completed behaviour assessment scale	·parents' self-reported level of worry about the doctor's visit positively correlated with their reports of the children's level of worry about the doctor's visit and the children's baseline pulse.	Anxiety level	The Child Drawing Hospital (Higher score indicates higher anxiety) Likert rate	Not a significant effect of type of activity on the children's level of anxiety, as depicted in their drawings	NO EFFECT				
				Level of fear	Fear Scale (5-points likert scale) (Higher score indicates higher level of fear)	1. Increase in fear for children in the medical play group, from baseline (1.44) to post-activity (1.67), suggesting that children in the medical play activity became more fearful after medical play 2. Increase in fear for children in the medical play activity from baseline (1.44) to after the doctor's visit (1.94), suggesting that children in the medical play activity were more fearful after the doctor's visit compared to their baseline fear 3. Decrease in fear for children who watched the medical information video from baseline (1.28) to post activity (1.11) and from baseline (1.28) to after the doctor's visit (1.10), suggesting that the medical information video was effective in reducing children's fear after the activity and after the doctor's visit.	NEGATIVE (MEDICAL PLAY) NEGATIVE (MEDICAL PLAY) POSITIVE (MEDICAL VIDEO) POSITIVE (MEDICAL VIDEO)	1.44 1.44 1.28 1.28	1.67 (post activity) 1.11 (post activity)	1.94 (after doctor's visit) 1.1 (after doctor's visit)	Relationship between variables
				Children's distress behaviours	Behaviour Observation Scale (BOS) (Higher score indicates less alert) Children's Hospital of Eastern Ontario Pain Scale (CHEOPS) (Higher score indicates more difficulty)	Children in the control play group regressed in their ratio, or alertness, from baseline (1.39) to post-activity (1.73) The video control group shifted from an active alert state at baseline (1.28) to a more reactive alert state (1.72) after watching the safari video. The decrease in alertness also occurred from baseline (1.28) to after the doctor's visit (1.67), suggesting that they were most alert before the activity The medical play group has higher mean than the rest of the activity groups on nurse's report on tension difficulty, suggests that children in the medical play group were more difficult than any other group while the nurse was performing triage procedures.	POSITIVE (TYPICAL PLAY) NEGATIVE (MEDICAL PLAY) NEGATIVE (NON-MEDICAL VIDEO) NEGATIVE (NON-MEDICAL VIDEO) NEGATIVE (MEDICAL PLAY)	1.94 1.28 1.28	1.72 (post activity) 1.72 (post activity)	1.67 (after doctor's visit)	
								FREQUENCIES			
Brown JC, 2012, USA	1. The medical play collage consisted of medical materials, such as cloth, tweezers, gauze, medical tape and small syringes without needles. 2. The control group received no intervention and continued doing whatever typical activity they were already doing	1. researchers	no significant association between parents talking about doctor visit to the children and parental perception about worry of child between control and intervention group	Children's behaviour and feelings	Behaviour Observation Scale (BOS) (Higher score indicates more alert) Faces Scale(6-points likert scale) (Higher score indicates more positive mood)	- The intervention group showed significant differences between time points pre-act and post-act and post-one and post-two. These differences were seen between pre-and post-one state (Z = -2.85, p = .004), pre-and post-one activity (Z = -2.83, p = .005), post-one and post-two activity (Z = -2.85, p = 0.14), pre-and post-one vocalization (Z= -2.83, p = .005), and post-one and post-two vocalization (Z = -2.83, p = .005). Behaviors declined after the intervention. Activity, alertness, vocalization, and affect had more activity and more vocalizations. After the intervention, levels of activity and vocalization went back down	State Activity Vocalization	POSITIVE POSITIVE POSITIVE	Inactive/alert: 3 (Pre) Active alert: 10 (Pre) Moderate: 11 (Pre) High: 1 (Pre) Low: 4 (Pre) High: 4 (Pre)	Inactive/alert: 1 (Post-one) Active alert: 14 (Post-one) Moderate: 9 (Post-One) High: 9 (Post-One) Low: 1 (Post-one) High: 9 (Post-one)	Active alert: 11 (Post-two) Moderate: 11 (Post-Two) High: 8 (Post-Two) Low: 7 (Post-two) High: 6 (Post-two)	Pre - before intervention Post one - right after intervention Post two - after doctor's visit
								MEAN SCORES			
Ottenheim M, 2018, Netherlands	On the day of the visit to the Teddy Bear Hospital every child brings their own teddy bear with an undetermined medical problem, again playing the role of the parent. With their teddy bear, the child will go through several rooms of the hospital mimicking a real hospital visit: the waiting room, the consulting room, the x- ray room, the operation room or the plaster room and finally the pharmacy. In these room the child and their teddy bear will be helped by several different doctors ans were supplied with educational materials around the theme "hospital", to prepare for the visit. The package included picture books, a stuffed animal, songs, colouring activities and explanatory posters.	These 'doctors' are actually student medical doctors playing the role of doctor	NA	1. Children's attitude towards visiting a hospital 2. Children's attitude towards doctors 3. Children's attitude towards feeling ill 4. Children's attitude towards hospitals	Seven-point likert scale (Higher score indicates more positive attitude)	Significant difference in anxiety level for doctors between the three measure moments. Gender, if children visited a hospital before or the interaction between the covariates did not differ significantly. The Bonferroni pairwise comparison showed significant differences between T2 (5.36) and T0 (5.00) (p=0.002), T2 (5.36) and T1 (5.00) (p=0.018) but not between T0 (5.00) and T1 (5.00) (p=1.00). Significant difference in anxiety level for falling ill between the three measure moments. Gender, if children visited a hospital before or the interaction between the covariates did not differ significantly. The Bonferroni pairwise comparison showed significant differences between T0 (2.72) and T1 (3.23) (p=0.000), T0 (2.72) and T2 (3.25) (p=0.000) but not between T1 (3.23) and T2 (3.25) (p=1.00). Gender had a significant effect on anxiety levels for going to the hospital (F with girls reporting on average higher levels of anxiety (low on the Likert scale) compared to boys.	POSITIVE POSITIVE	5.00 2.72 Boy: 4.42 Girl: 4.40	5.00 3.23 Boy: 4.83 Girl: 4.54	5.34 3.25 Boy: 4.63 Girl: 4.34	T0: before the lesson with the materials provided by the infoBeds. T1: after the lesson with the materials provided by the infoBeds but before the actual visit to the Teddy Bear Hospital. T2: after the visit to the Teddy Bear Hospital.
								FREQUENCY PERCENTAGE			
Ong L, 2018, Singapore	Teddy Bear Hospital is a simulated environment of a patient's visit to hospital. At each station or within each class at the primary school, the children were divided into 4 groups to participate in 4 concurrent stations in turns, before coming together for the final station. There are 4 stations depicting various clinical areas in a hospital, namely the Family Medicine Clinic, the Asthma clinic, the Orthopaedics Clinic and the Operation Theatre. Participants will learn about the various specialties depicted in each station through role-play as young doctors taking care of their teddy bears or stuffed toys. Each station will last 6 to 8 minutes. The entire Teddy Bear Hospital session (rotation through all 5 stations) takes approximately 45minutes to 1 hour.	The programme is fully run by the medical student volunteers.	Parents were not present during the TBH sessions	Childhood feelings towards healthcare services & willingness of health-seeking behaviour'	5-point likert scale (Higher score indicates more positive mood)	Children's feeling of happiness towards visiting the doctor increased from 82.5% to 97.4% (p-value =0.001) with the median changing from 2 to 3. Children who felt happy to visit the hospital increased from 70.2% to 73.4% (p-value =0.026) when measured on the same 5-point scale.	POSITIVE POSITIVE	82.50% 70.20%	94.40% 73.40%		
				Children's healthcare knowledge	5 health-care related questions	Children who answered all three knowledge-related questions correctly increased from 81 (32.1%) before attending TBH to 181 (73.4%) after attending TBH, with the median value improving from 2 to 3 (p-value <0.001)	POSITIVE	32.10%	73.40%		
								MEAN SCORES			
Dalley JS, 2016, Canada	45-minute tour ribbon spent most of their time learning about general medical equipment in the main exam room (e.g., needles, stethoscope, and otoscope). The tour also included a visit to the pediatric waiting room and physical therapy in the rehabilitation room. Children were taught coping strategies throughout the entire tour, including an initial introduction to the coping strategies at the beginning of the tour, followed by reminders and further rehearsal during each scheduled room of the tour. The tour script (see the Appendix) was based on the pediatric medical communication literature. The script utilized concrete language and short statements regarding the reason for medical procedures, the medical equipment used, and the sensations children would feel, with analogies included when applicable (e.g., using a stethoscope makes listening to your breath easier because it is louder, like turning up the volume on a TV). There is some consensus that children learn information best through demonstration [26] and therefore medical procedures were described and demonstrated by tour leaders on each child's stuffed animal	The tour was led by research assistants from the second author's Pediatric Pain, Health and Communication Lab and veterinary students from the Primary Healthcare Centre.	NA	Level of fear	The Children's Fear Scale (5-points likert scale) (Higher score indicates higher level of fear)	Children's fear ratings did not significantly change from pre to post intervention				
				Level of expected pain	The Faces Pain Scale – Revised (FPS-R [scale]) (Higher score indicates higher level of expected pain)	Mean expected pain ratings decreased significantly from Time 1 (M = 2.00, SD = 1.27) to post intervention at Time 2 (M = 1.64, SD = 1.16)	POSITIVE	2.00	1.64		
				Coping strategies	14-item coding system (Higher score indicates more coping strategies adopted)	Children reported a higher frequency of intervention-taught coping strategies at Time 2 (M = 0.90, SD = 0.72) versus Time 1 (M = 0.70, SD = 0.76), t(78) = −2.17, p = .036, and r = −.06, representing a significant, small effect	POSITIVE	0.7	0.9		
								MEAN SCORES			
Pontes JE, 2015, Brazil	Instructional therapeutic play sessions were carried out individually, using a doll, disposable syringes, needles, cotton, and bandages for venal puncture. The researcher showed the procedure in the doll and asked the child to participate and repeat the procedure at the end of the role play.	resecaher	NA	1. Children's body movement during procedure 2. Children's verbal expression during procedure 3. Children's emotional expression during procedure	Child's reaction checklist	Most reactions related to a greater acceptance of the procedure, such as staying still, smiling, collaborating, and being at ease, were more frequent in the experimental group. Fisher's exact test showed that the differences were significant for standing still (p=0.006) and collaborating spontaneously (p=0.001), with significance level at 5%. Some of the reactions that show less acceptance of the procedure were not observed in the experimental group, such as hitting, kicking, pushing, biting, pulling the hand of the professional, clinging to their parents, asking for cuddling and showing muscle stiffness, pallor, and perspiration. Fisher's test showed that differences were significant for pushing (p=0.006), clinging to their parents (p=0.006), asking for cuddling (p=0.024), and muscle stiffness (p=0.001).	POSITIVE				
								MEAN SCORES			
Leonhardt C, 2014, Germany	The visit to the TBH starts with registration when the child with his or her stuffed animal gives his or her name and describes the illness of the soft toy. Then the child can play in a wait- ing room until a contact person accompanies the child to his or her teddy doctor. Each teddy doctor wears a lab coat and has the typical doctor's equipment (e.g., stethoscope, clinical thermometer, syringes and adhesive tapes), in an examination room that is quite realistic, the teddy doctor carries out the physical examina- tion, depending on the individual illness of the soft toy. The children are encouraged to ask questions and take part in the examination. After the consultation, teddy may be referred to the teddy surgeons, to a dentist, to a place with ultrasound, scales or X-ray simulation (camera connected with a computer) or to the operating theatre. The teddy doctors converse with the children in an easy and responsive manner, depending on the child's verbal capability and emotional state. Finally, the children are given a prescription and go to the TBH pharmacy, where they are provided with tea, warming pans, ice bags, adhesive tapes or bandages from students from the School of Pharmacy. An important aspect of prescription for each child is the 'snuggle therapy' that is recommended for each individ- ual case. The child has the chance of loading into an ambulance and finally goes back to the waiting area, where the nursery teacher or parents wait.	teddy doctor (graduate medical students). The organizing committee is made up of approximately 20 stu- dents.	NA	Children's medical knowledge	Children's Medical Knowledge Scale (ChildMedKnow) (Higher score indicates better knowledge)	Children who visited the TBH had a significantly better knowledge of their body, health and disease (ChildMedKnow at time point 2) than the children from the control group (F(1, 118) = 144.07, p < 0.001). The effect size was large (partial η2 = 0.55). There was an influence of the baseline score as well (F(1, 118) = 213.15, p < 0.001, partial η2 = 0.65). Children who visited the TBH improved their knowledge of the different organs of the body significantly (F(1, 121) = 79.43, p < 0.001, partial η2 = 0.41). Children from the intervention group knew the organs of their body better after visiting the TBH (M = 6.5, SD = 1.1) than children in the control group (M = 3.1, SD = 1.2). Baseline ty- pe	POSITIVE POSITIVE	6.93	Intervention: 4.5 Control: 3.1		Time 1 - during the week prior to the intervention Time 2 - 1 week after visiting the TBH
						The ANCOVA also revealed here a significant baseline score as covariate (F(1, 121) = 13.05, p < 0.001, partial η2 = .46).					
								MEAN SCORES			
Burns-Nader S, 2013, USA	Child participants were randomly assigned to 1 of 4 groups: (a) a medical play group, where children were given a doll and authentic medical materials to play with (i.e., bandages, tongue depressors, stethoscope), (b) a medical information video group, in which children were shown a taped medical play session; (c) a typical play group, where children played a board game; or (d) a control group whose participants viewed a video on safari life.	researcher	NA	Children's mood	Mood Scale (5-points likert scale) (Higher score indicates less positive mood)	A significant interaction was found between time and type of activity on children's self-reported mood, F(3,363.122.31) = 3.63, P < .05. Paired samples t tests were conducted for post hoc comparisons within groups and revealed lower scores in mood (ie, shifting from the direction of very happy to more in the direction of happy) for children in the medical play group, from baseline to postactivity and from baseline to after the doctor's visit, suggesting the medical play activity lowered the children's mood after the activity (1.67) and after the doctor's visit (1.94). Baseline mean = 1.44 Higher scores in mood (ie, shifting from the direction of happy to more in the direction of very happy) for children who watched the medical information video from baseline to post activity and from baseline to after the doctor's visit, suggesting that the medical information video improved children's mood after the activity (1.11) and after the doctor's visit (1.1). Baseline mean = 1.28	NEGATIVE (MEDICAL PLAY) POSITIVE (MEDICAL VIDEO)	1.44 1.44 1.28 1.28	1.67 (Post-intervention) 1.11 (Post-intervention)	1.94 (Post-doctor's visit) 1.1 (Post-doctor's visit)	
				Children's behaviour during procedure	Nurse's rating on Children's behaviours (Higher score indicates more distress)	Significant effect for children's leg behaviors, F(3) = 2.85, P < .05, and the children's overall difficulty during the triage procedures, F(3) = 5.63, P < .01. Post hoc results for leg behaviors revealed significant difference between the medical play group and the nonmedical information video control, with the medical play group having a higher mean, suggesting that they were more likely to display distress behaviors with their legs than the nonmedical information video group during triage. The medical play group was also significantly different than the medical information video, typical play, and nonmedical information video groups on the nurses' overall rating of how difficult the children were during the triage procedures. The medical play group had a higher mean than the rest of the activity groups. This would suggest that children in the medical play group were more difficult than any other group while the nurse was performing triage procedures.	NEGATIVE (MEDICAL PLAY) NEGATIVE (MEDICAL PLAY)				
								FREQUENCY			
Victorine A, 2002, UK	The child will be taken through to Hygiene 101 where they are taught about cleanliness, hygiene and hand washing. After that, they will then attend the 'Talk of Fears' – where in a group of 5-6 children, they talk about their fears and group them in a pyramid. Working from the bottom of the pyramid, we talk around the fears using methods similar to systemic desensitization with relaxation and breathing exercises in between exposure. Children are encourage to think about their fears, how they would overcome them, then relax, seeing a picture of what the child fears, relax and eventually going through it and being able to talk about it and feel less scared about it. From there, the child is then taken through to 'Talk Consultation' where the role-play begins. Teddy doctor will take a history from poorly teddy and teddy's parent the child, then decide to perform some examination – inspection, palpation, percussion, auscultation, taking teddy's temperature, blood pressure, x-ray and CAT scan. Depending on the child's capability, the consultation can be modified to involve the parents – ie. child taking mum or dad to see teddy doctor, role play swap where teddy doctor becomes the patient and the child becomes the doctor to allow them hands- on experiences with medical equipments etc. After the consultation, teddy may be referred to the teddy	Teddy doctors were recruited from the medical school via MedSoc.	NA	1. Children's attitude towards doctor 2. Children's attitude towards hospital 3. Children' attitude towards TBH	7-point likert scale (Higher score indicates more positive mood)	1. Of the 57 children, 54 of them gave a score of 6 on the likert scales when asked if they would teddy bear research added 3 of them gave a score of 5. 2. When asked if they enjoyed Teddy Bear Hospital, 6 of the children gave a score of 5, 19 children gave a score of 6 and the balance 34 gave a score of 7. 3. Of the 57 children, when asked if Teddy Bear Hospital has helped them to relax on their fears, 50 of them answered yes and 2 not. Of the two that said not, both expressed no fear of doctors and hospital at all from the beginning. 4. All the 60 children answered yes when asked if they would attend Teddy Bear Hospital again, if teddy would attend Teddy Bear Hospital again, if the children will invite their friends to Teddy Bear Hospital and if teddy will bring teddy's friend to Teddy Bear Hospital.	POSITIVE	Likert score 7 (most happy and comfortable) Doctor: 0 Hospital: 0	Likert score 7 (most happy and comfortable) Doctor: 32 Hospital: 0		
								MEAN SCORES			
Bloch YH, 2008, Israel	Children were invited to attend a pre-simulated hospital where they would act as parents of their own teddy bears beforehand they were asked to think of a disease for their teddy bear. This activity was facilitated by the kindergarten teacher, but the choice of the illness or medical condition was the child's alone. All of the "teddy bear hospital" activities took place in the Geriatx University Medical Center's main courtyard. This allowed proximity to the hospital facilities, but in a pleasant atmosphere and without exposing the children to the possible risks of the wards, such as infectious diseases, other children's pain, etc. The acting physicians conducted a structured protocol of medical procedures including history taking, physical examinations, and various laboratory and diagnostic tests. The history was taken in a flexible and responsive manner according to the child's own verbal capability and using his or her own	Timely medical students at the Ben-Gurion University's Faculty of Health Sciences played the role of the physicians	NA	Children' s anxiety about hospitalization	One-item visual analog scale of anxiety about hospitalization (5 facial expressions) (Higher score indicates higher level of anxiety)	A repeated measures ANOVA revealed that time significantly interacted with group (F(1, 69) = 14.94, P <0.001). While baseline levels of anxiety were not different between groups (t(69) = 0.4, NS), children in the teddy bear hospital group reported significantly lower levels of anxiety than the control group at follow-up	POSITIVE	Teddy: 4.5 Control: 7.0	Teddy: 5.0 Control: 8.0		Pre and post TBH study

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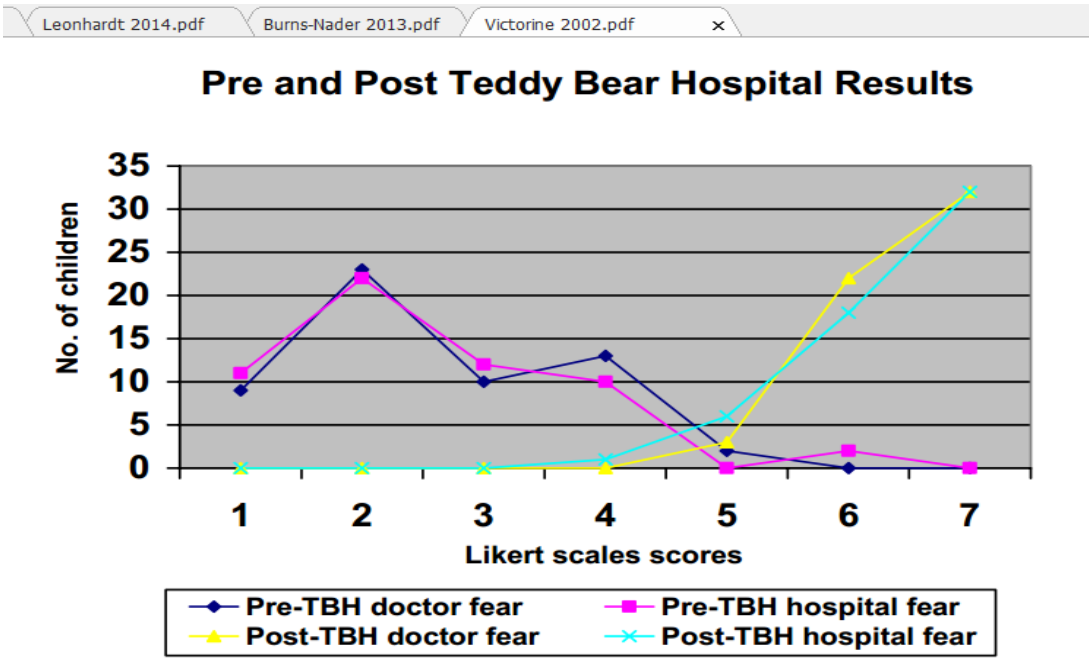
Burns Nader 2011.pdf

Table 5
Mean Scores (sd) and Between Group Differences for Nurse Rating of Child Distress Scale

Variable	Groups: Medical Play	Medical Play Video	Typical Play	Video Control	*F*	*p*
Facial	0.56 (.71)$_a$	0.39 (.61)$_a$	0.39 (.50)$_a$	0.33 (.59)$_a$	0.45	.72
Verbal	0.61 (.61)$_a$	0.56 (.62)$_a$	0.44 (.51)$_a$	0.33 (.49)$_a$	0.88	.46
Torso	1.28 (.46)$_a$	1.11 (.32)$_a$	1.17 (.38)$_a$	1.06 (.24)$_a$	1.25	.30
Touch	1.28 (.46)$_a$	1.17 (.38)$_a$	1.17 (.38)$_a$	1.00 (0)$_a$	1.87	.14
Legs	1.33 (.49)$_a$	1.11 (.32)$_{ab}$	1.17 (.38)$_{ab}$	1.00 (0)$_b$	2.85	.04*
Total	6.06 (1.8)$_a$	5.39 (1.88)$_a$	5.33 (1.33)$_a$	4.72 (1.18)$_a$	2.15	.10
Overall Stress	1.00 (1.14)$_a$	0.56 (1.25)$_a$	0.56 (.78)$_a$	0.28 (.46)$_a$	1.75	.17
Overall Pain	0.33 (1.19)$_a$	0.39 (.61)$_a$	0.50(.86)$_a$	0.39 (1.24)$_a$	0.09	.97
Overall Difficulty	0.56 (.78)$_a$	0.17 (.51)$_b$	0.00 (0)$_b$	0.00 (0)$_b$	5.63	.002**

Note. Mean scores that do not have the same subscript are significantly different at the .05 level. Lower scores are more optimal.
* p < .05, ** p < .0

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nhardt 2014.pdf Burns-Nader 2013.pdf Victorine 2002.pdf Yu

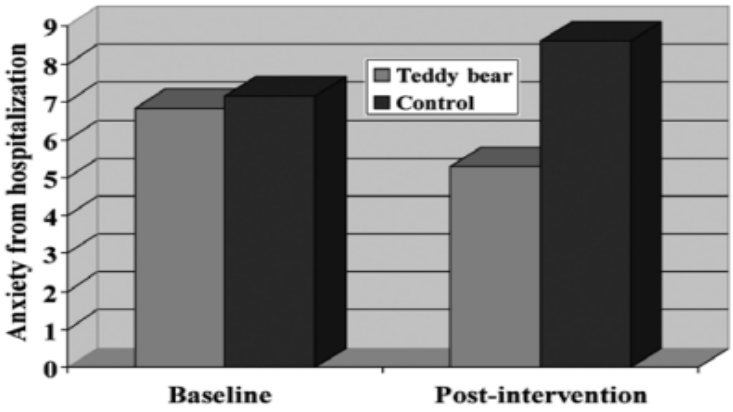


Figure 2. Effects of time of measurement on children's anxiety regarding hospitalization

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The effectiveness of pretend medical play in improving children's health outcomes and wellbeing: A systematic review

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The effectiveness of pretend medical play in improving children's health outcomes and wellbeing: A systematic review

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Strength and Limitations to this Study

1. This review examines the effectiveness of pretend medical play using the Teddy Bear Hospital (TBH) concept on children's health outcomes and wellbeing.

2. We follow established guidelines and identify 10 interventional studies to assess effectiveness.

3. The limitations include a lack of high-quality studies. The risk of bias for random sequence generation was unclear or high for nine of the 10 studies.

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Abstract

Background

The healthcare setting is stressful for many people, especially children. Efforts are needed to mitigate children's healthcare-related anxiety. Medical play using the Teddy Bear Hospital (TBH) concept can expose children to healthcare settings and help them develop positive experiences in these settings. In this role-playing game, children bring their soft toys and act as parents to the 'sick' teddies in a pretend hospital or clinic play setting. The objective of this systematic review is to evaluate the effectiveness of the TBH in improving children's health outcomes and wellbeing.

Methods

We searched the reference lists of included studies from four electronic databases (PubMed, CINAHL, Scopus, and Google Scholar) from inception until November 2020. We included pre-post, quasi-experimental, and case-control studies, as well as randomised controlled trials (RCTs) that discussed medical play using the TBH concept as an intervention. Studies that involved sick patients and used interventions unlike the TBH were excluded. We assessed the quality of the included studies using the Cochrane Collaborations' 'Risk of bias' tool.

Results

Ten studies were included in this systematic review. Five specifically investigated the TBH method, while the others involved the same concept of medical play. Only three studies were randomised controlled trials. All of the studies report more than one outcome – mostly positive outcomes. Two report lower anxiety levels post-intervention. Two found better healthcare knowledge, with one reporting increased feelings of happiness regarding visiting a doctor. Two studies found no change in anxiety or feelings, while another two found

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increased levels of fear and lowered mood after the medical play (which involved real medical equipment).

Conclusions

The practice of TBH has mostly positive outcomes, with lower anxiety levels and improved healthcare knowledge. Its effectiveness should be verified in future studies using a more robust methodology.

Systematic review registration: PROSPERO2019 CRD42019106355

Keywords: Teddy Bear Hospital, children, health knowledge, anxiety, education

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BACKGROUND

Children visit healthcare facilities for many reasons. Among these are regular check-ups, immunisation, or treatments. [1,2] These visits have been related to negative reactions among children.[2] Some studies report that approximately 30% of adults experience anxiety due to healthcare visits, to the extent that they are said to have 'white coat syndrome' and experience a rise in blood pressure during medical visits.[3] Like adults, children also fear the unknown. They tend to create fantasies and distort information due to a lack of accurate knowledge; this generates fear and anxiety.[4] It has been reported that children who are well-informed of a certain condition cope better and have less fear when facing the real situation.[5] Thus, various efforts have been made to ease children's anxiety and fear regarding the healthcare setting and healthcare personnel. The Teddy Bear Hospital (TBH) is one such effort that uses the concept of medical play. [6]

The TBH incorporates pretend hospital play. Children bring their teddy bears or soft toys to roleplay healthcare services. They also incorporate an element of medical play, which allows children to play with and explore medical equipment, mostly in the shape of toys but sometimes using real medical equipment. Medical play is a fun activity that incorporates medical themes. It is initiated by an adult but led by the child; its primary aim is to reduce anxiety in healthcare settings. [1,7]

In TBH role play, children act as the teddy bears' parents, bringing the sick teddy bear to visit a teddy doctor, who is usually played by a medical student or volunteer. The setting is in the form of a clinic or hospital where the play area is filled with pretend or real medical equipment,

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when appropriate. Most TBH games are aimed at healthy children, such as kindergarten students, or conducted during an open day in a specific healthcare centre for health education purposes. [5,8–12] The objective of this role-play intervention is to ease the child's anxiety in the healthcare setting and to simultaneously enhance their health knowledge. To our knowledge, there are currently no reviews available demonstrating that TBH programs have achieved these outcomes.

Therefore, the overall objective of this review is to determine the effectiveness of medical play in the TBH concept in improving children's health outcomes and wellbeing.

METHODS

Study design

This systematic review was conducted and reported based on the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines.[13] This review was registered with PROSPERO (CRD: 42019106355). The search strategy can be found in supplementary file 1 (S1).

Search strategy

We performed an electronic literature search in MEDLINE (PubMed), CINAHL, Scopus, and Google Scholar from inception to November 2020. A combination of Medical Subject Heading (MeSH) and free text terms were used for the keywords 'Teddy-Bear Hospital', 'Teddy-Bear Clinic', 'medical play', and 'child'. These keywords had to appear in conjunction in the title, abstract, or full text of the article. We also conducted an additional search on the references of relevant articles.

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Inclusion and exclusion criteria

We included all pre-post, quasi-experimental, and case-control studies, along with randomised controlled trials (RCTs), performed in both community and health settings and discussing the Teddy Bear Hospital or medical play interventions involving dolls or soft toys and medical equipment. The medical play interventions involving dolls or soft toys and medical equipment were included because they apply the same concept as the TBH. Furthermore, the included studies report on the primary outcomes, including health outcomes such as health knowledge and psychological impacts. We excluded studies that reported sick children in hospital, as well as studies that use interventions other than the TBH concept of soft toys and medical play equipment. As mentioned earlier, since most TBH studies are performed with healthy children, we excluded studies performed with sick children. The concept of TBH is focused on healthy children going for well-check visits, as these visits are common and the experiences may influence children's medical fears.[12] Furthermore, the psychology of sick, hospitalised children most likely differs from that of healthy children.[14]

Study selection

The articles obtained from the literature search were imported into EndNote©; duplicate articles were removed. The remaining articles were screened by two teams of reviewers, each composed of two reviewers: DR and AAR and NHS and CAT. Each reviewer independently screened the titles and abstracts of the articles and compared findings with the other team members. Any discrepancies regarding inclusion were discussed between the two team members, with the decision based on the consensus agreed after discussion. In cases where one team could not reach a consensus, the other team provided input. After a paper was

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identified as relevant, the full text was retrieved. These papers were independently reviewed again by each team member using a spreadsheet of the same format. The spreadsheets were later compared. If there were any discrepancies regarding the articles between the two reviewers in one team, the other team was consulted. The included articles were combined into one spreadsheet.

Data extraction

Data were extracted from the relevant studies by one reviewer (AAR) and checked by a second reviewer (CAT) on another spreadsheet. Disagreements at any stage were resolved through discussion with a third reviewer (NHS or DR). We extracted data on the following subjects: setting, objective, population, intervention/approach, measuring tools, outcomes of interest, N/N intervention, N/N control, results, and conclusion.

Quality assessment

The quality of the methodology used in the studies was assessed using The Cochrane Collaborations' 'Risk of bias' tool.[15] Each included study was assessed based on several characteristics: a) selection bias (randomisation and allocation concealment), b) performance bias (blinding of participants or personnel), c) detection bias (blinding of outcome assessment), d) attrition bias (incomplete outcome data, both long- and short-term), and e) reporting bias (selective reporting). All studies were critically appraised in duplicates by the two reviewers (AAR and CAT) and presented to the other research team members. We discussed any disagreements to reach a consensus and produce a combined assessment result.

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Data Analysis

The extracted data were analysed via narrative analysis. We did not perform meta-analysis because the tools for measuring outcomes varied among the studies; thus, the results could not be pooled.

RESULTS

Study selection

Our literature search yielded 16,306 studies from inception to November 2020. We identified 12,522 articles after excluding 3,784 duplicates. Of these, 12,473 were excluded. The excluded articles had titles and abstracts outside the scope of our study. The remaining 49 studies were available for retrieval in full text. Ten out of 49 studies were selected based on the inclusion and exclusion criteria. Studies were excluded because they investigated sick children in the hospital, because they did not use TBH or similar interventions, or because their outcomes did not align with the study objectives. The PRISMA flow diagram and the exclusion criteria are presented in Figure 1.

Description of included studies

Out of the 10 studies we included, three were RCTs, [1,16,17] three were non-randomised trials with a control group [5,8,18], and the remaining were pre-post studies. [9,10,19,20] These studies had population groups ranging from 30 to 543 participants aged 2–12. Six of the included studies involved TBH as an intervention,[5,8–12] while the remaining four studies used medical play interventions.[1,16–18] This medical play applied a similar concept; three studies involved dolls with real medical equipment,[1,17,18] while another study used medical play with a collection of medical equipment.[16] Four studies were conducted in North America (three in the United States of America[1,16,17] and one in Canada[20]), four in Europe (one in the United Kingdom,[10]

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one in the Netherlands,[9] and one in Germany[5]), and one each in Singapore, [19] Brazil, [18] and Israel. [8] The studies were conducted between 2002 and 2018 and involved mostly medical or veterinary students in TBH settings or researchers in medical play settings. An overview of the studies is given in supplementary file 2 (S2).

Quality of studies

The risk of bias for random sequence generation was unclear or high for 9 out of 10 studies; one study was an RCT.[1] All studies had a high risk of bias for allocation concealment and performance bias. Most studies did not conduct randomisation for their participants. Apart from that, concealing the intervention could not be done with TBH or medical play. Regarding detection bias, only 20% of the studies carried a low risk of bias, with the assessors blinded from the group arrangements of the interventions. [1,17] Three studies were found to have a high risk of reporting bias because a proportion of data was excluded from the results analysis.[9,10,19] All studies carried a low risk of bias for selective reporting; no other biases were found in the studies. Figure 2 shows the proportion of studies with a low risk, unclear risk, or high risk of bias. The assessment of biases in the individual studies is shown in supplementary file number 3 (S3).

Results of individual studies

Children's feelings

Overall, the studies showed mixed results. Seven studies measured the children's feelings, using the level of anxiety and fear as the outcomes. Four studies (one non-randomised trial with a control group and three pre-post studies) concluded that the children had more positive feelings with lower anxiety levels after the TBH.[8–10,19] Three more studies (two RCTs and one pre-post study) concluded otherwise – the RCTs[1,17] showed that medical play

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intervention had lowered the children's mood and increased the level of fear. The pre-post study[20] showed no effect of TBH on the children's level of fear.

Children's behaviour

Two RCTs found that the participants in the medical playgroup were more difficult during triage procedures than the children in the medical information group, the typical playgroup, or the control group.[1,17] Two other studies (one RCT and one non-randomised trial with a control group) showed positive outcomes for medical play intervention.[16,18] They reported the children as being more alert and demonstrating greater acceptance of the procedure (immunisation) after the intervention.[16,18] Thus, the results are mixed.

Children's healthcare knowledge

Two studies discuss the effects of the TBH on children's healthcare knowledge. Both of these studies (one non-randomised trial with a control group and one pre-post study) conclude that children had improved healthcare knowledge after the TBH intervention. [5,19] Therefore, these two studies investigating children's knowledge showed positive results.

A detailed summary of the results of individual studies is included in supplementary file number 4 (S4).

Others

Although we did not assess outcomes for parents, it is notable that one study reported that parents' level of worry significantly correlated with their children's perceived level of worry regarding seeing the doctor.[1] The other studies showed either no significant associations with parents talking to their children; and perception of the children's worry to see the doctor between the control and intervention group, or showed parents level of worry positively

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correlated with their perception of their children's level of worry .[1,16] The other eight studies either did not involve parents or did not report any findings on the parents.

DISCUSSION

TBH is a popular project that has been performed around the world, primarily to reduce children's fear and anxiety regarding healthcare professionals and hospitals through pretend medical play. The strength of our review is that it represents the first effort to summarise the effectiveness of TBH programs and the concept of medical play in improving children's health outcomes and wellbeing. Because the TBH has become a popular concept for educating children, a review examining its effectiveness is important.

In general, the studies included in this review show a high level of selection bias and performance bias. However, this finding is unavoidable due to the nature of the intervention, which involves children participating in physical activities related to the TBH intervention or medical play. Detection bias was high because most of the studies did not describe the methods for blinding the outcome assessors.

Our review suggests that medical play using the TBH concept has a positive effect on children's health outcomes and wellbeing, particularly for reducing anxiety and fear. This is probably because playful activities are important in child development and learning, especially regarding social interaction. Therefore, medical play helps them become more open to healthcare personnel and surroundings. [18,20] Exploration of medical items in medical play reportedly helps children express themselves more clearly, thereby improving psychological wellbeing, specifically in the healthcare setting.[16,17] This method of simulating

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a scenario is quite common and is used primarily for teaching in medical and healthcare environments.[21,22] For children specifically, the term 'experiential learning' is more commonly used; this practice has proven to be effective. For example, children actively participating in growing fruits and vegetables in their school gardens show greater improvements in nutrition knowledge while exhibiting increased preferences for eating fruits and vegetables, compared to those that were only taught nutrition but not exposed to gardening activities. [23,24]

Considering that medical play is supervised, this may constitute another factor in anxiety reduction. Another reason for this reduction is the non-threating environment of the TBH, which facilitates 'control over a possible threatening situation'.[8] Through medical play, TBH can also significantly improve children's healthcare knowledge. This is unsurprising – as mentioned earlier, children learn well via playful activities. Furthermore, the role-playing component enhances the effectiveness of learning.[5,18]

However, two studies conducted by Burns-Nader et al. found a contradictory effect of medical play intervention on children's health outcomes. This involved a reported increase in fear and a depressed mood by the children after the play therapy and after the doctor's visit (compared to baseline).[1,17] This discrepancy in the results suggests the need for further research to determine which components of medical play interventions affect the final outcomes of the intervention. The medical play in both studies involved real medical equipment without any toys – in contrast to the equipment used in a TBH setting, which consists mostly of toys made to resemble medical equipment.

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Further studies should explore whether the physical involvement of children in role-playing – or, rather, the medical information obtained through these activities – serves as the primary factor in reducing children's fear and anxiety.

We hypothesise that the lack of consistency in the studies' findings may result from having third-party personnel assess the children's responses, instead of having the children self-report outcomes. The children's distress behaviours were assessed by nurses during triage, which may have induced greater fear among children than in other studies, where the outcomes were usually reported by the children themselves or assessed by non-medical personnel.

Significant improvements in healthcare-related knowledge were seen in studies performed by Leonhardt et al. and Victorine et al.[5,10] This indicates that TBH may be an effective method in programmes aimed at raising children's awareness of health-related issues. For example, TBH can be used to teach children about healthy living to prevent childhood obesity, which is rising globally. Early educational intervention is important for preventing adverse health outcomes later in adulthood.[25] The implementation of these programmes, however, requires cooperation from various parties: healthcare professionals, parents, teachers, and authorities.

Further clinical implications of this study include the potential use of the TBH as a tool for teaching children about personal safety to prevent child sexual abuse, in view of the drastic rise in the number of child sexual abuse cases that is occurring nationally and globally.[26,27]

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Limitations

This review includes a relatively small number of studies; most are pre-post studies. This limitation may impact the generalisability of the findings for future implementations. There are also issues surrounding obtaining consent for the children – they are unable to consent to their involvement in these interventions. Furthermore, some studies found a negative effect on the children.

Implications of the review on practice and further research

We have found that the concept of the TBH – which involves pretend medical play and using toy or (occasionally) real medical equipment – lowers children's anxiety levels and improves the children's healthcare knowledge. Therefore, the TBH may constitute a potential public health strategy for improving health interventions, specifically as a health education tool for teaching pre-school and primary-school children about topics such as immunisation, a healthy lifestyle, or sexual and physical abuse.

Our review indicates that the quality of the studies was not satisfactory, as most of the studies were pre-post studies. Future studies should adopt a more robust methodology to examine the effects of the TBH concept of medical play intervention.

CONCLUSION

TBH interventions have a positive outcome for children's emotions and healthcare knowledge, according to most of the studies. Its effectiveness should be verified in future studies using a more robust methodology.

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List of abbreviations

TBH: Teddy Bear Hospital; PRISMA: Preferred Reporting Items for Systematic Reviews and Meta-Analyses; RCT: Randomised Controlled Trials

DECLARATIONS

Patient and Public Involvement Statement

No patients or members of the public were involved.

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Availability of data and materials

All data and materials analysed are included in this article and the supplemental files.

Competing interests

The authors declare no competing interests.

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Authors' contributions

AAR, CAT, NHS, and DR designed the study. RHS was responsible for conducting the literature search. All authors participated in the selection of the included studies. AAR, CAT, NHS, and DR were responsible for data extraction, study appraisal, data analysis, and results

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interpretation. All authors contributed to drafting the manuscript. The final manuscript was approved by all authors.

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REFERENCES

1. Burns-Nader, E. S. The effects of medical play on reducing fear, anxiety, and procedure distress in school-aged children going to visit the doctor. (University of Alabama Libraries, 2011).
2. Lerwick, J. L. Minimizing pediatric healthcare-induced anxiety and trauma. *World J. Clin. Pediatr.* **5**, 143–150 (2016).
3. Alves, L. M. M., Nogueira, M. S., Godoy, S. de, Hayashida, M. & Cárnio, E. C. Prevalence of white coat hypertension in primary health care. *Arq. Bras. Cardiol.* **89**, 28–35 (2007).
4. Santen, L. & Feldman, T. Teddy bear clinics: a huge community project. *MCN Am J Matern Child Nurs* **19**, 102–106 (1994).
5. Leonhardt, C., Margraf-Stiksrud, J., Badners, L., Szerencsi, A. & Maier, R. F. Does the 'Teddy Bear Hospital' enhance preschool children's knowledge? A pilot study with a pre/post-case control design in Germany. *J. Health Psychol.* **19**, 1250–1260 (2014).
6. Kaufman, J., Modak, M. & Moylan, S. the Teddy Bear Hospital In Australia. *J Paediatr Child Health* **48**, 541–542 (2012).
7. McCue, K. Medical Play: An Expanded Perspective. *Child. Health Care* **16**, 157–161 (1988).

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8. Bloch, Y. & Toker, A. Doctor, is my Teddy Bear Okay? A "Teddy Bear Hospital" as a Method to Reduce Children's Fear of Hospitalization. *Isr. Med. Assoc. J.* **10**, 597–9 (2008).
9. Ottenheim, M. & Sommeren, R. van. The Teddy Bear Hospital. How to influence the attitude of kindergarten children towards doctors, hospitals and falling ill. /paper/The-Teddy-Bear-Hospital.-How-to-influence-the-of-Ottenheim-Sommeren/778b811ab7b57a3da3a35d58a664ed3a64d7f774 (2018).
10. Victorine, A., Wong, Z., Geilani, J. & Charters, A. Teddy Bear Hospital (TBH)–reducing children's fear of doctors and hospital environment. *PEDIATRICS* **110**, e33 (2002).
11. Ong, L., Chua, K. H., Soh, J. Y. & Aw, M. M. H. Y. Healthcare encounters in young children: Impact of Teddy Bear Hospital, Singapore. *Asia Pac. Sch.* **3**, 24–30 (2018).
12. Dalley, J. S., Creary, P. R., Durzi, T. & McMurtry, C. M. An Interactive Teddy Bear Clinic Tour: Teaching Veterinary Students How to Interact with Young Children. *J Vet Med Educ* **44**, 302–315 (2017).
13. Moher, D., Liberati, A., Tetzlaff, J., Altman, D. G. & Group, T. P. Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. *PLOS Med.* **6**, e1000097 (2009).
14. Rennick, J. E. & Rashotte, J. Psychological outcomes in children following pediatric intensive care unit hospitalization: a systematic review of the research. *J. Child Health Care* **13**, 128–149 (2009).
15. Chapter 8: Assessing risk of bias in included studies. In: Higgins JPT, Green S (editors). Cochrane Handbook for Systematic Reviews of Interventions. Version 5.1.0. in.
16. Brown, J. C. Group medical play for reducing stress and improving mood in children going to visit the pediatrician. (University of Alabama Libraries, 2012).

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17. Burns-Nader, S., Hernandez-Reif, M. & Thoma, S. J. Play and Video Effects on Mood and Procedure Behaviors in School-Aged Children Visiting the Pediatrician. *Clin. Pediatr. (Phila.)* **52**, 929–935 (2013).

18. Pontes, J. E. D., Tabet, E., Folkmann, M. Á. dos S., Cunha, M. L. da R. & Almeida, F. de A. Therapeutic play: preparing the child for the vaccine. *Einstein São Paulo* **13**, 238–242 (2015).

19. Ong, L., Chua, K. H., Soh, J. Y. & Aw, M. M. H. Y. Healthcare encounters in young children: Impact of Teddy Bear Hospital, Singapore. *Asia Pac. Sch.* **3**, 24–30 (2018).

20. Dalley, J. S. & McMurtry, C. M. Teddy and I get a check up: A pilot educational intervention teaching children coping strategies for managing procedure-related pain and fear. *Pain Res. Manag.* (2015).

21. Jeffries, P. R., Bambini, D., Hensel, D., Moorman, M. & Washburn, J. Constructing Maternal-Child Learning Experiences Using Clinical Simulations. *J. Obstet. Gynecol. Neonatal Nurs.* **38**, 613–623 (2009).

22. Schreiber, J. *et al.* Experiential Learning With Children: An Essential Component of Professional Physical Therapy Education. *Pediatr. Phys. Ther.* **27**, 356–367 (2015).

23. Parmer, S. M., Salisbury-Glennon, J., Shannon, D. & Struempler, B. School Gardens: An Experiential Learning Approach for a Nutrition Education Program to Increase Fruit and Vegetable Knowledge, Preference, and Consumption among Second-grade Students. *J. Nutr. Educ. Behav.* **41**, 212–217 (2009).

24. Heim, S., Stang, J. & Ireland, M. A Garden Pilot Project Enhances Fruit and Vegetable Consumption among Children. *J. Am. Diet. Assoc.* **109**, 1220–1226 (2009).

25. Reilly, J. J. Childhood Obesity: An Overview: Childhood Obesity. *Child. Soc.* **21**, 390–396 (2007).

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26. Pereda, N., Guilera, G., Forns, M. & Gómez-Benito, J. The prevalence of child sexual abuse in community and student samples: A meta-analysis. *Clin. Psychol. Rev.* **29**, 328–338 (2009).

27. Singh, H. A., Yiing, W. W. & Nurani, H. N. K. Prevalence of childhood sexual abuse among Malaysian paramedical students. *Child Abuse Negl.* **20**, 487–492 (1996).

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Figures legends

Figure 1: PRISMA flowchart detailing the search strategy used.

Figure 2: The proportion of studies with low risk, unclear risk and risk of bias

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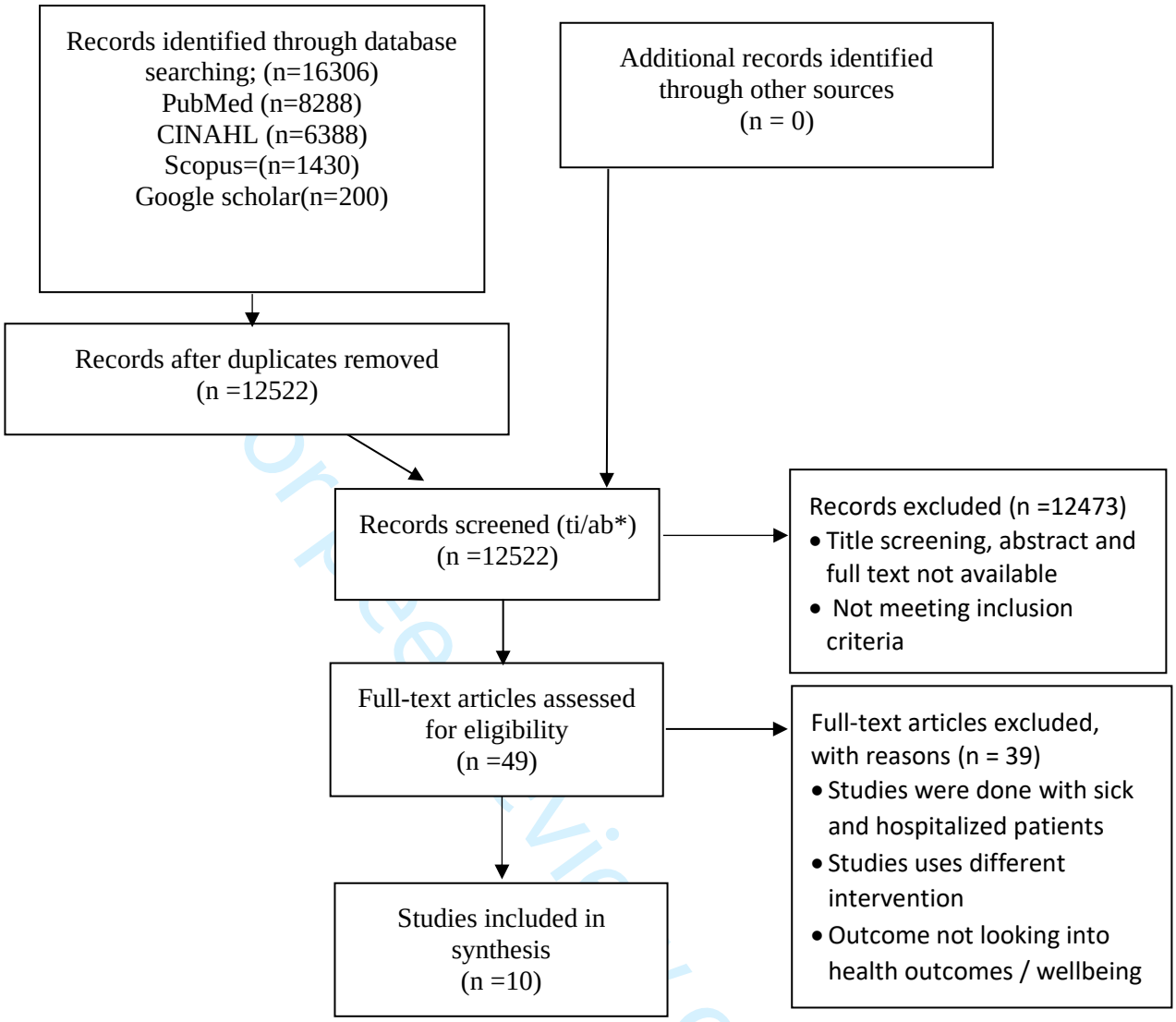


Figure 1: PRISMA flowchart detailing the search strategy used.

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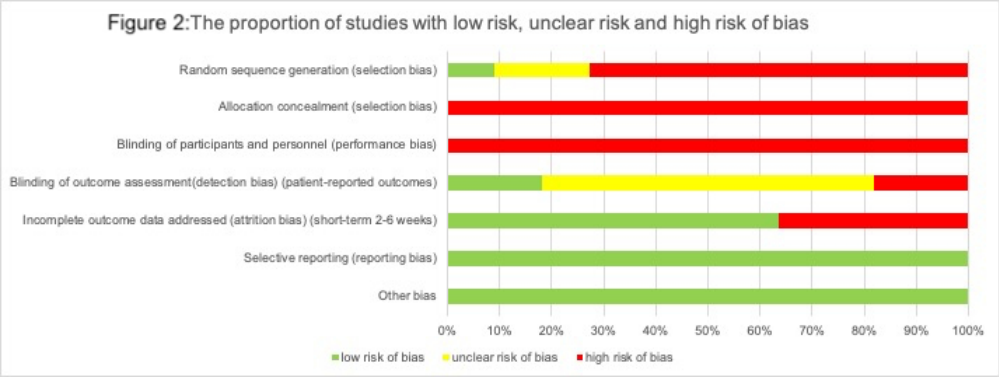


Figure 2: The proportion of studies with low risk, unclear risk and risk of bias

251x94mm (72 x 72 DPI)

Supplementary file number 1 (S1): **Search strategy (S1)**

1.	Pubmed	((((("Teddy bear hospital"[Title/Abstract] OR Teddy bear clinic*[Title/Abstract] OR "medical play"[Title/Abstract] OR ("Play and Playthings"[Mesh])) OR "teddy bear"[Title/Abstract] OR soft toy*[Title/Abstract] OR dolls[Title/Abstract])) AND ((((((("child"[MeSH Terms]) OR "child, preschool"[MeSH Terms]) OR child*[Title/Abstract] OR preschool child*[Title/Abstract] OR preschooler*[Title/Abstract] OR "play therapy/methods"[MeSH Major Topic]) OR pediatric*[Title/Abstract] OR boys[Title/Abstract] OR girls[Title/Abstract] OR kindergar*[Title/Abstract] OR toddler*[Title/Abstract] OR "pediatrics/education"[MeSH Major Topic]))))))	8288

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Supplementary file number 2 (S2): Overview of included studies.

Study (Author, Year, Country)	Population	Setting	Intervention	Outcome measured	Results	Outcome assessment
Randomised Controlled Trials						
Burns-Nader ES, 2011, USA (10)	Children aged 5.5 – 12 years	Paediatric clinic	3 intervention groups: 1. Medical play group (doll and authentic medical equipment) 2. Medical information video group 3. Typical play group	1. Children's anxiety level 2. Children's level of fear 3. Children's distress behaviours	• No effect of types of activity on children's anxiety level • Medical play group had increased level of fear after the activity and after doctor's visit • Medical play group was more difficult (higher mean score in nurses' overall rating of difficulty) during triage procedure as compared to medical information video group, typical play group and control group.	Self-report Self-report Third-party evaluation

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			Control: viewed video on safari life			
Burns-Nader S, 2013, USA (11)	Children aged 5.5 – 12 years	Paediatric clinic	3 intervention groups: 1. Medical play group (doll and authentic medical equipment) 2. Medical information video group 3. Typical play group Control: viewed video on safari life	1. Children's mood 2. Children's behaviours during procedures	• Medical play lowered children's mood after the activity and after the doctor's visit • Medical play group was more difficult (higher mean score in nurses' overall rating of difficulty) during triage procedure as compared to medical information video group, typical play group and control group.	Self-report Third-party evaluation
Brown JC, 2012, USA (9)	Children aged 3 - 5 years	Paediatric clinic	Medical collage group (medical play) Control: No intervention	1. Children's behaviours: - State - Affect - Activity - Vocalisation	• Medical collage (medical play) group was more alert, and had more activity and more vocalizations after the intervention. Levels of activity and vocalization went back down after doctor's visit	Third-party evaluation

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				- Fidgeting/Squirming 2. Children's feelings		Self-report
Non-randomised trials with controlled group						
Leonhardt C, 2014, Germany (5)	Children aged 4.5 - 5.5 years	Kindergarten	TBH Control: No intervention	Children's medical knowledge	• Children in TBH group had better knowledge than the control group one week after the intervention	Self-report
Bloch YH, 2008, Israel (6)	Children aged 3 – 6.5 years	TBH at healthcare centre	TBH Control: No intervention	Children's anxiety about hospitalisation	• Children in TBH group had lower levels of anxiety than the control group one week after the intervention	Self-report
Pontes JE, 2015, Brazil (12)	Children aged 3 - 6 years and 11 months	Paediatric clinic	Instructional therapeutic play (medical play) Control: No intervention	1. Children's body movement during procedure 2. Children's verbal expression during procedure 3. Children's emotional expression during procedure	• Instructional therapeutic play (medical play) group had greater acceptance of the procedure as compared to control group: - Staying still - Collaborating - Smiling - Being at ease	Third-party evaluation

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Pre-post Studies						
Ottenheim M, 2018, Netherlands (13)	Children aged 4 - 7 years	TBH at healthcare centre	TBH	1. Children's attitude towards visiting a hospital 2. Children's attitude towards doctors 3. Children's attitude towards falling ill 4. Children's attitude towards hospitals	• Reduced anxiety level towards doctors after TBH • Reduced anxiety level towards falling ill after introductory lesson and after TBH	Self-report
Ong L, 2018, Singapore (14)	Children aged 5 - 8 years	TBH at kindergarten and primary schools	TBH	1. Childhood feelings towards healthcare services 2. Willingness of health-seeking behaviour 3. Children's medical knowledge	• Children's feeling of happiness towards visiting the doctor increased after TBH • Children who felt happy to visit the hospital increased after TBH • Increased healthcare knowledge after TBH	Self-report
Dalley JS, 2016, Canada (15)	Children aged 5 - 10 years	TBH at healthcare centre	TBH	1. Children's level of fear 2. Children's level of expected pain 3. Children's coping strategies	• No effect on level of fear • Decreased mean expected pain ratings after TBH • Higher frequency of intervention-taught coping strategies after TBH	Self-report

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Victorine A, 2002, UK (7)	Children aged 2 – 12 years	TBH at hotel	TBH	1. Children's attitude towards doctor 2. Children's attitude towards hospital	• Increased proportion of children with positive feelings towards doctor and hospital after TBH **No further test was done to determine whether results were statistically significant*	Self-report

A summary of the author, year of publication, country of publication, population, setting, intervention, outcome measured, results and outcome assessment of each included studies.

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Supplementary file number 3 (S3): Quality Assessment of Studies.

Cochrane risk of bias		Ottenheim 2018		Lynn 2018		Dalley 2016		Pastor 2015		Leonhardt 2014		Burns 2013		Brown 2012		Burns-Nader 2011		Victorino 2002		Yuval 2008	
entry		Judgement	Support for judgement	Judgement	Support for judgement	Judgement	Support for judgement	Judgement	Support for judgement	Judgement	Support for judgement	Judgement	Support for judgement	Judgement	Support for judgement	Judgement	Support for judgement	Judgement	Support for judgement	Judgement	Support for judgement
Random sequence generation (selection bias)	D1	High Risk	No randomization - TRA kindergarten (age 4-7) participated in the Teddy Bear Hospital at Leiden University Medical Centre (Leiden, the Netherlands). Data was collected on 143 children attending elementary schools in the Leiden region.	High Risk	No randomization - cross sectional descriptive study Children who took part in TBH sessions between 28 March and 20 May 2016. All children aged 5 to 8 years old were recruited from kindergartens and primary schools which	High Risk	No randomization - A convenience sample was recruited from groups of children already scheduled to attend a Teddy Bear Clinic Tour at the Hill's Pet Nutrition Primary Healthcare Centre of the Ontario Veterinary College (OVC)	High Risk	No randomization - subjects were not randomly selected, and therefore this is not a true experimental study.	High Risk	No randomization - the allocation was not possible because the TBH is an established local project and any kindergarten in the area is free to decide whether it will take part or not. No randomization	Unclear Risk	The study did not specify how randomization was done - Children were randomly assigned to participate in a medical college activity or a control group.	Unclear Risk	The study did not specify how randomization was done - Children were randomly assigned to participate in a medical college activity or a control group.	Low Risk	Assignment to groups was determined using random sample numbers generated by a computer software program	High Risk	No randomization - The children in 1 to 3 classes were attending a scheduled, or walk-in, doctor's appointment at a general medical school pediatric clinic.	High Risk	No randomization - As a child comes in, a baby doctor in the reception area will register the child and help them fill in part one of the questionnaire. Participants are selected based on attendance.	High Risk	No randomization - Forty-one preschool children, age 4-6 years (mean ± 1 ± 0.7 years), from three kindergartens in Beer Sheva, Israel, took part in this study. Fifty preschool children, age matched and from a similar residential area, served as the control group.
	D2	High Risk	Participants were selected based on their attendance to Teddy Bear hospital. No randomization	High Risk	All the children from all the schools and kindergartens that has participated in TBH were recruited with the exception of children without parental consent. No randomization.	High Risk	A convenience sample was recruited from groups of children already scheduled to attend a Teddy Bear Clinic Tour at the Hill's Pet Nutrition Primary Healthcare Centre of the Ontario Veterinary College (OVC).	High Risk	subjects were not randomly selected, and therefore this is not a true experimental study. No randomization	High Risk	Pre-formulation was not possible because the TBH is an established local project and any kindergarten in the area is free to decide whether it will take part or not. No randomization	High Risk	Seventy-two caregiver-child dyads were recruited to participate, following institutional review board approval. The children (n = 33 males) were attending a scheduled, or walk-in, doctor's appointment at a pediatric clinic in a large urban region of the United States, for either a well-checkup (ie, school physical, vaccinations, etc) or for an ill-checkup (ie, respiratory infection, skin rash, etc). No randomization.	Unclear Risk	A total of 60 children ranging in age from three to	High Risk	No randomization. The children (n = 33 males) were attending a scheduled, or walk-in, doctor's appointment at a general medical school pediatric clinic.	High Risk	No randomization. As a child comes in, a baby doctor in the reception area will register the child and help them fill in part one of the questionnaire. Participants are selected based on attendance.	High Risk	No randomization. Forty-one preschool children, age 4-6 years (mean ± 1 ± 0.7 years), from three kindergartens in Beer Sheva, Israel, took part in this study. Fifty preschool children, age matched and from a similar residential area, served as the control group.
Allocation concealment (selection bias)	D3	High Risk	No allocation concealment - Questionnaire is given to children who participated in TBH	High Risk	No allocation concealment - Questionnaire is given to children who participated in TBH	High Risk	No allocation concealment - Questionnaire is given to children who participated in TBH	Unclear Risk	Method of concealment was not described - Children were distributed to an experimental group, with 30 children taking part in an activity, therapeutic play session before vaccination, and to a control group, with 30 children who did not receive any kind of previous emotional preparation.	High Risk	No allocation concealment - experimental group was recruited from kindergartens who registered to participate in TBH and control group from non-participating facilities.	High Risk	Method of concealment was not described	High Risk	Method of concealment was not described	High Risk	Method of concealment was not described	High Risk	No allocation concealment - Questionnaire is given to children who participated in TBH	High Risk	No allocation concealment - intervention group was recruited from kindergartens who participated in TBH and control group from a similar residential area
	D4	High Risk	The same questions were used in this, from pre-exposure to TBH, to post TBH lecture and post TBH participation. No allocation concealment.	Low risk	Each children were assigned a coded number without identifier.	Unclear risk	The allocation concealment method was not sufficiently described.	High Risk	Data were recorded in a Excel with information about identification of child and the reactions during vaccination. No allocation concealment was practiced	Low Risk	Children were assessed in small groups (up to 5 children) in the kindergarten during the week prior to the intervention and 1 week after visiting the TBH. No individual identifiers involved.	Low Risk	Child participants were randomly assigned to one of 4 groups: (a) a medical play group, where children were given adult and authentic medical materials to play with (ie, stethoscope, otoscope, tongue depressor, etc.); (b) a medical information video group, in which children were shown a taped medical play session; (c) a typical play group (i.e., played Connect Four), and (d) a control group whose participants viewed a video on safari life. Assignment to groups was determined using random sample numbers generated by a computer software program	Low Risk	The children were randomly assigned to participate in medical college activity or control group.	Low Risk	Child participants were randomly assigned to one of four groups: 1) a medical play group, where children were given adult and authentic medical equipment, 2) a medical information video group, in which children were shown a taped medical play session, 3) a typical play group (i.e., played Connect Four), and 4) a control group whose participants viewed a video on safari life. Assignment to groups was determined using random sample numbers generated by a computer software program	Low Risk	All the questionnaires are numbered which mean each participants are allocated a number and identifiers are not used.	Low Risk	The participants are divided into intervention group and control group.
Blinding of participants and personnel (performance bias)	D5	High Risk	No blinding - arranged for all participating schools a box with materials and a concrete lesson plan. Teachers with the intention that the c lesson was given before the visit to the Teddy Bear Hospital.	High Risk	No blinding - Questionnaire was administered right before TBH and immediately after TBH	High Risk	No blinding - Data were collected from participating children immediately before and immediately after the tour	High Risk	Blinding is not possible - children in experimental group have to participate in instructional therapeutic play sessions.	High Risk	No blinding - Children were assessed in small groups in the kindergarten during the week prior to the intervention and 1 week after visiting the TBH. Control children were interviewed twice as well and at a similar time interval.	High Risk	Blinding is not possible - children were either involved in a medical play group, a medical information video group, a typical play group (control group) whose participants viewed a video on safari life.	High Risk	Blinding is not possible - study group has to participate in medical college activity	High Risk	Blinding is not possible - children were either involved in a medical play group, a medical information video group, a typical play group (control group) whose participants viewed a video on safari life.	High Risk	No blinding - Every child was given a questionnaire to fill in, prior to attending Teddy Bear Hospital and after the experience.	High Risk	No blinding - a simple one-item visual analog scale of anxiety about hospitalization was assessed individually one day prior to the intervention and again a week after the intervention when the children were in kindergarten.
	D6	High Risk	No Blinding was applied . All the participants of TBH was included in selection with a prior lesson before participation. Despite that, a significant difference was observed between the intervention which is T2 . For question one, significant difference were seen between T2 and T1. T2 and T3, but no difference were seen between T1 and T3. Outcome is not influenced by the lack of blinding.	High Risk	Unease regarding the concept of TBH was provided to child ren immediately after TBH session to assess knowledge acquired and their changed feelings towards healthcare.	High Risk	The data was collected from participants before and immediately after the tour.	High Risk	After informing children and their legal guardians about the study and their participation, those selected for the experimental group were invited to play. No blinding	Low Risk	Children were assessed in small groups (up to 5 children) in the kindergarten during the week prior to the intervention and 1 week after visiting the TBH. No individual identifiers involved. Control children were interviewed twice as well and at a similar time interval (about 21 days). No blinding	Low Risk	Upon participating in their assigned activity, the children underwent a doctor's checkup where a trained nurse will conduct a behavioural assessment on the children.	Low Risk	A behavioural Observation Scale was used by researchers to evaluate the child's behaviour pre and post study group and control group activity participation.	Low Risk	The child's pulse reading (indication of anxiety) and Behavioural rating were conducted before and after the group activities were consulted in the clinical exam room.	High Risk	The child participants were required to fill in a 5 anxiometre pre and post intervention. No blinding was applied.	Low Risk	Assessment included a single one-item visual analog scale of anxiety about hospitalization. The scale included five facial expressions indicating a happy face on one side of the scale and a very distressed face on the other. Children had to choose the face that best described the way they would feel if they were required to be hospitalized [7]. This was assessed individually one day prior to the intervention and again a week after the intervention when the children were in kindergarten.
Blinding of outcome assessment (detection bias) (patient-reported outcomes)	D7	Unclear Risk	The study did not address blinding of outcome assessors	Unclear Risk	The study did not address blinding of outcome assessors	Unclear Risk	No mention about blinding - Two independent research assistants independently coded all of the participants' responses. Disagreements were resolved by discussion and consensus.	High Risk	Reactions were observed by researcher.	Unclear Risk	No mention about blinding - The interviews with the children were conducted by two trained students who were supervised by an experienced psychologist.	Low Risk	The nurse who rated the children's behaviour were unaware of the group assignments	High Risk	Reactions were observed by researcher. Behaviour Observation Scale)	Low Risk	A researcher, blind to the children's group assignment, rated the children's affect and behaviors. The developmental interview was administered to the parent by a researcher blind to the children's group assignment.	Unclear Risk	The study did not address blinding of outcome assessors	Unclear Risk	The study did not address blinding of outcome assessors
	D8	Unclear risk	Insufficient details for judgement	Unclear Risk	The study did not address this outcome	Unclear risk	Insufficient details for judgement	High Risk	The children's reaction were observed by the researcher themselves with no action at ing.	High Risk	The interviews with the children were conducted by two trained students who were supervised by an experienced psychologist.	High Risk	Because there will be trained nurse assigned to conduct the behavioural assessment on the children.	High Risk	The researchers evaluate the child non's behaviour by using Behavioural Observation Scale.	Low Risk	A researcher, blind to the children's group assignment, rated the children's affect and behaviors. before the assigned activity, during triage, and after the doctor's visit.	Low Risk	Questionnaires were used to assess the affect of TBH to reduce child's fear of doctors and hospitals.	Unclear Risk	Insufficient details for judgement
Blinding of outcome assessment (detection bias) (all-cause mortality)	D9	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	D10	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Incomplete outcome data addressed (attrition bias) (short-term 2-6 weeks)	D11	High Risk	The four questions of the questionnaire were analysed separately so no effect on the interpretation of the results are expected however missing data percentages ranges from 17.5% -19.89%. Data analysis was done only for completed questionnaire.	High Risk	We excluded children who had incomplete data, such as missing answers to any of the multiple-choice question in the survey. Children who completed the multiple-choice component but did not answer the open-ended questions were still included. (Excluded 42 out of 194 - 19.59%)	Low Risk	A total of 71 participants had parental consent and assented to participate; no data were excluded from the final analysis.	Low Risk	No data were excluded from the final analysis	Low Risk	Four reasons of in-appropriate agreed outlier data, 8 children were excluded from final analysis. (=16%)	Low Risk	No data were excluded from the final analysis	Low Risk	No data were excluded from the final analysis	Low Risk	No data were excluded from the final analysis	High Risk	Total 126 questionnaire, questionnaires numbered 86-126 were not found. Out of the 85 questionnaires, 28 of the questionnaire were not filled in fully. Therefore, only 57 fully completed questionnaires were considered for this study. (Excluded 47 out of 126 - =37%)	Low Risk	Three children on the intervention group (=5%) and four in the control group (8%) were excluded from the study because of one or more previous hospitalizations.
	D12	Low risk	There were missing data in terms of non balanced participation between each questions. But the questions were analysed separately and the insufficient details is not enough to produce clinically relevant impact on the study .	High Risk	We excluded children who had incomplete data, such as missing answers to any of the multiple choice question in the survey. Children who completed the multiple choice component but did not answer the open-ended questions were still included.	Low Risk	A total of 71 participants had parental consent and assented to participate; no data were excluded from the final analysis.	Low Risk	No incomplete data was detected	High Risk	Four reasons of in-appropriate agreed outlier data, 8 children were excluded from final analysis (N = 141). Attrition bias.	Unclear risk	The exclusion of data was not mentioned	Unclear risk	No incomplete data was detected	Low Risk	All data were included in final analysis	High Risk	All the questionnaires are numbered, unfortunately questionnaires numbered 86-126 were not found. Thus we have only 85 filled questionnaires. Out of the 85 questionnaires, 28 of the questionnaires were not filled in fully, some only completed part 1 and not part 2 while others were scribbled on. Therefore, only 57 fully completed questionnaires were considered for this study.	Low Risk	Three children on the intervention group and four in the control group were excluded from the study because of one or more previous hospitalizations.
Incomplete outcome data addressed (attrition bias) (long-term >6 weeks)	D13	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	D14	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Selective reporting (reporting bias)	D15	Low Risk	All outcomes of interests were reported	Low Risk	All outcomes of interests were reported	Low Risk	All outcomes of interests were reported	Low Risk	All outcomes of interests were reported	Low Risk	All outcomes of interests were reported	Low Risk	All outcomes of interests were reported	Low Risk	All outcomes of interests were reported	Low Risk	All outcomes of interests were reported	Low Risk	All outcomes of interests were reported	Low Risk	All outcomes of interests were reported
	D16	Low Risk	All the outcomes were reported in a prespecified way	Low Risk	All outcomes were reported in a prespecified way	Low Risk	All outcomes were reported in a prespecified way	Low Risk	All outcomes were reported after a statistical analysis through Fisher Exact Test.	Low Risk	All outcomes were reported in a prespecified way.	Low Risk	All outcomes were reported in a prespecified way	Low Risk	All outcomes were reported in a prespecified way	Low Risk	All outcomes were reported in prespecified way	Low Risk	All outcomes were reported in prespecified way	Low Risk	All outcomes were reported in prespecified way
Other bias	D17	Low Risk	None	High Risk	Marion M. Aw is the clinical mentor for the medical students in charge of the Teddy Bear Hospital, Singapore. Ong Lynn and Chua Khoon Hee were both involved in TBH as part of the organising committee during the course of this study.	Low Risk	None	Low Risk	None	Low Risk	None	Low Risk	None	Low Risk	None	Low Risk	None	Low Risk	None	Low Risk	None
	D18	Low Risk	None	Low Risk	None	Low Risk	None	Low Risk	None	Low Risk	None	Low Risk	None	Low Risk	None	Low Risk	None	Low Risk	None	Low Risk	None
Decided after discussion																					

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Supplementary file number 4 (list) - Detailed summary of the results of individual studies

Study (Author, Year, Country)	Intervention	Staff	Subjects	Outcomes	Tools	Results	Positive/Negative Outcome	Pre Scores	T1	T2	Remarks
								MEAN SCORES			
Burns-Nader 65, 2011, USA	·An medical play program, where children were given a doll and authentic medical equipment, 2) a medical information video group, in which children were shown a taped medical play session, 3) a typical play group (i.e., played Connect Four), or 4) a control group whose participants viewed a video on safari life.	·5 nurses completed behaviour assessment scale	·parents' self-reported level of worry about the doctor's visit positively correlated with their reports of the children's level of worry about the doctor's visit and the children's baseline pulse.	Anxiety level	The Child Drawing Hospital (Higher score indicates higher anxiety) Likert rate	Not a significant effect of type of activity on the children's level of anxiety, as depicted in their drawings	NO EFFECT				
				Level of fear	Fear Scale (5-points likert scale) (Higher score indicates higher level of fear)	1. Increase in fear for children in the medical play group, from baseline (1.44) to post-activity (1.67), suggesting that children in the medical play activity became more fearful after medical play 2. Increase in fear for children in the medical play activity from baseline (1.44) to after the doctor's visit (1.94), suggesting that children in the medical play activity were more fearful after the doctor's visit compared to their baseline fear 3. Decrease in fear for children who watched the medical information video from baseline (1.28) to post activity (1.11) and from baseline (1.28) to after the doctor's visit (1.10), suggesting that the medical information video was effective in reducing children's fear after the activity and after the doctor's visit.	NEGATIVE (MEDICAL PLAY) NEGATIVE (MEDICAL PLAY) POSITIVE (MEDICAL VIDEO) POSITIVE (MEDICAL VIDEO)	1.44 1.44 1.28 1.28	1.67 (post activity) 1.11 (post activity)	1.94 (after doctor's visit) 1.1 (after doctor's visit)	Relationship between variables
				Children's distress behaviours	Behaviour Observation Scale (BOS) (Higher score indicates less alert) Children's Hospital of Eastern Ontario Pain Scale (CHEOPS) (Higher score indicates more difficulty)	Children in the control play group regressed in their ratio, or alertness, from baseline (1.39) to post-activity (1.73) The video control group shifted from an active alert state at baseline (1.28) to a more reactive alert state (1.72) after watching the safari video. The decrease in alertness also occurred from baseline (1.28) to after the doctor's visit (1.67), suggesting that they were most alert before the activity The medical play group has higher mean than the rest of the activity groups on nurse's report on tension difficulty, suggests that children in the medical play group were more difficult than any other group while the nurse was performing triage procedures.	POSITIVE (TYPICAL PLAY) NEGATIVE (MEDICAL PLAY) NEGATIVE (NON-MEDICAL VIDEO) NEGATIVE (NON-MEDICAL VIDEO) NEGATIVE (MEDICAL PLAY)	1.94 1.28 1.28	1.72 (post activity) 1.72 (post activity)	1.67 (after doctor's visit)	
								FREQUENCIES			
Brown JC, 2012, USA	1. The medical play collage consisted of medical materials, such as cloth, tweezers, gauze, medical tape and small syringes without needles. 2. The control group received no intervention and continued doing whatever typical activity they were already doing	1. researchers	no significant association between parents talking about doctor visit to the children and parental perception about worry of child between control and intervention group	Children's behaviour and feelings	Behaviour Observation Scale (BOS) (Higher score indicates more alert) Faces Scale(6-points likert scale) (Higher score indicates more positive mood)	- The intervention group showed significant differences between time points pre-act and post-act and post-one and post-two. These differences were seen between pre-and post-one state (Z = -2.85, p = .004), pre-and post-one activity (Z = -2.83, p = .005), post-one and post-two activity (Z = -2.85, p = 0.14), pre-and post-one vocalization (Z= -2.83, p = .005), and post-one and post-two vocalization (Z = -2.83, p = .005). Behaviors declined after the intervention. Activity, alertness, vocalization, and affect had more activity and more vocalizations. After the intervention, levels of activity and vocalization went back down	State Activity Vocalization	POSITIVE POSITIVE POSITIVE	Inactive/alert: 3 (Pre) Active alert: 10 (Pre) Moderate: 11 (Pre) High: 1 (Pre) Low: 4 (Pre) High: 4 (Pre)	Inactive/alert: 1 (Post-one) Active alert: 14 (Post-one) Moderate: 9 (Post-One) High: 9 (Post-One) Low: 1 (Post-one) High: 9 (Post-one)	Active alert: 11 (Post-two) Moderate: 11 (Post-Two) High: 8 (Post-Two) Low: 7 (Post-two) High: 6 (Post-two)	Pre - before intervention Post one - right after intervention Post two - after doctor's visit
								MEAN SCORES			
Ottenheim M, 2018, Netherlands	On the day of the visit to the Teddy Bear Hospital every child brings their own teddy bear with an undetermined medical problem, again playing the role of the parent. With their teddy bear, the child will go through several rooms of the hospital mimicking a real hospital visit: the waiting room, the consulting room, the x- ray room, the operation room or the plaster room and finally the pharmacy. In these room the child and their teddy bear will be helped by several different doctors ans were supplied with educational materials around the theme "hospital", to prepare for the visit. The package included picture books, a stuffed animal, songs, colouring activities and explanatory posters.	These 'doctors' are actually student medical doctors playing the role of doctor	NA	1. Children's attitude towards visiting a hospital 2. Children's attitude towards doctors 3. Children's attitude towards feeling ill 4. Children's attitude towards hospitals	Seven-point likert scale (Higher score indicates more positive attitude)	Significant difference in anxiety level for doctors between the three measure moments. Gender, if children visited a hospital before or the interaction between the covariates did not differ significantly. The Bonferroni pairwise comparison showed significant differences between T2 (5.36) and T0 (5.00) (p=0.002), T2 (5.36) and T1 (5.00) (p=0.018) but not between T0 (5.00) and T1 (5.00) (p=1.00). Significant difference in anxiety level for falling ill between the three measure moments. Gender, if children visited a hospital before or the interaction between the covariates did not differ significantly. The Bonferroni pairwise comparison showed significant differences between T0 (2.72) and T1 (3.23) (p=0.000), T0 (2.72) and T2 (3.25) (p=0.000) but not between T1 (3.23) and T2 (3.25) (p=1.00). Gender had a significant effect on anxiety levels for going to the hospital (F with girls reporting on average higher levels of anxiety (low on the Likert scale) compared to boys.	POSITIVE POSITIVE	5.00 2.72 Boy: 4.42 Girl: 4.40	5.00 3.23 Boy: 4.83 Girl: 4.54	5.34 3.25 Boy: 4.63 Girl: 4.34	T0: before the lesson with the materials provided by the infoBeds. T1: after the lesson with the materials provided by the infoBeds but before the actual visit to the Teddy Bear Hospital. T2: after the visit to the Teddy Bear Hospital.
								FREQUENCY PERCENTAGE			
Ong L, 2018, Singapore	Teddy Bear Hospital is a simulated environment of a patient's visit to hospital. At each station or within each class at the primary school, the children were divided into 4 groups to participate in 4 concurrent stations in turns, before coming together for the final station. There are 4 stations depicting various clinical areas in a hospital, namely the Family Medicine Clinic, the Asthma clinic, the Orthopaedics Clinic and the Operation Theatre. Participants will learn about the various specialties depicted in each station through role-play as young doctors taking care of their teddy bears or stuffed toys. Each station will last 6 to 8 minutes. The entire Teddy Bear Hospital session (rotation through all 5 stations) takes approximately 45minutes to 1 hour.	The programme is fully run by the medical student volunteers.	Parents were not present during the TBH sessions	Childhood feelings towards healthcare services & willingness of health-seeking behaviour'	5-point likert scale (Higher score indicates more positive mood)	Children's feeling of happiness towards visiting the doctor increased from 82.5% to 97.4% (p-value =0.001) with the median changing from 2 to 3. Children who felt happy to visit the hospital increased from 70.2% to 73.4% (p-value =0.026) when measured on the same 5-point scale.	POSITIVE POSITIVE	82.50% 70.20%	94.40% 73.40%		
				Children's healthcare knowledge	5 health-care related questions	Children who answered all three knowledge-related questions correctly increased from 81 (32.1%) before attending TBH to 181 (73.4%) after attending TBH, with the median value improving from 2 to 3 (p-value <0.001)	POSITIVE	32.10%	73.40%		
								MEAN SCORES			
Dalley JS, 2016, Canada	45-minute tour ribbon spent most of their time learning about general medical equipment in the main exam room (e.g., needles, stethoscope, and otoscope). The tour also included a visit to the pediatric waiting room and physical therapy in the rehabilitation room. Children were taught coping strategies throughout the entire tour, including an initial introduction to the coping strategies at the beginning of the tour, followed by reminders and further rehearsal during each scheduled room of the tour. The tour script (see the Appendix) was based on the pediatric medical communication literature. The script utilized concrete language and short statements regarding the reason for medical procedures, the medical equipment used, and the sensations children would feel, with analogies included when applicable (e.g., using a stethoscope makes listening to your breath easier because it is louder, like turning up the volume on a TV). There is some consensus that children learn information best through demonstration [26] and therefore medical procedures were described and demonstrated by tour leaders on each child's stuffed animal	The tour was led by research assistants from the second author's Pediatric Pain, Health and Communication Lab and veterinary students from the Primary Healthcare Centre.	NA	Level of fear	The Children's Fear Scale (5-points likert scale) (Higher score indicates higher level of fear)	Children's fear ratings did not significantly change from pre to post intervention				
				Level of expected pain	The Faces Pain Scale – Revised (FPS-R [scale]) (Higher score indicates higher level of expected pain)	Mean expected pain ratings decreased significantly from Time 1 (M = 2.00, SD = 1.27) to post intervention at Time 2 (M = 1.64, SD = 1.16)	POSITIVE	2.00	1.64		
				Coping strategies	14-item coding system (Higher score indicates more coping strategies adopted)	Children reported a higher frequency of intervention-taught coping strategies at Time 2 (M = 0.90, SD = 0.72) versus Time 1 (M = 0.70, SD = 0.76), t(78) = −2.17, p = .036, and r = −.06, representing a significant, small effect	POSITIVE	0.7	0.9		
								MEAN SCORES			
Pontes JE, 2015, Brazil	Instructional therapeutic play sessions were carried out individually, using a doll, disposable syringes, needles, cotton, and bandages for venal puncture. The researcher showed the procedure in the doll and asked the child to participate and repeat the procedure at the end of the role play.	resecaher	NA	1. Children's body movement during procedure 2. Children's verbal expression during procedure 3. Children's emotional expression during procedure	Child's reaction checklist	Most reactions related to a greater acceptance of the procedure, such as staying still, smiling, collaborating, and being at ease, were more frequent in the experimental group. Fisher's exact test showed that the differences were significant for standing still (p=0.006) and collaborating spontaneously (p=0.001), with significance level at 5%. Some of the reactions that show less acceptance of the procedure were not observed in the experimental group, such as hitting, kicking, pushing, biting, pulling the hand of the professional, clinging to their parents, asking for cuddling and showing muscle stiffness, pallor, and perspiration. Fisher's test showed that differences were significant for pushing (p=0.006), clinging to their parents (p=0.006), asking for cuddling (p=0.024), and muscle stiffness (p=0.001).	POSITIVE				
								MEAN SCORES			
Leonhardt C, 2014, Germany	The visit to the TBH starts with registration when the child with his or her stuffed animal gives his or her name and describes the illness of the soft toy. Then the child can play in a wait- ing room until a contact person accompanies the child to his or her teddy doctor. Each teddy doctor wears a lab coat and has the typical doctor's equipment (e.g., stethoscope, clinical thermometer, syringes and adhesive tapes), in an examination room that is quite realistic, the teddy doctor carries out the physical examina- tion, depending on the individual illness of the soft toy. The children are encouraged to ask questions and take part in the examination. After the consultation, teddy may be referred to the teddy surgeons, to a dentist, to a place with ultrasound, scales or X-ray simulation (camera connected with a computer) or to the operating theatre. The teddy doctors converse with the children in an easy and responsive manner, depending on the child's verbal capability and emotional state. Finally, the children are given a prescription and go to the TBH pharmacy, where they are provided with tea, warming pans, ice bags, adhesive tapes or bandages from students from the School of Pharmacy. An important aspect of prescription for each child is the 'snuggle therapy' that is recommended for each individ- ual case. The child has the chance of loading into an ambulance and finally goes back to the waiting area, where the nursery teacher or parents wait.	teddy doctor (graduate medical students). The organizing committee is made up of approximately 20 stu- dents.	NA	Children's medical knowledge	Children's Medical Knowledge Scale (ChildMedKnow) (Higher score indicates better knowledge)	Children who visited the TBH had a significantly better knowledge of their body, health and disease (ChildMedKnow at time point 2) than the children from the control group (F(1, 118) = 144.07, p < 0.001). The effect size was large (partial η2 = 0.55). There was an influence of the baseline score as well (F(1, 118) = 213.15, p < 0.001, partial η2 = 0.65). Children who visited the TBH improved their knowledge of the different organs of the body significantly (F(1, 121) = 79.43, p < 0.001, partial η2 = 0.41). Children from the intervention group knew the organs of their body better after visiting the TBH (M = 6.5, SD = 1.1) than children in the control group (M = 3.1, SD = 1.2). Baseline ty- pe	POSITIVE POSITIVE	6.93	Intervention: 4.5 Control: 3.1		Time 1 - during the week prior to the intervention Time 2 - 1 week after visiting the TBH
						The ANCOVA also revealed here a significant baseline score as covariate (F(1, 121) = 13.05, p < 0.001, partial η2 = .46).					
								MEAN SCORES			
Burns-Nader S, 2013, USA	Child participants were randomly assigned to 1 of 4 groups: (a) a medical play group, where children were given a doll and authentic medical materials to play with (i.e., bandages, tongue depressors, stethoscope), (b) a medical information video group, in which children were shown a taped medical play session; (c) a typical play group, where children played a board game; or (d) a control group whose participants viewed a video on safari life.	researcher	NA	Children's mood	Mood Scale (5-points likert scale) (Higher score indicates less positive mood)	A significant interaction was found between time and type of activity on children's self-reported mood, F(3,363.122.31) = 3.63, P < .05. Paired samples t tests were conducted for post hoc comparisons within groups and revealed lower scores in mood (ie, shifting from the direction of very happy to more in the direction of happy) for children in the medical play group, from baseline to postactivity and from baseline to after the doctor's visit, suggesting the medical play activity lowered the children's mood after the activity (1.67) and after the doctor's visit (1.94). Baseline mean = 1.44 Higher scores in mood (ie, shifting from the direction of happy to more in the direction of very happy) for children who watched the medical information video from baseline to post activity and from baseline to after the doctor's visit, suggesting that the medical information video improved children's mood after the activity (1.11) and after the doctor's visit (1.1). Baseline mean = 1.28	NEGATIVE (MEDICAL PLAY) POSITIVE (MEDICAL VIDEO)	1.44 1.44 1.28 1.28	1.67 (Post-intervention) 1.11 (Post-intervention)	1.94 (Post-doctor's visit) 1.1 (Post-doctor's visit)	
				Children's behaviour during procedure	Nurse's rating on Children's behaviours (Higher score indicates more distress)	Significant effect for children's leg behaviors, F(3) = 2.85, P < .05, and the children's overall difficulty during the triage procedures, F(3) = 5.63, P < .01. Post hoc results for leg behaviors revealed significant difference between the medical play group and the nonmedical information video control, with the medical play group having a higher mean, suggesting that they were more likely to display distress behaviors with their legs than the nonmedical information video group during triage. The medical play group was also significantly different than the medical information video, typical play, and nonmedical information video groups on the nurses' overall rating of how difficult the children were during the triage procedures. The medical play group had a higher mean than the rest of the activity groups. This would suggest that children in the medical play group were more difficult than any other group while the nurse was performing triage procedures.	NEGATIVE (MEDICAL PLAY) NEGATIVE (MEDICAL PLAY)				
								FREQUENCY			
Victorine A, 2002, UK	The child will be taken through to Hygiene 101 where they are taught about cleanliness, hygiene and hand washing. After that, they will then attend the 'Talk of Fears' – where in a group of 5-6 children, they talk about their fears and group them in a pyramid. Working from the bottom of the pyramid, we talk around the fears using methods similar to systemic desensitization with relaxation and breathing exercises in between exposure. Children are encourage to think about their fears, how they would overcome them, then relax, seeing a picture of what the child fears, relax and eventually going through it and being able to talk about it and feel less scared about it. From there, the child is then taken through to 'Talk Consultation' where the role-play begins. Teddy doctor will take a history from poorly teddy and teddy's parent the child, then decide to perform some examination – inspection, palpation, percussion, auscultation, taking teddy's temperature, blood pressure, x-ray and CAT scan. Depending on the child's capability, the consultation can be modified to involve the parents – ie. child taking mum or dad to see teddy doctor, role play swap where teddy doctor becomes the patient and the child becomes the doctor to allow them hands- on experiences with medical equipments etc. After the consultation, teddy may be referred to the teddy	Teddy doctors were recruited from the medical school via MedSoc.	NA	1. Children's attitude towards doctor 2. Children's attitude towards hospital 3. Children' attitude towards TBH	7-point likert scale (Higher score indicates more positive mood)	1. Of the 57 children, 54 of them gave a score of 6 on the likert scales when asked if they would teddy bear research added 3 of them gave a score of 5. 2. When asked if they enjoyed Teddy Bear Hospital, 6 of the children gave a score of 5, 19 children gave a score of 6 and the balance 34 gave a score of 7. 3. Of the 57 children, when asked if Teddy Bear Hospital has helped them to relax on their fears, 50 of them answered yes and 2 not. Of the two that said not, both expressed no fear of doctors and hospital at all from the beginning. 4. All the 60 children answered yes when asked if they would attend Teddy Bear Hospital again, if teddy would attend Teddy Bear Hospital again, if the children will invite their friends to Teddy Bear Hospital and if teddy will bring teddy's friend to Teddy Bear Hospital.	POSITIVE	Likert score 7 (most happy and comfortable) Doctor: 0 Hospital: 0	Likert score 7 (most happy and comfortable) Doctor: 32 Hospital: 0		
								MEAN SCORES			
Bloch YH, 2008, Israel	Children were invited to attend a pre-simulated hospital where they would act as parents of their own teddy bears beforehand they were asked to think of a disease for their teddy bear. This activity was facilitated by the kindergarten teacher, but the choice of the illness or medical condition was the child's alone. All of the "teddy bear hospital" activities took place in the Geriatx University Medical Center's main courtyard. This allowed proximity to the hospital facilities, but in a pleasant atmosphere and without exposing the children to the possible risks of the wards, such as infectious diseases, other children's pain, etc. The acting physicians conducted a structured protocol of medical procedures including history taking, physical examinations, and various laboratory and diagnostic tests. The history was taken in a flexible and responsive manner according to the child's own verbal capability and using his or her own	Timely medical students at the Ben-Gurion University's Faculty of Health Sciences played the role of the physicians	NA	Children' s anxiety about hospitalization	One-item visual analog scale of anxiety about hospitalization (5 facial expressions) (Higher score indicates higher level of anxiety)	A repeated measures ANOVA revealed that time significantly interacted with group (F(1, 69) = 14.94, P <0.001). While baseline levels of anxiety were not different between groups (t(69) = 0.4, NS), children in the teddy bear hospital group reported significantly lower levels of anxiety than the control group at follow-up	POSITIVE	Teddy: 4.5 Control: 7.0	Teddy: 5.0 Control: 8.0		Pre and post TBH study

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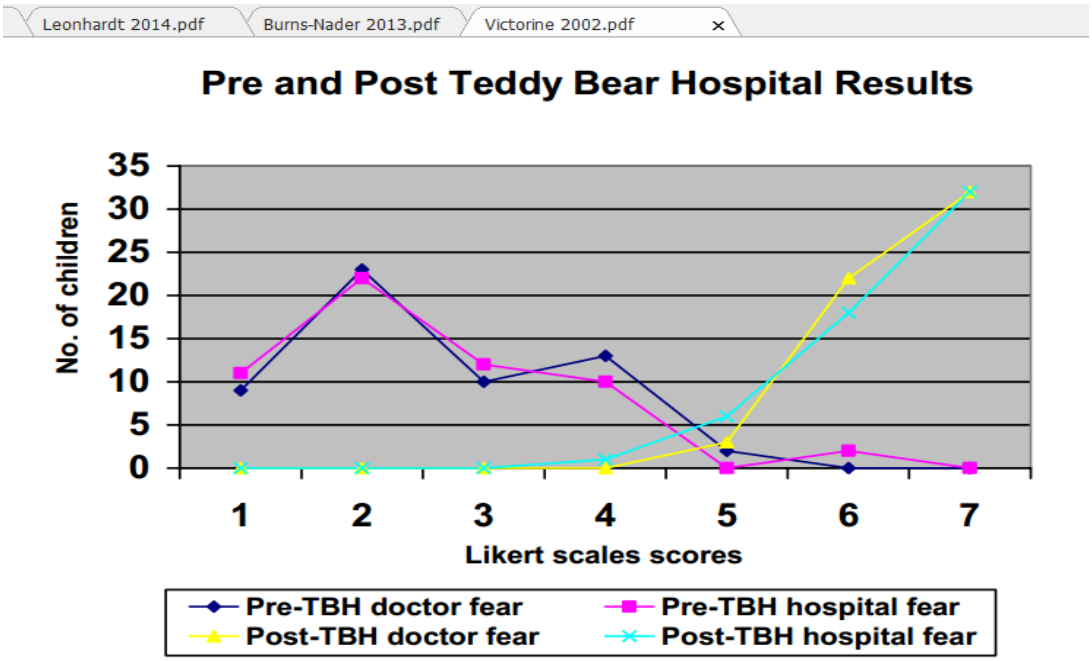
Burns Nader 2011.pdf

Table 5
Mean Scores (sd) and Between Group Differences for Nurse Rating of Child Distress Scale

Variable	Groups: Medical Play	Medical Play Video	Typical Play	Video Control	*F*	*p*
Facial	0.56 (.71)$_a$	0.39 (.61)$_a$	0.39 (.50)$_a$	0.33 (.59)$_a$	0.45	.72
Verbal	0.61 (.61)$_a$	0.56 (.62)$_a$	0.44 (.51)$_a$	0.33 (.49)$_a$	0.88	.46
Torso	1.28 (.46)$_a$	1.11 (.32)$_a$	1.17 (.38)$_a$	1.06 (.24)$_a$	1.25	.30
Touch	1.28 (.46)$_a$	1.17 (.38)$_a$	1.17 (.38)$_a$	1.00 (0)$_a$	1.87	.14
Legs	1.33 (.49)$_a$	1.11 (.32)$_{ab}$	1.17 (.38)$_{ab}$	1.00 (0)$_b$	2.85	.04*
Total	6.06 (1.8)$_a$	5.39 (1.88)$_a$	5.33 (1.33)$_a$	4.72 (1.18)$_a$	2.15	.10
Overall Stress	1.00 (1.14)$_a$	0.56 (1.25)$_a$	0.56 (.78)$_a$	0.28 (.46)$_a$	1.75	.17
Overall Pain	0.33 (1.19)$_a$	0.39 (.61)$_a$	0.50(.86)$_a$	0.39 (1.24)$_a$	0.09	.97
Overall Difficulty	0.56 (.78)$_a$	0.17 (.51)$_b$	0.00 (0)$_b$	0.00 (0)$_b$	5.63	.002**

Note. Mean scores that do not have the same subscript are significantly different at the .05 level. Lower scores are more optimal.
* p < .05, ** p < .0

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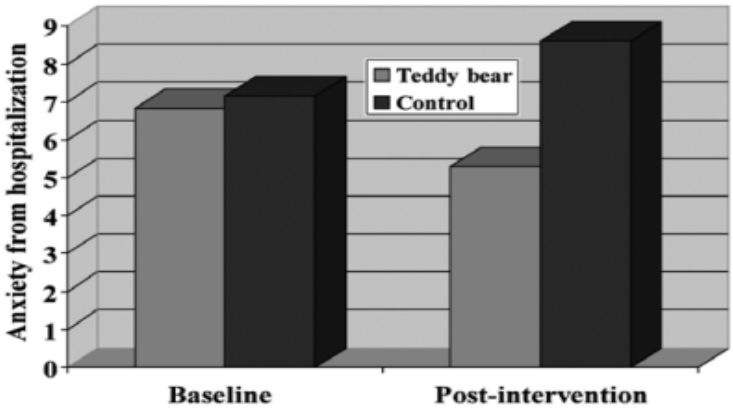


Figure 2. Effects of time of measurement on children's anxiety regarding hospitalization

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